FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Design and Implementation of a 10 Gbps High-Speed Transmitter

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Mestrado em Engenharia Eletrotécnica e de Computadores

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Resumo

Este trabalho apresenta o projeto, implementação física, e verificação pós-layout de um transmissor Current-Mode Logic (CML) de 10 Gb/s na tecnologia CMOS GlobalFoundries 55 nm LPx, visando atender aos requisitos elétricos da norma USB 3.2 Gen 2. O transmissor utiliza uma arquitetura segmentada com Feed-Forward Equalization (FFE) de três derivações, com pré-cursor, cursor principal e pós-cursor independentemente programáveis. Os requisitos de *preshoot* e *de-emphasis* do USB são traduzidos em coeficientes para cada uma das três derivações através de valores de corrente, implementados por fontes de corrente digitalmente controláveis, enquanto a terminação é implementada com um banco de resistências digitalmente controlável, permitindo ajuste para compensar variações de processo, tensão, temperatura e variações do processo de fabricação dos resistências.

Uma metodologia projeto modular e abutável é usada para o *layout*, desenvolvido utilizando posicionamento *common-centroid*, roteamento simétrico, dispositivos *dummy*, e *guard rings* contínuos para reduzir desajustes e efeitos dependentes do *layout*. O transmissor completo ocupa 176 μm $\times$ 82 μm. Simulações pré- e pós-extração são realizadas utilizando um canal de referência fornecido pela USB-IF com 21.6 dB de perda de inserção na frequência de Nyquist de 5 GHz. A extração parasítica afeta significativamente os tempos de transição de saída, o peso efetivo das derivações FFE, e a configuração da terminação, sendo necessário *tuning* das fontes de corrente e terminação para manter o desempenho em diferentes *corners* operacionais.

Os resultados mostram que a arquitetura segmentada fornece uma faixa de ajuste suficiente para recuperar pontos de operação compatíveis com USB 3.2 Gen 2 após a implementação física, mas também demonstram que os códigos de equalização e terminação necessários podem se tornar dependentes do processo e das condições de operação. Em particular, a equalização pós-extração não pode ser representada por um mapeamento fixo único em todos os *corners* PVT avaliados, o que destaca a importância dos mecanismos de calibração e a versatilidade da arquitetura proposta. O trabalho demonstra, portanto, uma metodologia completa de projeto para um transmissor configurável de 10 Gbps, desde o esquemático até ao *layout*, identificando a calibração automática, caracterização completa de *jitter* e Bit Error Rate (BER), e medição em silício como os passos principais necessários para validação experimental completa de conformidade.

Palavras-chave: Transmissor de alta velocidade, USB 3.2 Gen 2, Feed-Forward Equalization, Terminação, Projeto de circuitos integrados, Layout, Current Mode Logic.

Abstract

This dissertation presents the design, physical implementation, and post-layout verification of a 10 Gb/s Current-Mode Logic (CML) Transmitter (TX) targeting the USB 3.2 Gen 2 electrical requirements in GlobalFoundries 55 nm LPx CMOS technology. The transmitter employs a segmented design, with three-tap Feed-Forward Equalization (FFE), composed of independently programmable pre-cursor, main-cursor, and post-cursor paths. The USB preshoot and de-emphasis requirements are translated into tap coefficients and current targets, realized by digitally controllable current sources, while termination is implemented with a digitally controllable resistor bank, providing tuning to compensate for process, voltage, temperature, and resistor process variations.

A modular and abuttable layout methodology is developed using common-centroid placement, symmetric routing, dummy devices, and continuous guard rings to reduce mismatch and layout-dependent effects. The complete transmitter occupies 176 μm $\times$ 82 μm. Pre- and post-extraction simulations are performed using a USB-IF reference channel with 21.6 dB insertion loss at the 5 GHz Nyquist frequency. Parasitic extraction significantly affects the output transition times, effective FFE tap weighting, and termination configuration, and tuning is required to maintain performance across different process corners.

The results show that the segmented architecture provides sufficient tuning range to recover USB 3.2 Gen 2-compatible operating points after physical implementation, but also demonstrate that the required equalization and termination codes can become dependent on process and operating conditions. In particular, the desired post-extraction equalization cannot be represented by a single fixed preset mapping across all evaluated PVT corners, which highlights the importance of calibration mechanisms and the versatility of the proposed architecture. The work therefore demonstrates a complete schematic-to-layout methodology for a configurable 10 Gb/s transmitter while identifying automatic calibration, complete jitter and BER characterization, and silicon measurement as the principal steps required toward full experimental compliance validation.

Keywords: High-speed transmitter, USB 3.2 Gen 2, Feed-Forward Equalization, Termination, Integrated-circuit design, Layout, Current-Mode Logic.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) provide a global framework to achieve a better and more sustainable future for all. It includes 17 goals[1] to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

The specific Sustainable Development Goals mentioned have the following names:

SDG 7 Ensure access to affordable, reliable, sustainable and modern energy for all

SDG 9 Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

SDG	Target	Contribution	Performance Indicators and Metrics
7	7.3	The characterization and consideration of transmitter power consumption contribute to the development of energy-conscious high-speed communication electronics. Maintaining multi-gigabit data transmission while controlling circuit power is relevant to improving the energy efficiency of electronic communication systems.	Post-layout power consumption of 24.42 mW for EQ3 at the nominal condition; 10 Gb/s operating data rate; comparison of power across operating configurations/corners.
9	9.5	The development and physical implementation of a configurable 10 Gb/s CMOS transmitter contributes to research and technological innovation in high-speed integrated circuits and communication interfaces. The work investigates programmable equalization and termination techniques and validates their operation after parasitic extraction and across process and operating variations.	10 Gb/s data rate; complete schematic-to-layout implementation; USB 3.2 Gen 2 electrical-performance evaluation; operation across PVT and resistor corners; $176\,\mu\text{m} \times 82\,\mu\text{m}$ layout area.

This work contributes primarily to SDG 9 – Industry, Innovation and Infrastructure, Target 9.5, by advancing research and technological development in high-speed integrated communication circuits. The design, physical implementation, and post-layout verification of a configurable 10 Gb/s CMOS transmitter demonstrate the application of integrated-circuit design techniques to

[1] https://sdgs.un.org/goals

improve the performance and robustness of modern serial interfaces. Its contribution can be assessed through technical metrics such as the achieved 10 Gb/s data rate, the complete schematic-to-layout implementation, the 176 µm $\times$ 82 µm occupied area, and the verification of equalization, termination, and signal integrity across process and operating variations. The work also relates to SDG 7 – Affordable and Clean Energy, Target 7.3, through its consideration of energy efficiency in high-speed electronic communication. Although the transmitter is not directly associated with renewable-energy generation, reducing the energy required for high-data-rate communication is relevant to the broader objective of improving the efficiency of electronic systems. In this context, the principal performance metric is the transmitter power consumption, with the representative post-layout EQ3 configuration consuming 24.42 mW at 10 Gb/s, corresponding to approximately 2.44 pJ/bit. Together, these metrics provide measurable indicators of the work's contribution to technological innovation while also considering the energy cost associated with increasing communication bandwidth.

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List of Acronyms

AC Alternating Current
BEOL Back-End-of-Line
BER Bit Error Rate
CML Current-Mode Logic
CMOS Complementary Metal-Oxide-Semiconductor
CTLE Continuous-Time Linear Equalization
DC Direct Current
DCD Duty Cycle Distortion
DE De-emphasis
DFE Decision Feedback Equalization
DJ Deterministic Jitter
DRC Design Rule Checking
DRM Design Rules Manual
EM Electromigration
EMI Electromagnetic Interference
ESD Electrostatic Discharge
FEOL Front-End-of-Line
FF Fast-Fast
FFE Feed-Forward Equalization
FIR Finite Impulse Response
IC Integrated Circuit
ISI Inter-Symbol Interference
LFSR Linear Feedback Shift Register
LOD Length of Diffusion
LVS Layout Versus Schematic
M3 Metal 3
M4 Metal 4
M5 Metal 5
MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor
NRZ Non-Return-to-Zero

PAM2 2-Level Pulse Amplitude Modulation
PAM4 4-Level Pulse Amplitude Modulation
PDK Process Design Kit
PDN Pull-Down Network
PHY Physical Layer
PRBS Pseudo-Random Bit Sequence
PS Preshoot
PUN Pull-Up Network
PVT Process, Voltage, and Temperature
RJ Random Jitter
RMS Root Mean Square
RVT Regular Threshold Voltage
RX Receiver
SCR Silicon-Controlled Rectifier
SI Signal Integrity
SLVT Super Low Threshold Voltage
SNR Signal-to-Noise Ratio
SS Slow-Slow
SST Source-Series Termination
STI Shallow Trench Isolation
TJ Total Jitter
TT Typical-Typical
TX Transmitter
UI Unit Interval
USB Universal Serial Bus
USB-IF USB Implementers Forum
VMD Voltage-Mode Driver
WPE Well Proximity Effect

Chapter 1

Introduction

1.1 Context and Motivation

The continuous growth in data throughput demanded by modern electronic systems has made high-speed communication links a fundamental enabler of contemporary digital platforms. Applications ranging from consumer peripherals to data-center interconnects require sustained data rates in the multi-gigabit-per-second range, while simultaneously satisfying strict constraints on power consumption, Electromagnetic Interference (EMI), and physical interconnect complexity. As semiconductor technology has scaled, transistor switching speeds have improved considerably; however, the electrical performance of interconnects has not kept pace with this evolution, turning the communication channel itself into the dominant bottleneck of many high-performance systems.

Among the standards that rely on high-speed serial signaling, the Universal Serial Bus (USB) has become one of the most widely adopted interfaces for data transfer between electronic devices, owing to its broad industry support, backward compatibility, and continuous evolution toward higher data rates. The USB 3.2 Gen 2 revision, commercially marketed as SuperSpeed USB, specifies a maximum data rate of 10 Gbps, achieved through Non-Return-to-Zero (NRZ) signaling transmitted over a differential pair of conductors. Meeting this data rate places considerable demands on the Transmitter (TX) circuitry responsible for driving the signal onto the channel, particularly with respect to Signal Integrity (SI), output impedance control, and equalization capability, all of which must be guaranteed across Process, Voltage, and Temperature (PVT) variation.

This dissertation addresses the design and implementation of a TX driver targeting the USB 3.2 Gen 2 electrical specification, encompassing both the circuit-level design of the driver and its physical layout implementation in a 55 nm Complementary Metal-Oxide-Semiconductor (CMOS) technology node.

1.2 Problem Statement

As channel data rates increase, the frequency-dependent losses inherent to practical electrical interconnects become increasingly significant, giving rise to Inter-Symbol Interference (ISI) that closes the eye diagram observed at the Receiver (RX) and ultimately increases the Bit Error Rate (BER) of the link. To compensate for these losses, the TX driver must incorporate equalization, typically implemented as a multi-tap Finite Impulse Response (FIR) filter structure known as Feed-Forward Equalization (FFE), capable of pre-distorting the transmitted waveform in accordance with the Preshoot (PS) and De-emphasis (DE) settings defined by the USB 3.2 specification.

Beyond the equalization capability itself, the driver must simultaneously satisfy a demanding set of electrical requirements, including a tightly controlled differential output impedance, a configurable output voltage swing, and low TX-induced jitter, all while operating at a Unit Interval (UI) of 100 ps. These requirements must be met not only under nominal conditions, but across the full range of PVT corners and passive-device process variation to which the fabricated Integrated Circuit (IC) will be subject. Achieving this level of robustness requires careful circuit-level design of the driver's output signals generating mechanisms, as well as a physical layout methodology capable of preserving the electrical symmetry and matching properties required for differential, high-speed operation, where layout-induced parasitics can have a first-order impact on overall performance.

The central problem addressed in this dissertation is therefore twofold: first, the design of a segmented, digitally configurable Current-Mode Logic (CML) driver architecture capable of realizing the equalization and impedance requirements of the USB 3.2 Gen 2 standard; and second, the development of a modular, abuttable layout capable of preserving this performance once physical, layout-dependent effects are taken into account.

1.3 Objectives

The main objective of this dissertation is the design, implementation, and verification of a 10 Gbps TX driver targeting the USB 3.2 Gen 2 normative electrical specifications. This overarching objective can be decomposed into the following specific goals:

- Study and characterize the fundamental signal integrity mechanisms relevant to high-speed serial links, including ISI, jitter, and channel loss, and the equalization techniques used to compensate for them;
- Define and design a CML based driver architecture capable of implementing a 3-tap FFE structure, with independently configurable pre-cursor, main-cursor and post-cursor taps;
- Derive the tap current requirements for a representative set of equalization presets, covering the PS and DE ranges defined by the USB 3.2 specification;
- Size and bias the driver's differential pairs and current mirrors to guarantee correct operation, in saturation, across the full range of output voltage and tap current combinations;

- Design a modular, abuttable standard-cell layout for the basic building block of the driver, the tap unit, accounting for layout-dependent effects;
- Evaluate the driver against the relevant USB 3.2 Gen 2 normative electrical specifications both at schematic level and after layout parasitic extraction, across relevant PVT and resistor process corners.

1.4 Main Design Requirements

The design requirements adopted throughout this work are derived directly from the normative USB 3.2 Gen 2 10 Gbps transmitter electrical specifications, which specifies targets for the TX driver's output voltage swing, differential output impedance, equalization capability, and signal quality metrics, ensuring the interoperability of the transmitter with compliant receivers.

In addition to these electrical targets, the driver is designed to be digitally configurable, so that the output swing, equalization coefficients, and termination impedance can each be tuned independently to compensate for PVT variation and to support multiple equalization presets without requiring changes to the underlying circuit topology.

1.5 Methodology

The work presented in this dissertation follows the standard top-down IC design methodology, beginning with the definition of target specifications derived from the USB 3.2 Physical Layer (PHY) requirements, and progressively refining these into a fully verified physical implementation.

Behavioral and schematic-level design and simulation were carried out using Synopsys Custom Compiler, targeting the GlobalFoundries 55 nm LPx CMOS technology, with transistor-level simulations were performed using Spectre and HSPICE.

Following schematic-level validation, the physical layout of the driver was implemented with a focus on modularity, device matching, and abuttability, and subsequently verified through Design Rule Checking (DRC) and Layout Versus Schematic (LVS) checks against the foundry Design Rules Manual (DRM). Parasitic elements introduced by the layout were then extracted and back-annotated into simulation, allowing the driver's performance to be re-evaluated with layout-induced effects taken into account, and compared against both the pre-layout results and the normative USB 3.2 Gen 2 compliance targets.

1.6 Main Contributions

The main contributions of this dissertation can be summarized as follows:

- The design of a segmented CML TX driver architecture, composed of independently sized pre-cursor, main-cursor, and post-cursor tap stages, capable of realizing a representative set

of programmable equalization presets spanning the normative USB 3.2 Gen 2 PS and DE ranges through purely digital current-mirror configuration;

- A systematic derivation of the relationship between the FFE tap coefficients and the normative PS/DE equalization metrics, enabling direct translation of standard compliance targets into tap current requirements;
- A modular, abuttable tap-unit standard-cell layout methodology, incorporating common-centroid device placement, dummy-based shielding against layout-dependent effects, and a shared guard-ring strategy, allowing the driver to be reconfigured or resized with minimal area penalty;
- Pre- and post-layout verification of the proposed driver against the USB 3.2 Gen 2 normative specification, across PVT and resistor process corners.

1.7 Dissertation Organization

Besides this introductory chapter, this dissertation is organized as follows. Chapter 2 reviews the fundamental concepts underlying high-speed serial links, covering communication linka rchitectures, differential signaling, NRZ encoding, SI metrics, equalization techniques, TX driver architectures, and the IC layout considerations relevant to high-speed, matched analog design.

Chapter 3 presents the proposed TX driver architecture, detailing the derivation of the equalization tap coefficients, the circuit-level implementation and sizing of the driver, and the layout methodology adopted for the driver's building blocks.

Chapter 4 presents the verification methodology and the pre- and post-layout simulation results obtained for the proposed driver, evaluated against the USB 3.2 Gen 2 normative specifications.

Finally, Chapter 5 summarizes the work developed throughout this dissertation, discusses the fulfillment of the proposed objectives, identifies the main limitations of the current implementation, and outlines directions for future work.

Chapter 2

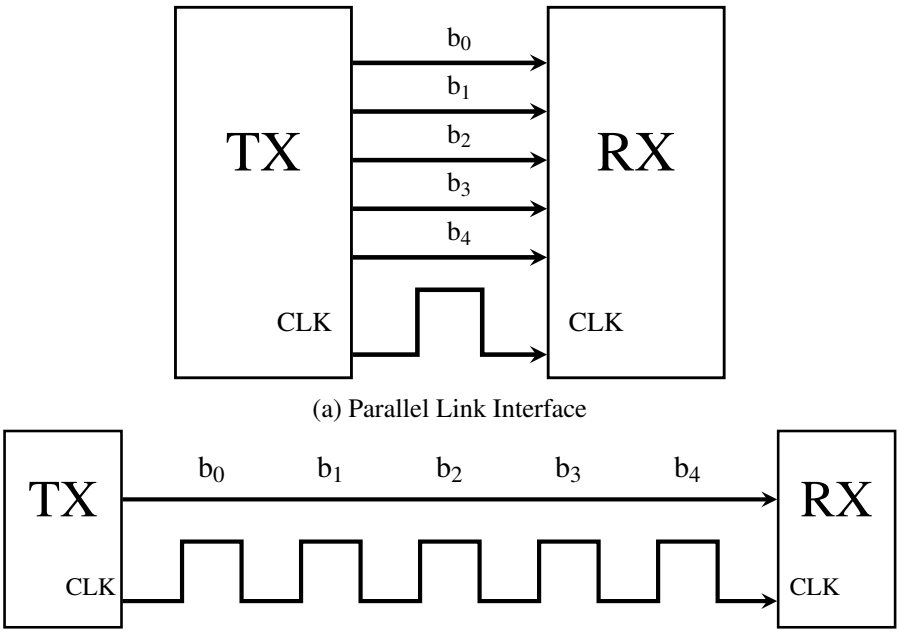
Background and Fundamental Concepts

2.1 Communication Links

The continuous growth in data throughput requirements of modern electronic systems has made high-speed communication links a fundamental enabler of contemporary digital platforms. Applications such as high-performance computing, mass storage, consumer peripherals, and data center interconnects require sustained data rates in the multi-gigabit-per-second range while maintaining strict constraints on power consumption, EMI, and physical interconnect complexity [1], [2]. As semiconductor technology has scaled, the switching speed of transistors has increased significantly. However, the performance of interconnects has not scaled at the same pace. Consequently, communication between ICs and system components has become a dominant bottleneck [3], [4]. In many modern systems, overall performance is limited not by logic speed but by the ability to reliably transmit data across electrical interconnects, motivating the development of high-speed link architectures optimized for signal integrity and energy efficiency.

2.1.1 Parallel and Serial Communication Links

Historically, short-reach communication between ICs relied on parallel buses, in which multiple data bits are transmitted simultaneously over several physical signal lines, typically synchronized by a shared clock. Parallel interfaces allow high aggregate throughput at moderate data rates but suffer from inherent scalability limitations as clock frequencies increase. As data rates rise, timing skew caused by mismatched interconnect lengths, variations in loading, and clock distribution complexity becomes increasingly difficult to control. In addition, wide parallel buses exacerbate crosstalk and simultaneous switching noise, which directly degrade signal integrity and timing margins [3], [5]. As a consequence, beyond a few gigabits per second, system performance becomes limited by interconnect effects rather than transistor speed. The parallel and serial arrangements are compared in Figure 2.1, highlighting the reduction in physical signal lines achieved by serialization.



(a) Parallel Link Interface

(b) Serial Link Interface

Figure 2.1: Communication link interfaces

High-speed serial links address these limitations by transmitting data sequentially, one bit at a time, over one or a small number of physical channels. By reducing the number of interconnects, serial architectures significantly simplify routing, eliminate inter-lane skew, and improve scalability with technology scaling. However, with the decrease of the number of physical channels, the data rate per channel must increase to maintain the same aggregate throughput. This requires higher signaling frequencies, which results in tighter margins for signal intergrity [4], discussed in Section 2.2.

2.1.2 Single-Ended and Differential Signaling

In addition to the choice between parallel and serial transmission, the signaling scheme used on the physical channel plays a critical role in determining link robustness. In single-ended signaling, information is conveyed as a voltage referenced to a common ground. The use of complementary conductors in differential transmission is illustrated in Figure 2.2. While simple to implement, single-ended communication is highly sensitive to noise, supply fluctuations, and EMI, particularly at high data rates. Differential signaling has therefore become the dominant communication scheme in high-speed serial links. In differential transmission, information is encoded in the voltage difference between two complementary signals [3], [7]. The output signal is known as the differential-mode voltage, expressed in (2.1).

$$V_{DM} = V_{D+} - V_{D-} \tag{2.1}$$

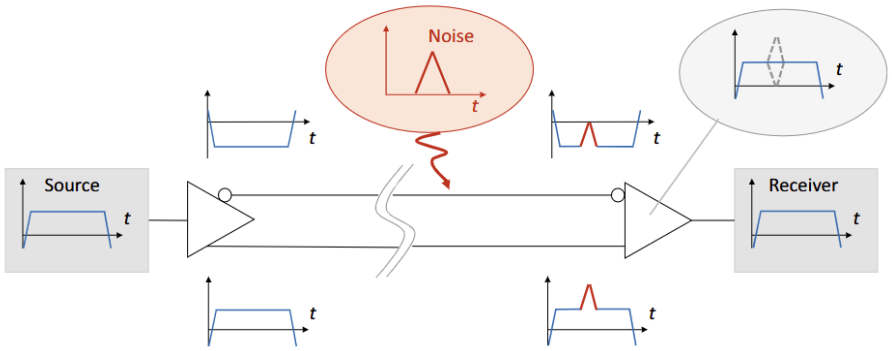


Figure 2.2: Differential signaling. Reproduced from [6]

This approach suppresses common-mode noise, reduces sensitivity to ground potential variations, and significantly lowers electromagnetic emissions [3], [7]. Furthermore, differential signaling allows smaller voltage swings while preserving noise margins, enabling faster edge rates and reduced dynamic power consumption, which balances out the downside of having two physical channels and added circuit complexity.

2.1.3 Non Return to Zero Encoding

NRZ is a digital line coding technique in which the transmitted signal is held constant at one of two voltage levels throughout the entirety of each bit period, without returning to a zero reference level between consecutive bits [8], [9]. An example of this two-level waveform is shown in Figure 2.3, illustrating the high and low logic levels and the transitions between them.

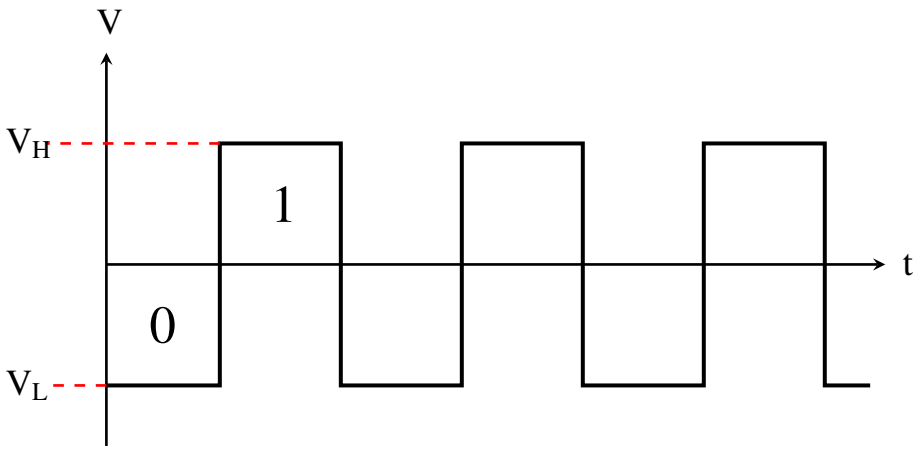


Figure 2.3: NRZ waveform

Since NRZ encodes exactly one bit per transmitted symbol, its minimum required channel bandwidth is given by the Nyquist frequency, corresponding to half of the bit rate [8]:

$$f_{Nyquist} = \frac{R_b}{2}, \tag{2.2}$$

where R_b is the bit rate. For a 10 Gb/s NRZ link such as USB 3.2 Gen 2, this corresponds to a Nyquist frequency of 5 GHz. The frequency-domain characteristics of an NRZ signal are illustrated by its power spectral density in Figure 2.4.

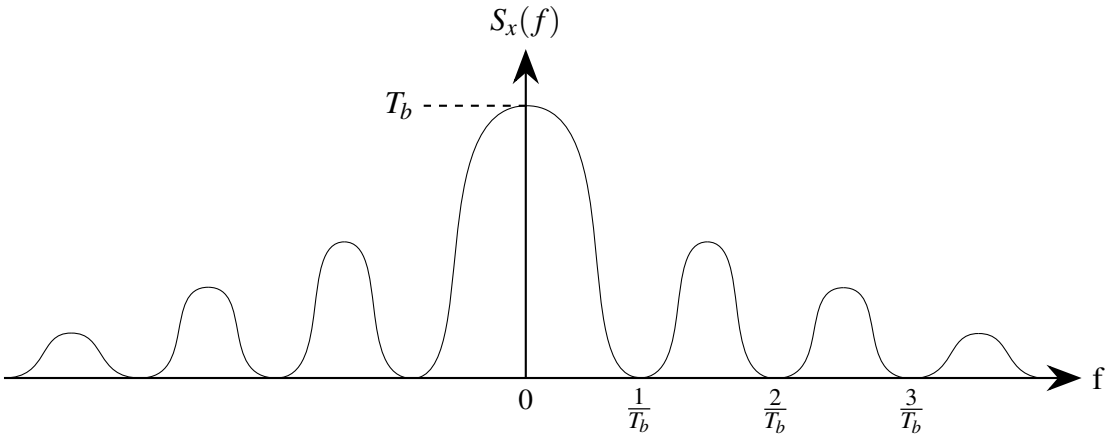


Figure 2.4: Power spectral density of a NRZ signal

In terms of modulation, NRZ is equivalent to 2-Level Pulse Amplitude Modulation (PAM2), since both encode binary data using two distinct voltage levels. As a PAM2 scheme, NRZ signaling is characterized by two symbol levels, two possible transitions (rising and falling), a single eye diagram per UI, and a maximum average transition density of 50%, corresponding to the case of alternating bit patterns [8]. Transitions between the two voltage levels in a NRZ/PAM2 signal are smooth, rather than instantaneous, due to the finite bandwidth of the TX, channel, and receiver. Figure 2.5 compares the voltage levels used in NRZ and 4-Level Pulse Amplitude Modulation (PAM4) signaling.

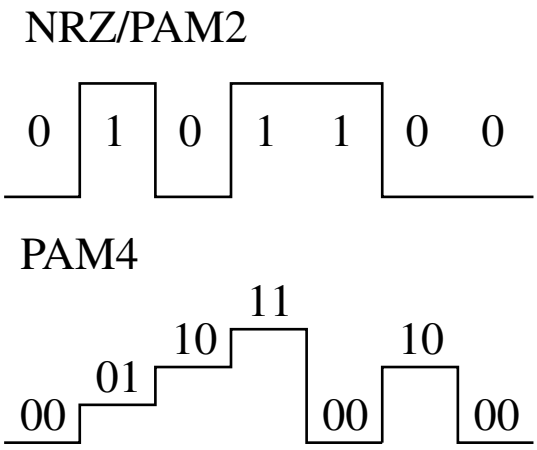


Figure 2.5: NRZ and PAM4

Compared to multi-level modulation schemes such as PAM4, NRZ/PAM2 offers superior error correction performance, greater resistance to noise, and requires a lower Signal-to-Noise Ratio

(SNR) to achieve a given BER [8], [9], owing to its simpler encoding, with only two symbol levels.

The main disadvantage of NRZ/PAM2 lies in its limited spectral efficiency: since only one bit is conveyed per symbol, achieving higher data rates requires proportionally higher channel bandwidth. This is a significant limitation in high-speed communication systems, especially when compared to multi-level modulation schemes such as PAM4, which encode multiple bits per symbol and can therefore achieve higher data rates within the same channel bandwidth [8], [10], at the cost of reduced noise margin and increased receiver complexity.

2.1.4 USB Standard

The USB is an industry standard for digital data transmission between electronic devices, first introduced in 1996. It is developed and maintained by the USB Implementers Forum (USB-IF), a consortium of companies that promotes, supports, and certifies compliance with the USB standard across the industry. The standard specifies the architecture of the communication link established between connected devices, encompassing both the physical interfaces used and the communication protocols that govern data exchange over them.

Since its introduction, USB has undergone several major revisions, progressively increasing its supported data rates: from USB 1.0, released in 1996 and supporting data rates of up to 12 Mb/s, to the most recent USB 4 standard, released in 2019, which supports data transfer rates of up to 40 Gb/s. Across all of these revisions, USB has consistently relied on differential signaling to transmit data over a pair of conductors, as discussed in Section 2.1.2, owing to its inherent noise immunity and suitability for high-speed operation.

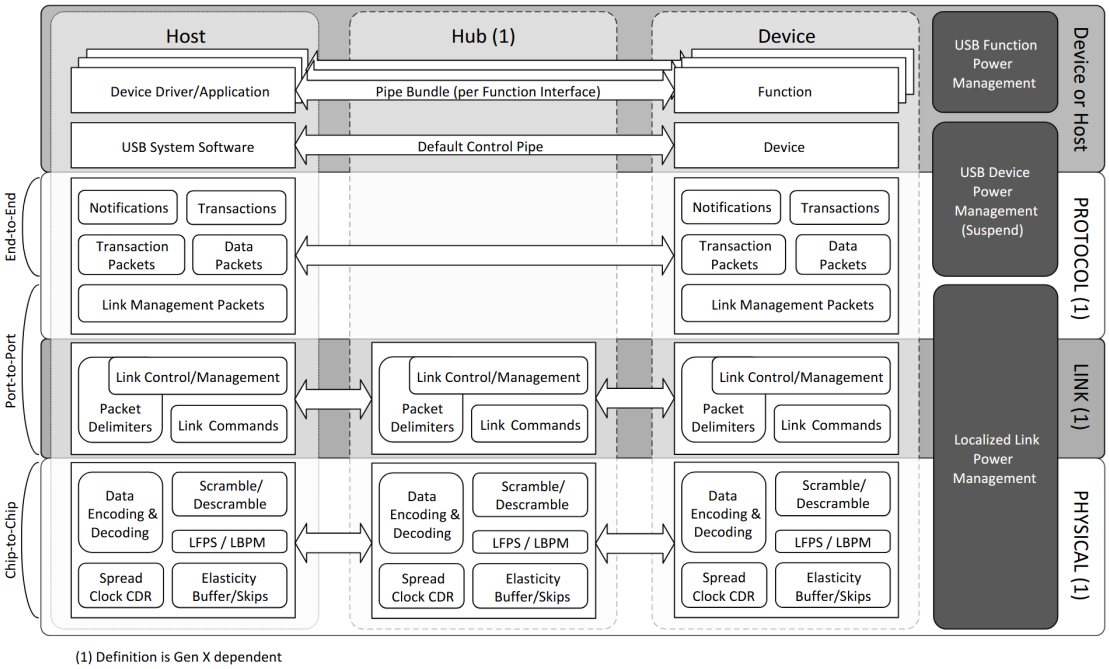


Figure 2.6: USB 3.2 layers. Reproduced from [11]

Architecturally, USB operates through three primary, interdependent layers: the PHY, the link layer, and the protocol layer, as illustrated in Figure 2.6. The PHY is responsible for the electrical transmission and reception of data over the physical medium, the link layer manages framing, flow control, and link-level management functions, and the protocol layer handles end-to-end communication between the host and device applications.

2.1.4.1 USB 3.2 PHY Layer

The work presented in this dissertation targets compliance with the USB 3.2 Gen 2 standard, commercially marketed as SuperSpeed USB, introduced with USB 3.1 in 2013 and subsequently incorporated into the USB 3.2 specification.. The USB 3.2 Gen 2 standard specifies a maximum data rate of 10 Gb/s, achieved through the use of NRZ signaling, discussed in Section 2.1.3, transmitted over a differential pair of conductors.

The USB 3.2 SuperSpeed PHY defines the underlying signaling technology used to implement this data rate, and is responsible for the encoding and decoding of data, as well as its transmission and reception over the physical channel. In addition to defining the signaling technology itself, the PHY layer specifies the electrical requirements that a compliant TX and receiver must satisfy, discussed in Section 3.1.

2.2 Signal Integrity

When a signal is transmitted through an electrical channel, it inevitably undergoes various forms of degradation, arising from the physical properties of the interconnect itself as well as from the surrounding electrical environment. This degradation can manifest as attenuation, distortion, noise, and crosstalk, all of which contribute to compromising the integrity of the transmitted signal by the time it reaches the receiver [7], [9]. This section discusses the main mechanisms of signal degradation relevant to high-speed serial links, and how to characterize their impact on the transmitted signal.

2.2.1 Inter-Symbol Interference

ISI is a form of signal distortion in which the energy associated with one transmitted symbol interferes with subsequent symbols, causing the signal to spread beyond its allotted time interval. Specifically, the energy transmitted during a given UI, the time interval defined for sampling an individual bit, spreads into neighboring UIs, so that the waveform observed at the receiver during any given bit period is no longer determined solely by the symbol transmitted in that interval, but also carries residual contributions from adjacent symbols.

This superposition of energy from neighboring symbols can result in an incorrect interpretation of the transmitted data at the receiver: the bit sampled at the decision point may be misread due to the overlap of adjacent symbols with the intended sample, leading to an increased BER. Fundamentally, ISI arises from the bandwidth-limited and frequency-dependent response of the channel,

whereby different spectral components of the transmitted signal experience different attenuation and phase delay, causing each symbol to spread into neighboring UIs [9].

2.2.2 Transmission Lines and Channel Characterization

At multi-gigabit data rates, the physical interconnect between the TX and receiver can no longer be treated as an ideal connection, but must instead be considered as a transmission line whose electrical properties influence the propagated signal. In practice, the channel exhibits a frequency-dependent response that tends to attenuate the higher-frequency components of the transmitted waveform more strongly than its lower-frequency content, effectively introducing a low-pass behavior, as illustrated in Figure 2.7.

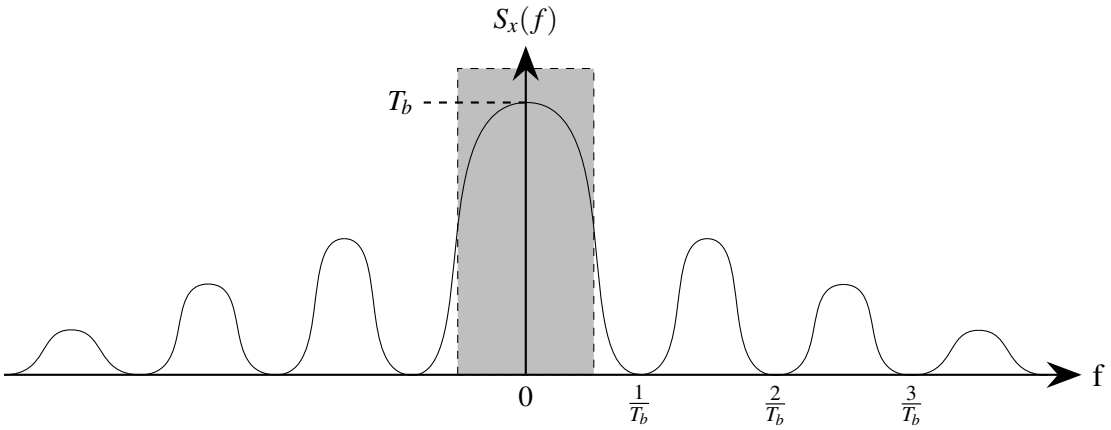


Figure 2.7: Power spectral density of a NRZ signal with a low-pass filter applied

Since these high-frequency components are responsible for preserving sharp transitions in an NRZ signal, their attenuation leads to slower edges, waveform distortion, and ultimately increased ISI. Consequently, characterizing the transmission channel in the frequency domain is essential for evaluating its effect on signal integrity and determining the impact it has at the TX. The following subsections therefore introduce the principal channel-characterization losses and metrics.

2.2.2.1 Insertion Loss

Insertion loss refers to the reduction in signal power that occurs as a signal propagates through a channel, and quantifies how much weaker a signal becomes after a given transmission line, or other channel element, is inserted into the signal path. This power reduction arises from several distinct physical loss mechanisms associated with the channel, such as dielectric losses, conductor losses, and reflection losses [7].

Dielectric losses occur when a portion of the signal's energy is absorbed by the dielectric material surrounding the channel's conductors; this loss mechanism becomes progressively more significant as frequency increases, since the dielectric absorption coefficient itself grows with frequency. Conductor losses occur when a portion of the signal's energy is dissipated as heat within

the channel conductors themselves, as a direct consequence of their finite electrical resistance, which similarly increases at high frequency due to the skin effect[9].

Reflection losses occur when a portion of the signal's energy is reflected back toward the TX due to impedance mismatches or discontinuities along the channel, rather than being delivered to the receiver [7].

Taken together, these loss mechanisms all increase with frequency, which is particularly problematic for high-speed serial links: since the high-frequency spectral content of the transmitted signal is primarily responsible for producing sharp, well-defined signal transitions [8], [9], the frequency-dependent nature of insertion loss disproportionately attenuates precisely the components of the signal needed to preserve fast, clean edges, directly motivating the need for a compensation mechanism such as TX equalization, discussed in Section 2.3.

2.2.2.2 Return Loss

Return loss is a measure of the fraction of the transmitted signal that is reflected back toward the TX due to impedance mismatches or discontinuities along the channel [7]. These reflected signals travel back along the channel and can subsequently interfere with the forward-propagating transmitted signal, introducing additional distortion beyond that caused by insertion loss alone.

2.2.2.3 Impedance Matching

To minimize the losses caused by signal reflections, the impedance discontinuities that give rise to them should be minimized wherever possible. In practice, this is achieved through impedance matching, that is, by designing the output impedance of the transmitting circuit to be as close as possible to the characteristic impedance of the channel. When these two impedances are equal, the channel presents no discontinuity to the signal at the TX interface, and the fraction of signal power reflected back toward the TX is minimized [7], [12].

2.2.2.4 Jitter

Jitter refers to the deviation of signal transitions from their ideal, intended positions in time. It can be caused by a variety of mechanisms, including impedance mismatches, crosstalk between neighboring signal paths, and frequency-dependent losses such as the dielectric losses described in Section 2.2.2.1. Regardless of its origin, jitter ultimately results in errors during the bit sampling process at the receiver, since transitions that occur earlier or later than expected reduce the effective timing margin available for correct sampling, thereby increasing the BER [4]. The deviation of actual signal transitions from their ideal timing positions is illustrated in Figure 2.8.

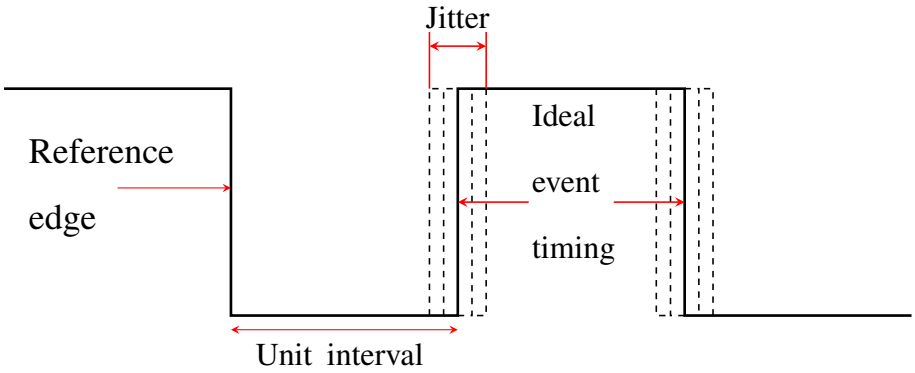


Figure 2.8: Jitter example

Jitter can be decomposed into two main components: Deterministic Jitter (DJ) and Random Jitter (RJ). Deterministic jitter is bounded and predictable, exhibiting a known, quantifiable peak-to-peak amplitude, and arises from specific, identifiable mechanisms within the system, such as crosstalk and impedance mismatches. Random jitter, in contrast, is unbounded and unpredictable in nature, and is typically well described by a Gaussian probability distribution [13]; it arises from electrical noise sources within the system's components, such as thermal noise.

2.2.2.5 Slew Rate

Slew rate refers to the speed at which a signal transitions between its two voltage levels, expressed as the rate of change of voltage with respect to time, dV/dt. Slew rate is inherently data-dependent, since the rate at which a given transition occurs depends on the preceding bit pattern and the resulting initial conditions of the signal. If the slew rate is too slow relative to the UI, the transmitted signal may fail to reach its intended voltage level before the next sampling instant, directly contributing to ISI [8] as described in Section 2.2.1. The effect of transition slope on signal rise and fall behavior is illustrated in Figure 2.9.

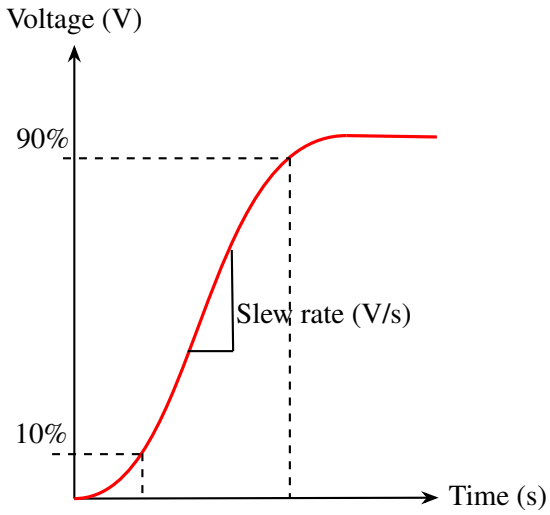


Figure 2.9: Slew rate example

In differential signaling, asymmetry between the rise and fall times of a signal can additionally give rise to Duty Cycle Distortion (DCD), in which the signal crosses the decision threshold at different relative times for rising and falling edges. This timing asymmetry introduces additional timing errors at the receiver, compounding the effects of both ISI and jitter, and further degrading overall SI [4].

2.2.3 Scattering Parameters and Channel Characterization

For the purposes of high-frequency characterization, an electrical channel can be modeled as an N-port network, where N denotes the number of physical ports through which the network interfaces with the surrounding circuitry. In the simplest case of a two-port network, the input and output ports are characterized by their respective voltages and currents, which can be related to one another through a set of linear equations describing the network's behavior. The voltage and current variables associated with a two-port network are illustrated in Figure 2.10.

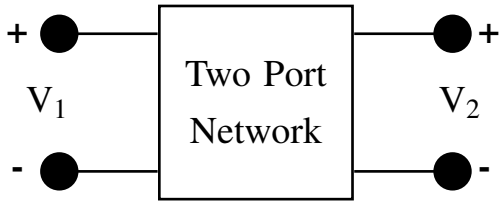


Figure 2.10: Two-port network representation

At the high frequencies relevant to multi-gigabit serial links, it is often more convenient to describe such a network in terms of traveling waves rather than voltages and currents directly. Scattering parameters (S-parameters) provide this alternative description, characterizing the behavior of a network in terms of the incident and reflected waves present at each of its ports. For a

given network, the relationship between the incident and reflected waves at each port can be fully described by an S-parameter matrix. Figure 2.11 shows the incident and reflected traveling waves used to describe a two-port network.

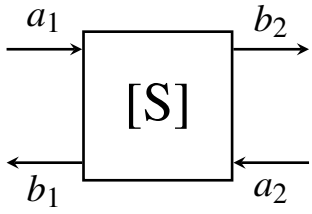


Figure 2.11: S-parameter representation of a two-port network

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

For a two-port network, such as a transmission line or channel segment, four S-parameters are defined:

- S_{11}: Input reflection coefficient, describing the fraction of the wave incident at the input port that is reflected back toward the source
- S_{21}: Forward transmission coefficient, describing the fraction of the incident wave at the input port that is transmitted through the network to the output port
- S_{12}: Reverse transmission coefficient, describing the fraction of the wave incident at the output port that is transmitted back through the network to the input port
- S_{22}: Output reflection coefficient, describing the fraction of the wave incident at the output port that is reflected back toward the load

Together, these four coefficients provide a complete characterization of a two-port channel's frequency response, capturing both its insertion loss, primarily reflected in S_{21}, and its return loss, primarily reflected in S_{11} and S_{22}, as discussed in Sections 2.2.2.1 and 2.2.2.2.

2.2.4 Signal Integrity Metrics

2.2.4.1 Eye Diagram

An eye diagram is a graphical representation of a digital signal's voltage as a function of time, constructed by overlaying multiple UIs of the signal onto a single common time axis. The resulting pattern resembles an open eye, with the open region at its center representing the ideal sampling window available to the RX. Eye diagrams provide a compact and visually intuitive means of assessing SI [7], since they reveal several critical SI parameters simultaneously, including an estimate of BER, SNR, and the available timing and voltage margins. The construction of an eye diagram from overlapping signal intervals is illustrated in Figure 2.12.

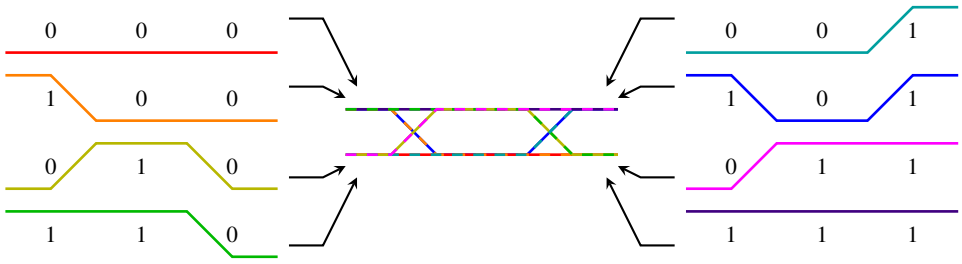


Figure 2.12: Eye diagram construction

These metrics can be extracted directly from the geometry of the eye diagram, by analyzing its vertical and horizontal openings, as well as the shape and symmetry of the signal transitions that form its boundaries.

The horizontal opening, or width, of the eye indicates the timing margin available for reliable sampling, that is, the duration for which the signal remains valid around the ideal sampling instant. Jitter can be estimated by analyzing the degree of closure of this horizontal opening.

The vertical opening, or height, of the eye indicates the voltage difference between the lowest observed logical-high level and the highest observed logical-low level; it reflects the overall clarity of the signal, with its size directly related to the noise present in the system and the resulting SNR, and is strongly correlated with the achievable BER.

The leading and trailing edges of the eye diagram correspond to the rising and falling transitions of the signal between bit periods. The region where these transitions cross one another, near the edges of the eye, highlights the variation present across individual bit transitions, providing a visual indication of the total jitter affecting the signal and its impact on overall signal quality.

Eye amplitude refers to the peak-to-peak voltage of the signal, defining the full voltage swing between a logical ‘1’ and a logical ‘0’. For an ideal, undistorted signal, the eye amplitude and eye height are approximately equal; however, insertion loss and other channel impairments often cause a noticeable reduction in eye height relative to the nominal transmitted amplitude, as energy from the signal is lost or redistributed into neighboring symbols.

An eye mask defines a predefined keep-out region within the eye diagram, specifying the minimum acceptable eye opening required for compliance with a given standard. Any portion of the received signal that falls within this keep-out region constitutes an eye mask violation, indicating non-compliant signal integrity.

Rise and fall times define the speed at which the signal transitions between its two voltage levels, its slew rate. Excessively slow transitions blur the contour of the eye diagram and reduce the clarity of the resulting eye opening. The threshold, or crossing percentage, refers to the signal level at which a receiver can reliably distinguish between a logical ‘1’ and a logical ‘0’ while maintaining an acceptable BER, and defines the effective decision point used when interpreting the eye diagram.

Taken together, eye diagrams allow several distinct signal integrity impairments, including insertion loss, ISI, crosstalk, reflections and impedance mismatches, and jitter and timing errors,

to be identified and, to some extent, distinguished from one another through visual inspection of the resulting eye shape.

2.3 Equalization

Equalization is a signal-processing technique in which the transmitted data signal is deliberately distorted using a transform that approximates the inverse of the channel's frequency response. The goal of equalization is to flatten the overall frequency response of the TX-channel-receiver system out to the Nyquist frequency, discussed in Section 2.1.3, thereby removing, or substantially reducing, the time-domain ISI introduced by the channel [9], [14], as described in Section 2.2.1. Equalization can, in principle, be implemented at the TX, at the RX, or split between both, depending on the specific link architecture and standard requirements. The frequency-dependent attenuation that motivates equalization is illustrated in Figure 2.13.

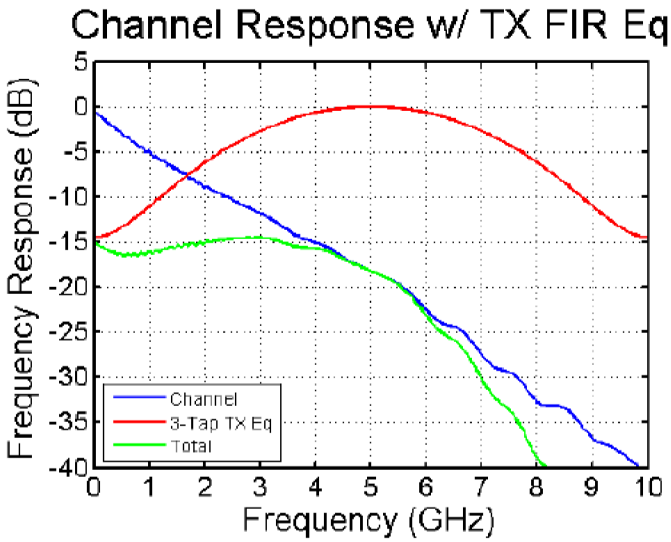


Figure 2.13: Channel response. Reproduced from [15]

As discussed in Section 2.2.2.1, practical electrical channels typically exhibit progressively greater loss at higher frequencies. This behavior is particularly problematic for high-speed serial links, since the high-frequency spectral components of the transmitted signal are primarily responsible for producing sharp signal transitions and, consequently, for determining the eye opening observed at the receiver. Equalization is therefore used specifically to compensate for these high-frequency losses, restoring the spectral balance of the signal and preserving adequate eye opening at the sampling point.

2.3.1 Transmitter-Side Equalization

TX-side equalization operates by pre-distorting the transmitted signal in a manner designed to invert the anticipated distortion introduced by the channel [9], [16], a technique commonly referred

to as transmit pre-emphasis. In practice, since the overall signal swing available at the TX is bounded, this pre-distortion is realized by selectively boosting the high-frequency transitions of the signal relative to its low-frequency, steady-state content; the practical effect of this boosting, given a fixed total output swing, is an attenuation of the steady-state levels of the transmitted signal relative to its transition levels, in order to compensate for the high-frequency losses introduced by the channel, as discussed in the context of PS and DE in Section 3.3.

The distortion intentionally introduced at the TX is subsequently reverted, at least in part, at the receiver, through RX-side equalization techniques such as Continuous-Time Linear Equalization (CTLE) or Decision Feedback Equalization (DFE), which further shape the received signal to restore an adequately open eye at the sampling point [9].

TX-side equalization is most commonly implemented using a multi-tap FIR structure, referred to as FFE. This FIR structure operates by applying a set of delayed and scaled copies of the original signal, and combining them with the original signal using an appropriate set of weights [14], [16], or tap coefficients.

This weighted combination is performed specifically to counteract the effects of ISI caused by channel dispersion [9]: since a dispersive channel spreads the energy of each transmitted symbol into neighboring unit intervals, as discussed in Section 2.2.1, the FFE pre-compensates for this effect by effectively adding the anticipated dispersed portion of a symbol's energy back into the signal ahead of time, at the TX, rather than allowing it to be lost to the channel's frequency response.

For Gen 2 operation, the USB 3.2 specification defines a three-tap FIR filter structure for the TX equalizer.

2.3.1.1 FFE and the 3-Tap FIR Architecture

High-speed serial links such as USB 3.2 transmit data over channels that behave as low-pass filters, attenuating the high-frequency content of the signal far more than its low-frequency content. As the transmitted bit rate increases, the channel's limited bandwidth causes each pulse, resulting in ISI and a closed eye diagram at the RX and severely limits the maximum achievable data rate. Equalization is the general technique used to counteract this effect, either at the receiver, at the TX, or through a combination of both. The three-tap FIR architecture, with precursor, main, and postcursor coefficients, is shown in Figure 2.14.

FFE is one of the principal techniques used to combat ISI, and can be implemented at either end of the link. Conceptually, an FFE stage generates a number of delayed copies of the input signal, scales each copy by a weighting coefficient, and sums the results to produce the equalized output. An FFE implementation looks for all intents and purposes like a digital FIR filter. When applied at the TX, this process is commonly referred to as pre-empahsis or de-emphasis, and is implemented directly in the driver by scaling the current, and hence the voltage swing, associated with each delayed version of the data.

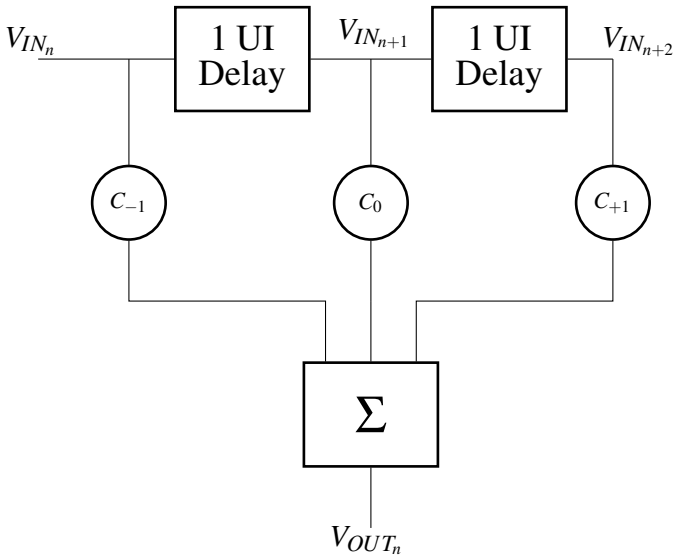


Figure 2.14: 3-tap TX Equalizer Structure

Because the FFE structure is functionally a FIR filter, its behaviour can be described using the standard discrete-time convolution sum [16]. For a filter with N taps and coefficients c_k, the equalized output sample $y[n]$ is given by

$$y[n] = \sum_{k=-M}^{P} c_k x[n-k] \tag{2.3}$$

where $x[n]$ is the input data sequence, M is the number of post-cursor taps and P is the number of pre-cursor taps. Each tap introduces a delay of one UI relative to its neighbour, so that the filter effectively looks both ahead of and behind the current symbol being transmitted FFE creates a number of delayed versions of the input signal that are added back to the signal with the proper weights. Typically, the delay is 1 UI in duration.

The FIR structure proposed by the USB 3.2 specification is a 3-tap filter.In this structure, three consecutive symbols in the data stream are each scaled by their own coefficient and summed to produce the driver output for the current symbol. Denoting the coefficients as c_{-1}, c_0 and c_1, Equation (2.3) reduces to

$$y[n] = c_{-1}x[n+1] + c_0 x[n] + c_1 x[n-1] \tag{2.4}$$

Each of the three taps plays a distinct role in shaping the transmitted waveform. One tap handles pre-cursor ISI, one post-cursor ISI, and one current-cursor ISI. The main, or cursor, tap c_0 operates on the symbol currently being transmitted and is responsible for setting the nominal output swing of the driver. The pre-cursor tap c_{-1} operates on the symbol that has not yet been transmitted, and is used to boost or attenuate the leading edge of the current symbol in anticipation of teh transition that follows [16]; this is the mechanism responsible for the preshoot behaviour introduced earlier in this section. The post-cursor tap c_1 operates on the previously transmitted

symbol, and is used to cancel the residual energy left behind by that symbol's transition [9], [16]; this is the mechanism responsible for de-emphasis. Because the taps act on immediately adjacent symbols, the polarity of c_{-1} and c_1 relative to c_0 depends on whether consecutive symbols represent a transition or a repeated bit, which is what allows the filter to selectively boost transition edges while leaving steady-state levels comparatively unaffected.

A defining property of this class of equalizer is that the total weight distributed across the taps is fixed, so that increasing the magnitude of the pre- and post-cursor taps necessarily reduces the magnitude of the main tap, and vice versa. This behaviour is captured by the normalization constraint introduced later in this section (Equation (3.11)), and reflects a fundamental trade-off in FFE design: stronger equalization sharpens transition edges and opens the eye at the receiver, but does so at the cost of a reduced main-tap amplitude and, consequently, reduced overall signal swing.

2.4 Transmitter Drivers

Data generated by digital circuitry, such as the serialized bitstream produced by a serializer, is typically represented as voltage levels corresponding to binary logic states. These voltage levels, together with the digital encoding scheme used to represent them, are generally not directly suitable for transmission over a physical channel, due to limitations in SI, power efficiency, and compatibility with the electrical characteristics of the channel itself. The TX driver serves as the interface between the digital logic domain and the physical transmission medium, converting the digital logic signal into an analog waveform that can be reliably transmitted over the channel while meeting the required SI and power efficiency specifications [9].

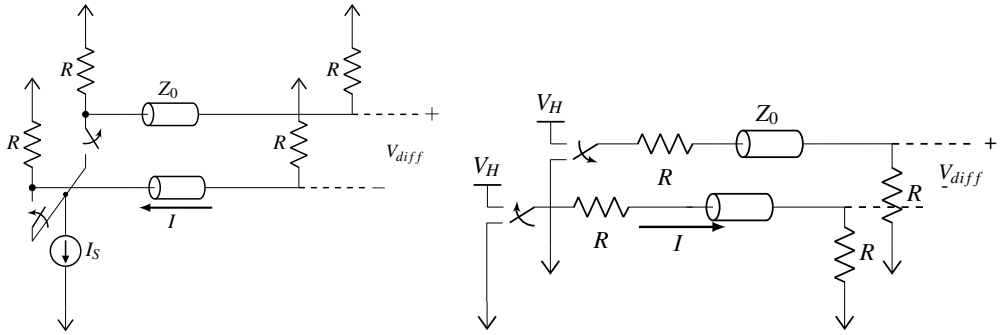
Beyond this basic conversion function, the TX driver is also responsible for modifying the transmitted signal to compensate for channel losses, through equalization techniques that account for the frequency-dependent attenuation and dispersion introduced by the channel [9], [10], as discussed in Section 2.3. This pre-distortion of the transmitted waveform enables the RX to correctly interpret the received signal and recover the original data with minimal errors, despite the non-ideal frequency response of the channel.

As data rates increase, the performance margins within which the TX driver must operate become significantly tighter. Higher data rates demand faster rise and fall times, lower TX-generated jitter, higher driver bandwidth, and more stringent control over SI metrics such as eye opening and BER. In addition to these speed-related requirements, it is generally desirable for the TX driver to be designed for low power consumption, since power efficiency is a key design constraint in most modern high-speed serial interfaces.

2.4.1 Driver Architectures

The two principal controlled-impedance driver architectures can be represented by the simplified functional models shown in Figure 2.15. A current-mode driver uses a controlled current source together with a parallel termination and is therefore analogous to a Norton source, whereas a

voltage-mode driver uses a low-impedance voltage source together with series termination and is analogous to a Thévenin source.



(a) Current-mode functional model (b) Voltage-mode functional model

Figure 2.15: Differential driver functional models

Current-mode signaling can be implemented using different circuit topologies, including push-pull structures such as those commonly associated with CML drivers, widely employed in high-performance serial links because their differential current-steering structure supports high-speed operation with relatively small voltage excursions.

In a conventional CML driver, an approximately constant tail current is steered between two complementary output branches according to the input data. The load or termination resistances perform the current-to-voltage conversion, producing the differential output signal. From an equivalent-circuit perspective, the high output impedance of the current-steering stage is combined with a parallel resistance chosen to match the transmission line.

For the ideal matched link, let $R = Z_0$ denote the single-ended characteristic impedance and let I_{CM} denote the total steered current. With double termination, the differential voltage at the receiver assumes the levels

$$V_{d,1} = +\frac{I_{\mathrm{CM}}R}{2}, \qquad V_{d,0} = -\frac{I_{\mathrm{CM}}R}{2}. \tag{2.5}$$

The resulting differential peak-to-peak swing is therefore

$$V_{d,\mathrm{pp}} = V_{d,1} - V_{d,0} = I_{\mathrm{CM}}R, \tag{2.6}$$

and the current required to produce a specified differential swing is

$$I_{\mathrm{CM}} = \frac{V_{d,\mathrm{pp}}}{R}. \tag{2.7}$$

Thus, the output swing of a CML driver is directly controlled by the steered current and the termination resistance. Since the current source remains biased while the driver is enabled, the output stage exhibits an essentially static power component, approximately given by

$$P_{\mathrm{CM}} \approx V_{DD} I_{\mathrm{CM}}. \tag{2.8}$$

A Voltage-Mode Driver (VMD) generates the transmitted signal by actively switching its output nodes between two voltage levels rather than steering a continuously biased current source. The output stage therefore contains both pull-up and pull-down paths and behaves as a low-impedance voltage source. In high-speed serial links, VMDs are commonly implemented using a Source-Series Termination (SST) architecture, in which the resistance of the conducting output device is combined with an explicit series resistor to obtain

$$R_{\text{out}} = R_{\text{SS}} + R_{\text{on}} \approx Z_0, \tag{2.9}$$

where R_{SS} is the explicit series resistance and R_{on} is the on-resistance of the conducting transistor [3], [17].

Let V_S denote the voltage excursion applied by each side of the ideal voltage-mode source. Because the source resistance is matched to the receiver termination, the received amplitude is reduced by the corresponding voltage division. The differential receiver levels are

$$V_{d,1} = +\frac{V_S}{2}, \qquad V_{d,0} = -\frac{V_S}{2}, \tag{2.10}$$

and consequently

$$V_{d,\text{pp}} = V_S. \tag{2.11}$$

For the same received differential swing, the ideal current drawn by a VMD depends on the termination configuration. With single-ended receiver termination, the required current is

$$I_{\text{VM,SE}} = \frac{V_{d,\text{pp}}}{2R}, \tag{2.12}$$

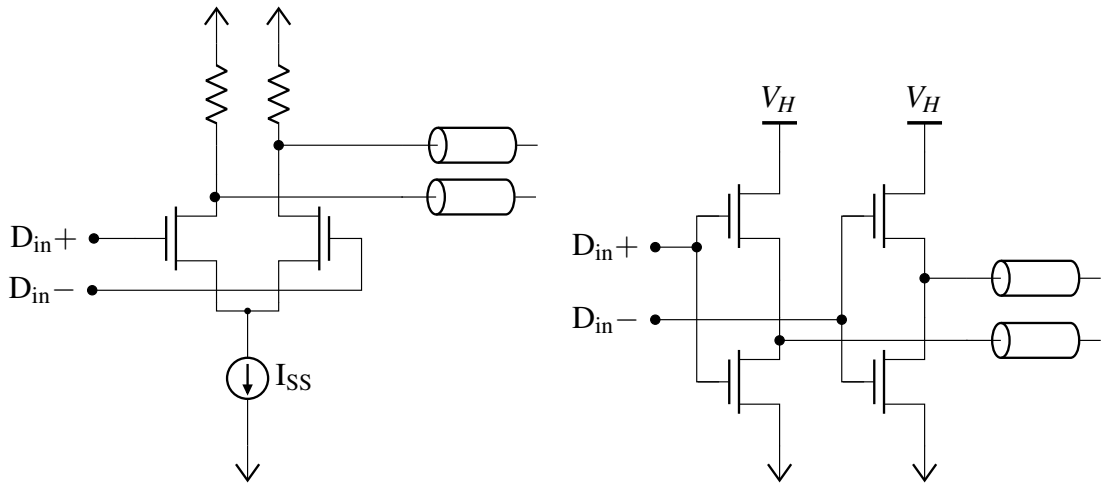
whereas with a differential receiver termination it becomes

$$I_{\text{VM,diff}} = \frac{V_{d,\text{pp}}}{4R}. \tag{2.13}$$

Voltage-mode implementation also depends on the required output swing. For relatively low swings, typically below approximately 400–500 mV_{pp}, an all-NMOS output stage can be employed. For larger voltage swings, complementary NMOS/PMOS output stages are generally preferred because they provide increased output range without requiring excessive gate overdrive. Representative transistor-level implementations of the two driver architectures are illustrated in Figure 2.16.

2.4.2 Comparison Between Architectures

The choice between current-mode and VMDs involves trade-offs in power consumption, termination accuracy, output swing, and equalization complexity. A useful first comparison can be obtained by considering the ideal current required to generate the same received differential peak-to-peak swing, $V_{d,\text{pp}}$, in a channel with single-ended characteristic impedance Z_0.



(a) CML topology (b) VMD topology

Figure 2.16: TX driver topologies

Table 2.1: Ideal driver current required for a given received differential swing. Reproduced from [18].

Driver / RX termination	Driver current	Normalized current
Current-mode / single-ended	$\frac{V_{d,\mathrm{pp}}}{Z_0}$	1
Current-mode / differential	$\frac{V_{d,\mathrm{pp}}}{Z_0}$	1
Voltage-mode / single-ended	$\frac{V_{d,\mathrm{pp}}}{2Z_0}$	0.5
Voltage-mode / differential	$\frac{V_{d,\mathrm{pp}}}{4Z_0}$	0.25

As shown in Table 2.1, the ideal voltage-mode architecture can require one-half of the current of a current-mode driver with single-ended receiver termination and only one-quarter with differential receiver termination. Thus, under the assumptions of the simplified models, a VMD with differential termination provides the potential for a fourfold reduction in output-stage current for the same received differential swing [3], [17]. This comparison describes the output stage itself and should not be interpreted as a fourfold reduction in the total TX power, since the overall consumption also includes the pre-driver, impedance-control circuitry, clocking, and equalization hardware.

The nature of the power consumption also differs between the two architectures. In a conventional CML driver, the tail current remains approximately constant while the transmitter is enabled and is largely independent of the transmitted data pattern. Consequently, a significant fraction of the driver power is static. VMDs eliminate this continuously biased tail-current source and therefore reduce the static bias requirement. However, they should not be regarded as having zero static power: current can still flow through the conducting output path and channel termination for a

valid output state. Moreover, charging and discharging the relatively large output and pre-driver capacitances introduces a dynamic power component that increases with switching activity and data rate.

Current-mode architectures generally provide simpler output-impedance control. CML drivers behaves as a high-impedance current source in parallel with its termination resistance, such that the impedance presented to the channel is determined predominantly by the termination network. Although integrated resistor tolerances can still require calibration, the impedance is comparatively weakly dependent on the instantaneous state of the current-steering devices. VMDs instead rely on the series combination of R_{SS} and the transistor on-resistance, as expressed in (2.9). Since R_{on} varies with process, supply voltage, temperature, and transistor overdrive, voltage-mode output impedance is intrinsically more sensitive to PVT variation and typically requires an impedance-calibration mechanism.

The two architectures also differ in their suitability for TX equalization. In a current-mode driver, FFE can be implemented naturally by connecting multiple weighted current-steering branches to the same output nodes. The main, pre-, and post-cursor contributions are then summed in the current domain, and their coefficients can be controlled directly through their respective branch currents. This makes multi-tap equalization conceptually straightforward, although the resulting power consumption depends on whether the total available current is redistributed among the taps or additional current branches are introduced [14].

In a VMD, equalization is commonly implemented by segmenting the output stage into multiple unit cells and assigning groups of segments to different FIR taps. Since each enabled segment contributes not only to the transmitted voltage but also to the effective source impedance, the equalization and termination settings are coupled. Maintaining an approximately constant number of active segments can preserve the desired output impedance while the distribution of those segments among the main, precursor, and postcursor taps is changed [17]. Additional segments may also be required to provide sufficient impedance-calibration range. This segmentation increases area and can increase pre-driver power because the preceding circuitry must be capable of driving the maximum enabled load. Complementary high-swing implementations are additionally sensitive to NMOS/PMOS timing skew.

Output-swing requirements introduce a further trade-off. CML supports high-speed operation with relatively small internal voltage excursions, but the voltage headroom required by the current source and switching devices makes very large output swings increasingly difficult as CMOS supply voltages are reduced. VMDs avoid a stacked tail-current source and can therefore exploit a larger fraction of the available supply voltage, particularly when complementary pull-up and pull-down devices are used. On the other hand, obtaining both a large swing and accurate impedance matching requires careful control of the nonlinear transistor resistance.

From a linearity perspective, CML drivers generally have an advantage because the output is generated by steering an approximately constant current through resistive loads. In a VMD, the output depends on the on-resistance of the pull-up and pull-down transistors, which varies with their terminal voltages and becomes more nonlinear as the output swing increases. Consequently,

high-swing VMDs require careful control of device operating region and NMOS/PMOS matching, while CML linearity is mainly limited by the available voltage headroom.

Consequently, neither architecture is universally superior. Current-mode drivers offer robust high-speed operation, relatively straightforward impedance control, and a natural implementation of multi-tap current-domain equalization, at the expense of continuous bias current and therefore higher static power. VMDs offer substantially better theoretical current efficiency and can make more efficient use of the available voltage headroom, but place greater demands on output-impedance calibration, segmentation, pre-driver design, and NMOS/PMOS timing matching. The architecture must therefore be selected considering the required output swing, energy budget, termination accuracy, PVT robustness, and equalization requirements of the target link.

2.5 State of the Art in High-Speed CML Transmitters

The previous sections introduced the operating principles and main trade-offs of CML and voltage-mode transmitter architectures. To position the architecture developed in this dissertation, this section reviews representative high-speed transmitters that employ CML output stages, transmitter-side equalization, or techniques intended to reduce the power penalty associated with conventional current-mode drivers. Particular attention is given to implementations operating around 10 Gbps, since they provide the most meaningful comparison with the target data rate of this work.

One of the earlier examples of a programmable equalized CML transmitter was presented by Rylyakov and Rylov [19]. Implemented in 90 nm CMOS, the transmitter operates at 10 Gbps and employs a four-tap FFE in which delayed data streams are combined with digitally programmable current weights at the output stage. The fabricated circuit demonstrated error-free operation with a $2^{31} - 1$ Pseudo-Random Bit Sequence (PRBS) at 10 Gbps and 125 °C, while driving an approximately 0.9 V peak-to-peak differential signal. The total transmitter power reported under this operating condition was 174 mW. Although the power consumption is high compared with more recent implementations, this work established the usefulness of segmented current steering and programmable multi-tap equalization for compensating channel loss.

A closely related architecture was later demonstrated by Weiss *et al.* [20]. The transmitter was implemented entirely using differential CML and incorporated a three-tap FIR filter for transmitter pre-distortion. The tap currents were independently adjustable and summed at a common output load, illustrating the natural correspondence between current-mode summation and FFE implementation. Measurements demonstrated a BER below 10^{-12} at 11 Gbps over a 5 m RG58 cable and at 9 Gbps over a motherboard channel presenting approximately 24.3 dB attenuation at 4.5 GHz. This result is particularly relevant to the present work because it demonstrates that a small number of programmable current-mode taps can provide substantial compensation for strongly attenuating electrical channels.

The continuous static current associated with CML, however, has motivated alternative output-stage implementations. Chen *et al.* [12] proposed a switchable CML driver in 130 nm CMOS

in which switched PMOS devices replace the conventional resistive pull-up structure. The architecture was developed to reduce the current required to generate a given differential output swing while simultaneously providing impedance matching. Post-layout simulation results reported a differential swing of approximately 1.6 V, a total current of 10 mA, and approximately 4 ps peak-to-peak jitter. Relative to the conventional topology considered by the authors, the proposed structure reduced static current by approximately 75%. This approach directly addresses the static-power disadvantage of conventional CML, although it does not provide the programmable multi-tap equalization required in the present application.

A different method for improving transmitter efficiency was investigated by Yi *et al.* [21]. Their 65 nm CMOS transceiver combines a CML main driver with a voltage-mode capacitive pre-emphasis equalizer. The equalization path supplies additional current predominantly during data transitions, avoiding the low-frequency current penalty associated with a conventional current-mode equalization tap. The fabricated transmitter operates at 9 Gbps with a channel swing of 40 mV peak-to-peak differential and consumes 2.8 mW. The low transmitter power is enabled in part by the very small channel swing and by a receiver specifically designed for high input sensitivity. Consequently, the reported power cannot be directly compared with a high-swing, conventionally terminated transmitter, but the work demonstrates the potential efficiency advantage of moving part of the equalization function away from continuously biased current-mode taps.

Technology scaling has also allowed conventional current-mode structures to operate at reduced supply voltage. Lv and Zhang [22] presented a 12.5 Gbps CML driver in 28 nm CMOS operating from a 1.05 V supply. The driver combines a four-stage cascaded main path with an adjustable current-mode pre-emphasis circuit. Post-layout simulation gives an output differential swing of approximately 604 mV, an overall power consumption of 28.9 mW, and an equalization gain reaching approximately 11.48 dB. This implementation illustrates that current-mode pre-emphasis remains practical in scaled technologies, although the equalizer and output architecture differ from the independently weighted symbol-spaced three-tap structure adopted in this work.

A particularly relevant comparison is the 10 Gbps driver presented by Choi *et al.* [23], since it was also implemented in 55 nm CMOS and operates from a 1.2 V supply. The architecture employs a two-tap CML driver combined with a reconfigurable capacitive-peaking network. Rather than obtaining the complete high-frequency boost by increasing the tap currents, capacitor banks provide additional pre-emphasis without a proportional increase in static current. Post-layout simulations were performed using a channel with 12.3 dB insertion loss at the 5 GHz Nyquist frequency. The reported receiver eye has a height of 160 mV and a width of 0.92 UI, with approximately 8 ps peak-to-peak jitter. The complete driver, including the pre-driver and timing circuitry, draws 36.25 mA from a 1.2 V supply. The work demonstrates an effective method for extending the equalization capability of CML without relying exclusively on additional static tap current.

These implementations show that no single transmitter topology simultaneously optimizes all relevant metrics. Conventional current-mode FFE provides a particularly direct implementation of symbol-spaced equalization because the weighted tap currents can be summed at the output

node. The same property also allows the output swing and equalization strength to be programmed through segmented current sources. Its principal disadvantage is that increasing the available current range generally increases static power. Capacitive, voltage-mode, and hybrid equalization techniques can reduce this penalty, but typically introduce different constraints on output swing, termination, receiver sensitivity, calibration, or circuit complexity.

The architecture developed in this dissertation therefore does not target the highest reported data rate or the minimum absolute transmitter power. Instead, it targets a specific USB 3.2 Gen 2 operating point and combines three features that are important for this application: a three-tap current-domain FFE, digitally programmable tap currents, and digitally programmable output termination. The three-tap structure provides independent pre-cursor, main-cursor, and post-cursor control, while the segmented current and resistor banks provide the tuning range required to compensate for process and layout-induced variation. In addition, the implementation is evaluated after parasitic extraction and across PVT and resistor-process corners rather than only at a nominal schematic operating point. The resulting design therefore prioritizes configurability and post-layout robustness within a conventional CML framework.

Table 2.2 summarizes the main characteristics of the representative implementations discussed above. The numerical values should be interpreted together with the verification method and application conditions, since the reported power, swing, channel loss, and circuit boundaries differ substantially between publications.

Table 2.2: Representative high-speed CML and CML-based transmitter implementations.

Work	Tech.	Rate	TX equalization	Supply	Reported characteristic
Rylyakov and Rylov [19]	90 nm	10 Gb/s	4-tap FFE	1.2/1.65 V	0.9 Vppd, 174 mW
Weiss *et al.* [20]	130 nm	10 Gb/s	3-tap FIR	1.5 V	BER $< 10^{-12}$
Chen *et al.* [12]	130 nm	10 Gb/s	–	3.3 V	1.6 Vppd, 10 mA
Yi *et al.* [21]	65 nm	9 Gb/s	VM pre-emphasis + CML	1.0 V	40 mVppd, 2.8 mW TX
Lv and Zhang [22]	28 nm	12.5 Gb/s	Current-mode pre-emphasis	1.05 V	604 mVppd, 28.9 mW
Choi *et al.* [23]	55 nm	10 Gb/s	2-tap + capacitive peaking	1.2 V	160 mV eye, 0.92 UI

2.6 IC Design Methodology

The design of an analog IC follows a structured, top-down methodology, beginning with a set of target specifications and features and progressively refining these into a fully verified physical implementation.

The design process begins with the definition of a set of specifications and features that the final circuit must satisfy, derived from the target application and, in the context of this work, from the relevant USB 3.2 physical layer requirements. Starting from these specifications, functional, behavioral-level models of the various circuit blocks are used to further narrow down the design space, informing decisions on system architecture, circuit topology, and device sizing. At this stage, other process-related features are also selected, such as the threshold voltage flavor of the

transistors to be used, as well as the type of passive devices, such as capacitors or inductors, that best suit the design requirements.

Once these architectural and topological decisions have been established, the system is partitioned, depending on its complexity, into a set of sub-circuits, or macro-blocks, each of which is designed and validated independently according to its own operational requirements within the context of the overall system. These macro-level schematics are then progressively broken down into detailed schematics, in which each circuit element is modeled using the process-specific device models provided by the foundry through the Process Design Kit (PDK).

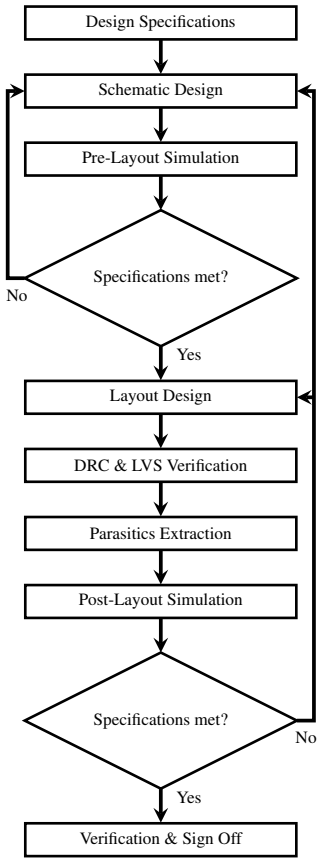


Figure 2.17: IC Design Flowchart

At this level, the schematics are simulated and validated against the target specifications, and the design is iteratively refined, adjusting topology and device sizing as needed, until the desired electrical performance is achieved at the schematic level.

Once the schematic-level design has been validated, the physical layout process begins. During this stage, device placement and interconnect routing are carried out, translating the schematic into the physical geometries that will define the fabrication masks, as discussed in Section 2.7. Throughout this process, DRC and LVS verification are performed to ensure that the resulting layout complies with the geometric and electrical design rules specified by the foundry in the DRM, and that the physical implementation is electrically equivalent to the validated schematic.

After the layout has successfully passed both DRC and LVS verification, the parasitic elements introduced by the physical implementation, such as parasitic resistances and capacitances associated with interconnect and device layout, are extracted from the physical design. A post-extraction validation of the design is then performed, in which the extracted parasitics are included in simulation, in order to characterize their impact on the overall system performance relative to the original schematic-level results.

If any of the target specifications are not met at this post-extraction stage, the design flow returns to the layout and, if necessary, the schematic stages, and the process is iteratively refined until the desired performance is achieved with the extracted parasitics included. This iterative schematic-layout-extraction loop, illustrated conceptually in Figure 2.17, is characteristic of high-speed analog and mixed-signal design, where layout parasitics can have a first-order impact on circuit performance.

2.7 Layout

Layout design constitutes the final main step of the IC design flow, carried out once the schematic-level design has been validated through pre-layout simulation and confirmed to meet the required functionality and performance specifications. During the layout stage, the schematic representation of the circuit is translated into a physical representation, defining the geometric shapes that will appear on the photomasks used during fabrication. The masks are created thorugh Front-End-of-Line (FEOL) processing, for the individual devices, and Back-End-of-Line (BEOL) processing, for the multi-layered interconnects [6].

This physical translation is not ideal, and inevitably introduces non-idealities, commonly referred to as parasitics, as well as layout-dependent effects that are difficult to predict accurately at the schematic level. These effects can alter the electrical characteristics and performance of the designed circuit relative to its pre-layout simulation results [2], [6], making careful layout design essential for high-speed circuits, where performance margins are tight.

Layout-dependent effects can be broadly classified into two categories: manufacturing-related and environmental. Manufacturing-related effects include dopant concentration gradients, oxide thickness gradients, and mechanical stress, all of which are difficult to predict and simulate accurately, since they depend on the specific fabrication process and the physical layout of the design. Environmental effects primarily include temperature gradients across the die, which are similarly difficult to account for during the design phase, since they depend on the operating conditions and power dissipation profile of the final system [6].

Dopant gradients arise as a manufacturing imperfection resulting from non-uniformities in the ion implantation process, such as ion beam gradients and beam angle variations across the wafer. Temperature gradients, in contrast, are an environmental factor, encompassing both external temperature variations in the operating environment and self-heating effects caused by power dissipation within the circuit itself. Mechanical stress arises from both manufacturing and packaging

imperfections, including material density variations across the die and thermal effects introduced during die-attach and wire-bonding processes.

2.7.1 Device Sizing and Matching

Device mismatch, that is, the deviation in electrical characteristics between devices that are nominally identical by design, can be classified into two main types: random mismatch and systematic mismatch. Systematic mismatches arises from layout-dependent effects and can, in principle, be predicted and compensated for through appropriate layout techniques, whereas random mismatches arises from inherently stochastic variations in the fabrication process [6].

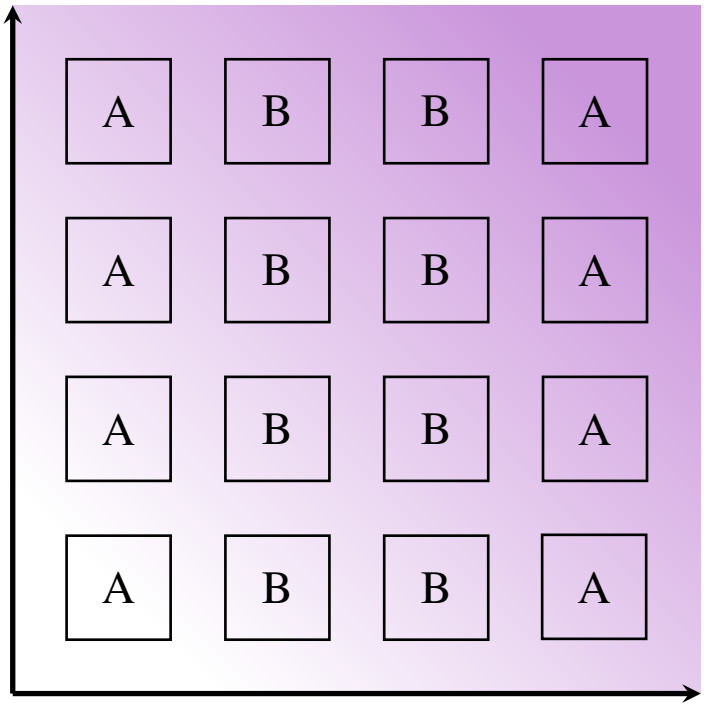


Figure 2.18: Wafer gradient impact on devices

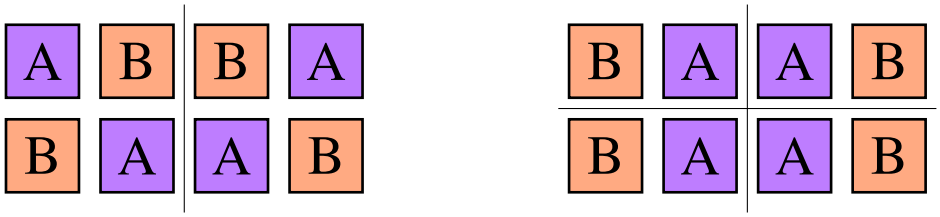
Systematic mismatch can be further divided into mask-related mismatch and gradient-related mismatch. Mask-related mismatch arises from mechanisms that affect all devices of a given size equally, and must therefore be accounted for at the schematic design stage rather than through layout techniques alone; a representative example is a systematic variation in the effective width or length of a device relative to its drawn dimensions. Gradient mismatch, in contrast, is caused by spatial process gradients across the die, and primarily affects devices that are physically separated from one another; matched devices should therefore be placed as close together as possible to minimize this effect. A representative example of gradient mismatch is the variation in threshold voltage caused by dopant concentration gradients across the die. The influence of spatial process gradients on different device placements is illustrated in Figure 2.18.

Random mismatch arises from statistical fluctuations in processing conditions or material properties at the device level, and its magnitude is highly dependent on the physical dispersion, or spatial distribution, of the matched devices across the layout.

Each type of mismatch can be mitigated through specific layout strategies. Mask-related mismatch can be compensated by constructing each device as a parallel combination of multiple unit-sized devices, and by surrounding the active devices with dummy devices to ensure that all unit devices experience similar local surroundings during fabrication [6], [24]. Gradient mismatch can be compensated by maximizing the size of the unit devices used, maximizing the area-to-perimeter ratio of each device (favoring square-like geometries), and placing matched devices as close together as possible. Random mismatch can be compensated by ensuring that matched devices share identical size, orientation, and, where possible, similar local placement, while also maximizing the spatial dispersion of the overall device distribution across the available layout area [2], [6], [24].

More generally, distributing matched devices as evenly as possible across the layout helps reduce a wide range of unwanted effects, by ensuring that all devices are exposed to similar local conditions. Dummy devices are also essential in enabling an abuttable, modular layout design approach, since they allow standard-cell-like blocks to be placed directly adjacent to one another while preserving matching conditions at the block boundaries, which in turn helps reduce the overall area of the layout.

2.7.1.1 Matching Techniques



(a) Interdigitation (b) Common centroid

Figure 2.19: Matching Patterns

Both gradient and random mismatch can be further reduced through dedicated layout techniques that introduce symmetry into the device placement, such as interdigitation, common centroid, and the use of dummy devices [24]. Interdigitation is a technique in which the matched devices are interleaved with one another along a single axis of symmetry. Common centroid is a technique in which the matched devices are arranged such that their geometric centroids coincide, using two orthogonal axes of symmetry; this arrangement creates symmetry in the layout and minimizes the impact of process gradients on the matched devices [2], [6]. The interdigitated and common-centroid placement patterns are compared in Figure 2.19.

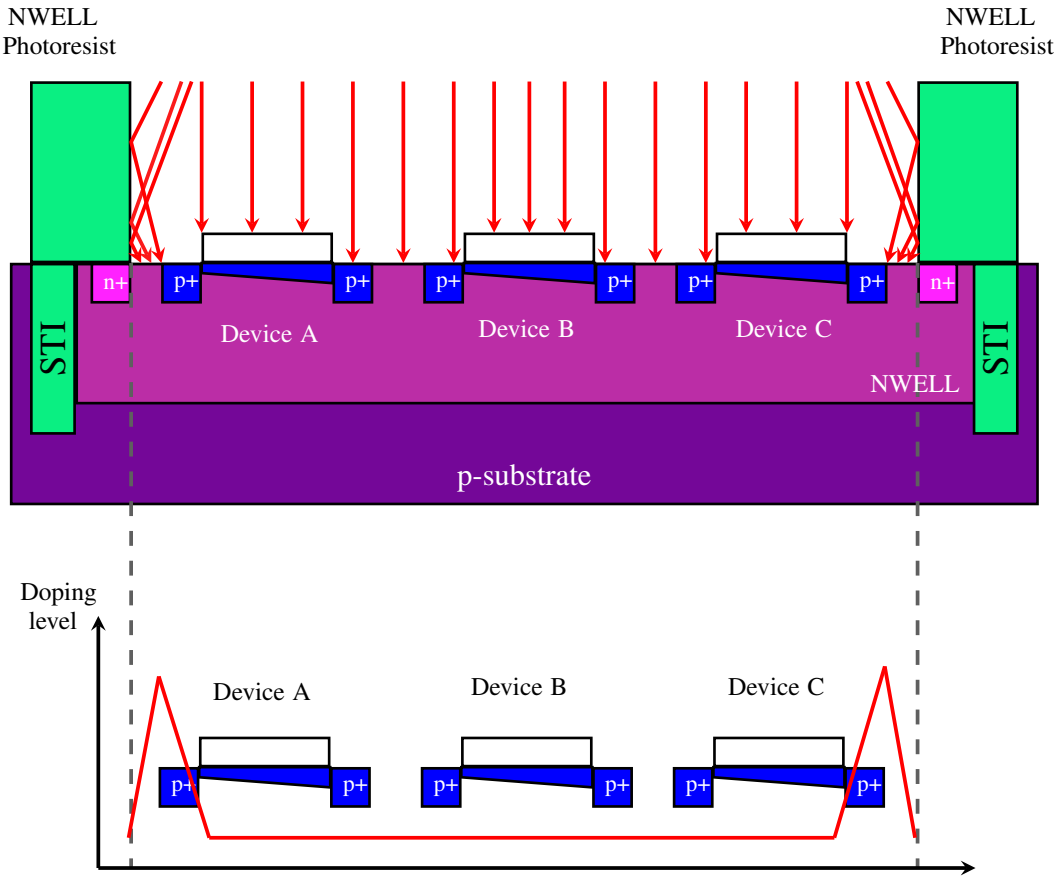


Figure 2.20: Well Proximity Effect (WPE) effect on the devices. Adapted from [25]

2.7.1.2 Well Proximity Effect

An additional effect that must be considered during layout is the WPE, which impacts the electrical characteristics of devices, such as threshold voltage, placed close to a well edge. This is caused by the lateral non-uniformity in the well doping process, during which, in the ion implantation process, high-energy ions reflect from the well photoresist and concentrate on the edges of the well, creating dopants gradient [2], [6]. This difference in dopant concentration causes mismatches between devices located near the wall edge. To counteract this effect, dummy devices can be placed around the matched core devices, providing sufficient separation between the core devices and the nearest well edge, and ensuring that all core devices experience equivalent local conditions, which helps minimize mask-related mismatch. The device geometry associated with the well proximity effect is illustrated in Figure 2.20.

2.7.1.3 Length of Diffusion Effect

Length of Diffusion (LOD) effect, also referred to as the Shallow Trench Isolation (STI) stress effect, is a manufacturing-induced effect in which mechanical stress is introduced at the edges of the active (diffusion) area during the formation of the STI regions that isolate adjacent devices.

This stress can cause variations in the threshold voltage of transistors located near the edges of their diffusion area, relative to devices with a larger surrounding active area, and is therefore an important consideration in the layout of matched analog devices [2], [6]. The relationship between diffusion geometry and the LOD effect is illustrated in Figure 2.21.

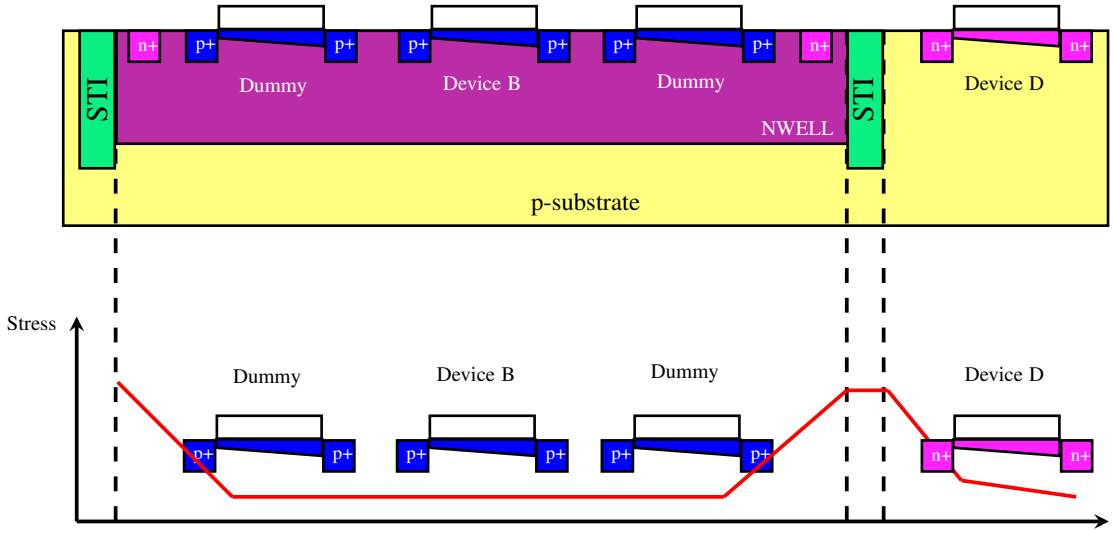


Figure 2.21: LOD effect on the devices. Adapted from [25]

The origin of the LOD effect lies directly in the STI formation process itself. As the wafer cools following STI formation, the oxide-filled trenches become compressive, inducing mechanical stress on the adjacent silicon active area. The magnitude of this stress is strongly influenced by the width of the STI region, that is, the spacing between neighboring active areas: devices with narrower active-to-active spacing experience a stronger stress effect than devices surrounded by wider isolation regions.

The compressive stress introduced by LOD affects electron and hole mobility asymmetrically: it improves hole mobility while decreasing electron mobility, and therefore has an opposite effect on PMOS and NMOS device performance for a given active area geometry. Because this effect is strongly dependent on the distance between a device and the edge of its active area, LOD can be mitigated by increasing the distance between the core devices and the surrounding well edge. In practice, this is achieved by surrounding the core devices with dummy devices [6], which increase the effective active area surrounding the matched devices and reduce the LOD-induced variation between them.

2.7.2 Routing and Signal Integrity

Routing interconnects must be carefully sized and spaced in order to minimize the parasitic effects they introduce, since these parasitics directly affect the signal integrity and reliability of the circuit. Parasitic capacitances and inductances appear between adjacent interconnects, altering signal propagation characteristics such as rise and fall times, and potentially introducing unwanted signal reflections and crosstalk between neighboring signal paths. In addition, interconnects introduce

parasitic series resistance along the signal path, which causes unwanted voltage drops, commonly referred to as IR drop, that can affect signal timing, introduce additional RC delay [6], [7], and reduce the effective switching speed of the devices being driven.

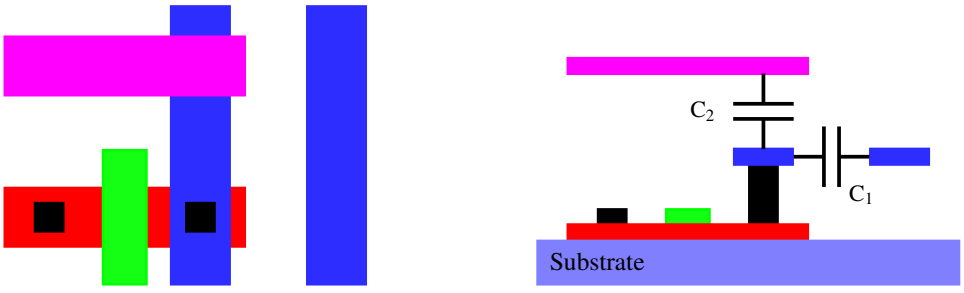
2.7.2.1 IR Drop

IR drop refers to the reduction in voltage available at a load node, caused by the voltage drop across the parasitic resistance of the interconnect supplying it. The magnitude of the IR drop is directly proportional to the current flowing through the interconnect and to its parasitic resistance, and, if left unmitigated, can lead to timing violations and signal integrity degradation [6]. IR drop can be reduced by increasing the width of the interconnect, thereby lowering its resistance, or by using lower-resistance routing materials where available.

IR drop is particularly critical in power delivery networks, where the combination of high current densities and long interconnect lengths can result in significant voltage drops across the network, degrading both the performance and long-term reliability of the entire system [3]. To mitigate IR drop in the power delivery network routing, designers can employ wider interconnects, use lower-resistance metal layers, and carefully place vias to minimize the resistive path between the supply source and the load devices.

2.7.2.2 Parasitic Capacitances

To minimize parasitic capacitance, interconnects should be routed as short as possible and placed as far as practical from neighboring interconnects. Where interconnects on different metal layers are routed in close proximity or overlap vertically, the voltage differential and signal activity present on the overlapping layers should be carefully considered and analyzed, since capacitive coupling between them can introduce unwanted crosstalk or timing-dependent noise [6], [7]. The coupling geometry between nearby metal interconnects is illustrated in Figure 2.22, in both top and cross-sectional views.



(a) Top view (b) Cross section

Figure 2.22: Parasitic capacitances between different metal layers

Another consideration to be taken into account is the introduction of a parasitic Miller capacitance on devices with metal tracks connected to their drain and source terminals in close proximity. This parasitic capacitance results in the Miller effect, which makes the input capacitance of the

device appear much larger than its actual value, and leads to a decrease in the switching speed of the device, limiting its bandwidth and causing unwanted delays, reducing overall performance.

2.7.2.3 Electromigration

The dimensions of an IC interconnect cannot be determined solely from its parasitic resistance and capacitance. The amount of current carried by the metal must also be considered in order to ensure long-term reliability. One of the main current-related reliability mechanisms in metallic interconnects is Electromigration (EM), which results from the transfer of momentum from conducting electrons to the metal atoms. Over time, this process produces a net transport of material along the interconnect, potentially creating regions of material depletion and accumulation. Material depletion can eventually form voids that increase the local resistance and may ultimately produce an open circuit [6], [26], [27]. The severity of EM is strongly related to the current density in the interconnect. For a metal line with width w_m and thickness t_m, the cross-sectional area available for current conduction is

$$A_m = w_m t_m, \tag{2.14}$$

and the instantaneous current density is therefore defined as [6]

$$J(t) = \frac{i(t)}{A_m} = \frac{i(t)}{w_m t_m} \tag{2.15}$$

where $i(t)$ is the current flowing through the interconnect. The physical current density J is expressed in $\mathrm{A/m^2}$, although $\mathrm{A/cm^2}$ or $\mathrm{MA/cm^2}$ are also commonly used for IC interconnects. Since the thickness of a given metal layer is fixed by the fabrication process, current-carrying capability is also commonly expressed as a current per unit width. Defining this width-normalized quantity as J_w gives

$$J_w(t) = \frac{i(t)}{w_m} = J(t) t_m \tag{2.16}$$

Consequently, for a given metal layer and current, increasing the routing width directly reduces the current density. This relationship explains why electromigration constraints often impose a minimum interconnect width for nets carrying large currents. Higher metal layers may also provide greater current-carrying capability because they are generally thicker and therefore present a larger cross-sectional area [2], [6]. For time-varying currents, a single instantaneous value is insufficient to characterize the electrical and thermal stress applied to the interconnect. Three current quantities are commonly considered: the peak, average, and Root Mean Square (RMS) currents. For a periodic, unidirectional current waveform $i(t)$ with period T, these quantities can be written as

$$I_{\text{peak}} = \max_{0 \le t < T} i(t) \tag{2.17}$$

$$I_{\text{avg}} = \frac{1}{T} \int_0^T i(t)dt \tag{2.18}$$

$$I_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T i^2(t)dt} \tag{2.19}$$

The corresponding current densities follow directly from the interconnect cross-sectional area:

$$J_{\text{peak}} = \frac{I_{\text{peak}}}{w_m t_m} \tag{2.20}$$

$$J_{\text{avg}} = \frac{I_{\text{avg}}}{w_m t_m} \tag{2.21}$$

$$J_{\text{RMS}} = \frac{I_{\text{RMS}}}{w_m t_m} \tag{2.22}$$

These quantities represent different forms of stress on the interconnect. The average current density is associated with the long-term net transport of metal caused by EM, whereas the RMS current density is particularly relevant to Joule self-heating because the dissipated power is proportional to the square of the current [28]. The peak current desnity characterizes the maximum instantaneous electrical stress and can become particularly relevant for short-duration or high-frequency current pulses. For this reason, technology design rules may specify different EM limits for average, RMS, and peak current depending on the type and duration of the waveform. The relationship between these quantities can be illustrated by considering a periodic rectangular current pulse of amplitude I_{peak}, active for a time t_{on} during each period T. Defining the duty cycle as

$$D = \frac{t_{\text{on}}}{T} \tag{2.23}$$

the average and RMS currents become

$$I_{\text{avg}} = D I_{\text{peak}} \tag{2.24}$$

$$I_{\text{RMS}} = \sqrt{D} I_{\text{peak}} \tag{2.25}$$

Since the interconnect geometry is unchanged, the same relationships apply directly to current density:

$$J_{\text{avg}} = D J_{\text{peak}} \tag{2.26}$$

$$J_{\text{RMS}} = \sqrt{D} J_{\text{peak}} \tag{2.27}$$

Equivalently, if the average current density is kept constant while the duty cycle is reduced,

$$J_{\text{peak}} = \frac{J_{\text{avg}}}{D} \tag{2.28}$$

$$J_{\text{RMS}} = \frac{J_{\text{avg}}}{\sqrt{D}} \tag{2.29}$$

This distinction is particularly important for high-speed digital interconnects. [28] show that two current waveforms having the same average current density can exhibit substantially different RMS and peak current densities as their duty cycles change [28]. Since the temperature rise caused by Joule heating increases approximately with the square of the current density, a low-duty-cycle waveform can satisfy an average-current constraint while still producing significant thermal stress. The resulting temperature increase is itself detrimental to electromigration lifetime, making current density and operating temperature coupled reliability considerations. Current density can additionally be reduced by distributing the total current among multiple parallel interconnect paths. If a total current I_{tot} is divided approximately equally among N identical metal tracks, the current carried by each track is

$$I_{\text{path}} \approx \frac{I_{\text{tot}}}{N} \tag{2.30}$$

and the corresponding current density becomes

$$J_{\text{path}} \approx \frac{I_{\text{tot}}}{N w_m t_m} \tag{2.31}$$

Therefore, increasing the metal width, selecting a thicker metal layer, or dividing the current among several parallel paths reduces the current density experienced by each individual interconnect. These principles are fundamental when routing high-current nets, and the routing interconnects should be designed according to the EM limits defined by the technology DRM.

2.7.2.4 Differential Signaling and Routing Symmetry

For systems employing differential signaling, such as the TX driver discussed in this dissertation, the layout floorplan must be carefully designed to ensure symmetry between the two complementary signal paths, so that both paths experience equivalent parasitic effects and the differential characteristics of the signal are preserved[10]. This consideration is particularly important for high-speed signals, where differences in parasitic capacitance, resistance, or coupling between the two paths can lead to significant signal degradation, duty-cycle distortion, and timing mismatches between the differential signal pair [6], [7].

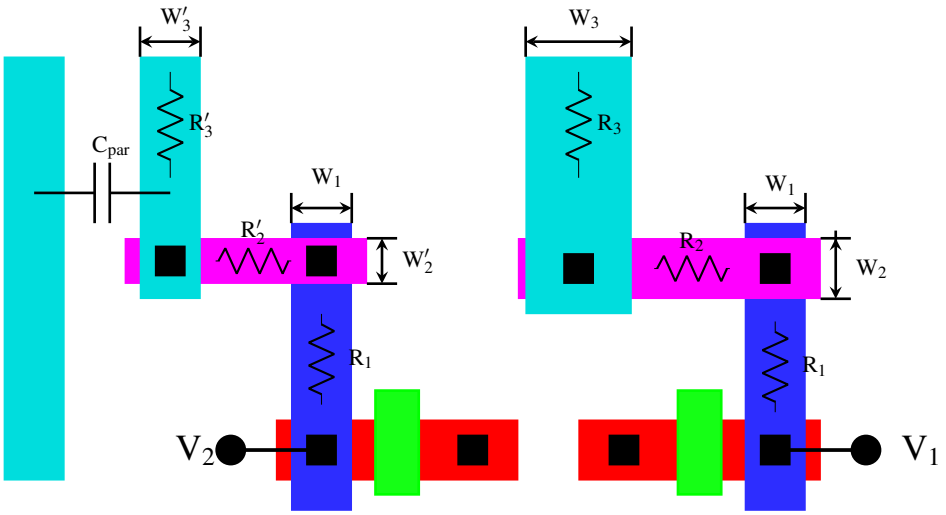


Figure 2.23: Example showing asymmetric differential routing

Figure 2.23 shows an example of an asymmetrical differential routing, where the two complementary signal paths, V_1 and V_2, are routed with different interconnect widths and suffer from different parasitic capacitances from neighbouring interconnects. Signal V_1 is routed through 3 different metal layers, with widths W_1, W_2, and W_3, which introduces respective parasitic resistance R_1, R_2, and R_3. Signal V_2, on the other hand, is routed through the same three metal layers, with the first layer having same width as V_1's first metal track, W_1, but with smaller widths for the second and third metal layers, W'_2 and W'_3 respectively. This decrease in metal width leads to an increase in parasitic resistance for the second and third metal layers, R'_2 and R'_3. A parasitic capacitance, C_{par}, is also introduced on net V_2, from the close proximity of its third metal layer to another metal in the same level of the metal stack carrying a different signal.

The signals' nets RC constants, τ_1 and τ_2, can be defined as follows:

$$\tau_1 = (R_{device} + R_1 + R_2 + R_3) \cdot C_{device} \tag{2.32}$$

$$\tau_2 = (R_{device} + R_1 + R'_2 + R'_3) \cdot (C_{device} + C_{par}) \tag{2.33}$$

where R_{device} and C_{device} are the parasitic resistance and capacitance of the device being driven by the signal. With the increase in parasitic resistance and capacitance on the V_2 path, the RC constant τ_2 becomes larger than τ_1, leading to a slower signal transition on V_2 compared to V_1. This asymmetry can result in timing mismatches for the differential signal pair, ultimately degrading the overall performance of the circuit [7], [10].

2.7.2.5 Latch-Up

Latch-up is a reliability phenomenon in bulk CMOS technologies caused by the parasitic structures inherently formed by the well, substrate, and source/drain diffusions. Adjacent NMOS and PMOS devices create parasitic NPN and PNP bipolar transistors which, together with the finite substrate

and well resistances, form a PNPN structure equivalent to a Silicon-Controlled Rectifier (SCR) [1], [6], [24].

Under normal conditions, this parasitic structure remains inactive. However, transient substrate or well currents can create voltage drops large enough to forward bias one of the parasitic bipolar junctions. The resulting current may activate the complementary transistor and establish a regenerative positive-feedback path between V_{DD} and V_{SS}. Once triggered, a large current can flow through the chip, potentially causing malfunction or permanent damage [1], [6]. An example of the resulting latch-up mechanism is shown in Figure 2.24.

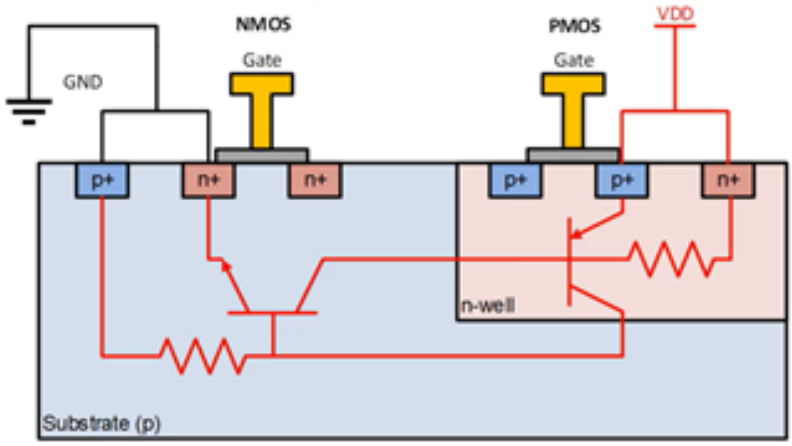


Figure 2.24: Example of a latch-up event. Reproduced from [29].

Fast-switching output stages are particularly susceptible because voltage overshoot, undershoot, capacitive coupling, or current injection from pad and Electrostatic Discharge (ESD) structures can disturb the local substrate and well potentials [1], [3]. Latch-up susceptibility is reduced by lowering the effective substrate and well resistances through closely spaced and sufficiently numerous well and substrate contacts. Guard rings are also commonly used to collect injected carriers before they can trigger the parasitic structure. A p^+ guard ring connected to V_{SS} can be placed around NMOS devices, while an n^+ well guard ring connected to V_{DD} can protect PMOS devices [1], [6].

2.8 Chapter Summary

This chapter presented the fundamental concepts required for the design and analysis of a high-speed serial-link TX. The discussion began with the evolution from parallel to serial communication links and introduced the advantages of differential signaling for high-speed data transmission. NRZ encoding was then examined in terms of its time- and frequency-domain characteristics, followed by an overview of the USB standard and the relevant USB 3.2 physical-layer requirements considered throughout this work.

The main SI limitations associated with multi-gigabit communication were subsequently discussed. ISI was related to the bandwidth-limited and frequency-dependent behavior of the transmission channel, while insertion loss, return loss, impedance discontinuities, jitter, and finite slew rate were introduced as important mechanisms that degrade the transmitted waveform. Transmission-line behavior and scattering parameters were presented as tools for characterizing the communication channel, and the eye diagram was introduced as a practical means of evaluating the resulting voltage and timing margins. Based on these limitations, TX and RX equalization techniques were discussed, with particular emphasis on TX-side feed-forward equalization. The operation of a finite-impulse-response equalizer was described through delayed and weighted replicas of the transmitted data, including the use of pre-cursor, main-cursor, and post-cursor taps to compensate for channel-induced ISI.

The chapter also introduced the output-driver architectures considered for high-speed TXs. CML and VMD were described in terms of their operating principles, output termination, differential swing, and power consumption. Their main trade-offs were then compared, showing that CML provides straightforward impedance control and implementation of multi-tap FFE at the expense of continuous static current, whereas voltage-mode and SST drivers can achieve improved power efficiency but require greater attention to output-impedance calibration and equalizer segmentation.

Finally, the discussion was extended from schematic-level design to physical implementation. The IC design methodology and the influence of layout parasitics were introduced, followed by device-sizing and matching techniques intended to reduce systematic and random mismatch. Layout-dependent effects such as the WPE and LOD effect were considered, together with routing-related limitations including IR drop, parasitic capacitance, EM, and latch-up. Particular attention was given to differential-routing symmetry, since unequal interconnect resistance and capacitance between complementary signal paths can introduce different RC time constants and consequently degrade the timing and waveform symmetry of high-speed differential signals.

These concepts establish the theoretical and physical-design basis for the TX architecture and circuit implementation presented in the following chapters.

Chapter 3

Proposed Transmitter Architecture and Development

3.1 USB 3.2 Transmitter Specifications

Building on the general transmitter requirements, Table 3.1 summarizes the normative electrical specifications defined by the USB 3.2 Gen 2 standard for 10 Gb/s operation, which constitute the design targets adopted throughout this work.

The UI is normatively defined with a nominal value of 100 ps, bounded between a minimum of 99.97 ps and a maximum of 100.03 ps, corresponding to the small allowable deviation from the nominal 10 Gb/s bit rate. The differential output voltage swing is specified with a nominal value of $1\,V_{pp}$, within an allowable range of $0.8\,V_{pp}$ to $1.2\,V_{pp}$, while the Direct Current (DC) differential output impedance is specified with a nominal value of $100\,\Omega$, within a normative range of $72\,\Omega$ to $120\,\Omega$. The Alternating Current (AC) coupling capacitor placed between the TX and RX must fall within a capacitance range of 75 nF to 265 nF.

The equalization capability of the transmitter is bounded by the normative PS and DE limits: PS is constrained between 1.2 dB and 3.2 dB, while DE is constrained between 2.1 dB and 7 dB. Finally, the standard specifies limits on the resulting eye diagram and on jitter: the eye height must fall between 70 mV and 1200 mV, while DJ, RJ, and Total Jitter (TJ) are bounded to a maximum of 0.530UI, 0.141UI, and 0.671UI, respectively.

Table 3.1: USB 3.2 Gen 2 (10 Gbps) normative transmitter specifications

Parameter	**Min**	**Nominal**	**Max**
UI	99.97 ps	100 ps	100.03 ps
Differential output voltage swing	0.8 V_{pp}	1 V_{pp}	1.2 V_{pp}
DC differential impedance	72 Ω	100 Ω	120 Ω
AC coupling capacitance	75 nF	—	265 nF
PS	1.2 dB	—	3.2 dB
DE	2.1 dB	—	7 dB
TX eye height at TP4[(1)]	70 mV	—	1.2 V
DJ	—	—	0.530 UI
RJ	—	—	0.141 UI
TJ	—	—	0.671 UI

[(1)] Measured after the prescribed reference channel and receiver equalization function.

These specifications collectively define the design envelope within which the transmitter driver presented in this dissertation must operate, and are referenced throughout the remainder of this work as the compliance targets against which the proposed design is evaluated.

3.2 Overall Driver Architecture

At a top level, the transmitter driver is composed of three parallel CML stages, one for each tap of the FIR filter described in Section 3.3: the pre-cursor, main-cursor, and post-cursor stages.

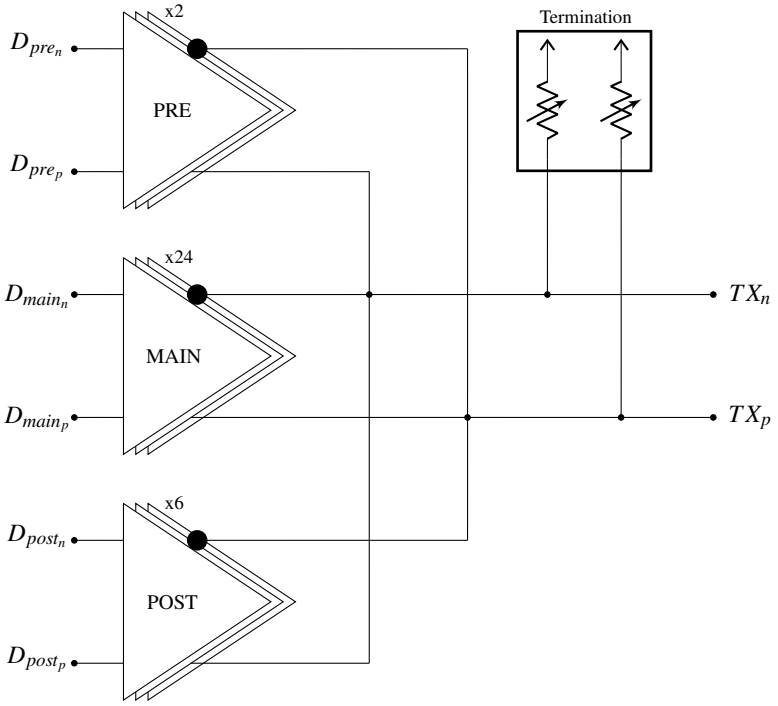


Figure 3.1: High level CML TX driver diagram

All three stages' outputs are connected to either one of two common output nodes, TX_P and TX_N, which are in turn connected to the termination and the channel. Due to the equalization coefficients used, the pre and post taps have their output polarities inverted relative to the main tap.

The transmitter driver is designed to have a tunable current source in order to control the output swing and equalization applied by the transmitter. Additionally, it requires a tunable termination to account for PVT variations. To achieve these requirements, a segmented driver architecture is chosen. Each of three stages is composed of a parallel array of identical basic segments, referred to as *tap units*, each consisting of a differential pair with a current mirror bank. Each tap unit is able to sink a maximum current of 1.2 mA. The number of tap units allocated to each stage is determined by the equalization coefficients needed for the defined equalization settings range.

The analytical equalization requirements indicate a maximum nominal main-tap current of 24 mA, corresponding to the EQ0 configuration. This requires a minimum of 20 tap units, or 120 independently controllable 200 µA current mirrors. In the final implementation, the main tap was expanded to 24 parallel tap units, providing 144 current-mirror segments and a maximum available current capacity of 28.8 mA. The additional four tap units provide digital tuning margin to compensate for PVT and layout-induced variations while preserving the nominal EQ0 operating point.

Finally, the output impedance control of the driver, corresponding to the on-die termination resistance, is likewise implemented digitally, as a *resistor bank* composed of a set of parallel resistors, each providing a fixed resistance value. The resistors within the bank are enabled or disabled such that the total output impedance corresponds to the parallel combination of the active resistors, allowing the termination to be calibrated and account for PVT and channel variations.

The combination of the three tap stages with the digitally calibrated termination forms the complete high-level structure of the CML transmitter driver, illustrated in Figure 3.1.

3.3 Transmitter Equalization

The USB 3.2 specification specifies the TX equalization settings as a range for both PS and DE. As discussed in Section 3.1, PS is bounded between a minimum of 1.2 dB and a maximum of 3.2 dB, while DE is bounded between a minimum of 2.1 dB and a maximum of 7 dB. Both parameters are defined as ratios of the differential output voltage measured at different instants of the transmitted waveform, corresponding to the settling behaviour of the 3-tap FIR filter around a data transition, as given by

$$PS(\text{dB}) = 20\log_{10}\left(\frac{V_c}{V_b}\right), \qquad DE_{\text{mag}}(\text{dB}) = 20\log_{10}\left(\frac{V_b}{V_a}\right). \tag{3.1}$$

where V_a, V_b and V_c are the differential output voltages measured at the instants shown in Figure 3.2.

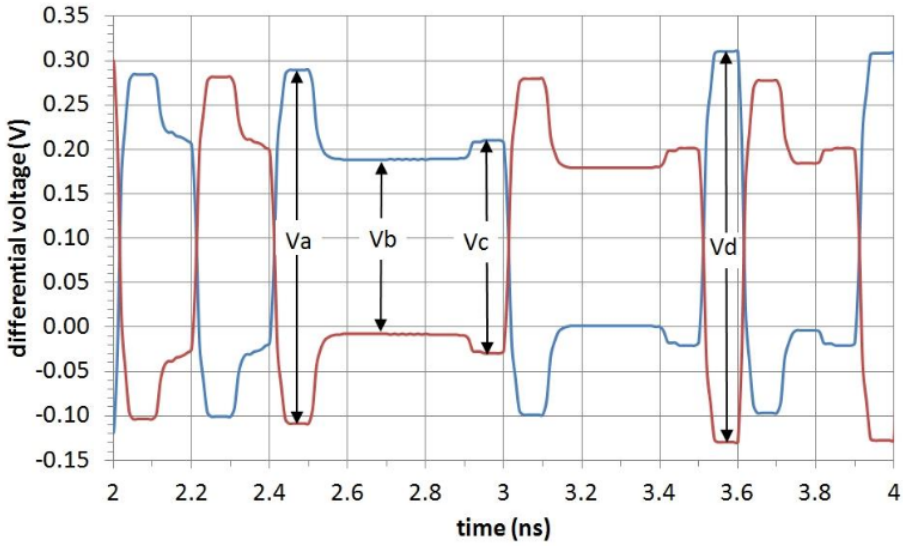


Figure 3.2: Example output waveform provided in the USB 3.2 specification. Reproduced from [11]

Since equalization is defined as a ratio of voltages, each of the output voltages V_a, V_b and V_c can be expressed as a function of the equalizer's tap coefficients, c_{-1}, c_0 and c_1, together with the defined output swing, V_{swing}:

$$V_a = C_a \cdot V_{swing} = (c_{-1} + c_0 - c_1) V_{swing} \tag{3.2}$$

$$V_b = C_b \cdot V_{swing} = (c_{-1} + c_0 + c_1) V_{swing} \tag{3.3}$$

$$V_c = C_c \cdot V_{swing} = (-c_{-1} + c_0 + c_1) V_{swing} \tag{3.4}$$

Substituting these expressions into Equation (3.1), the PS and DE ratios can be written directly in terms of the tap coefficients, as shown in Equation (3.5):

$$PS\,(\mathrm{dB}) = 20\log\left(\frac{-c_{-1} + c_0 + c_1}{c_{-1} + c_0 + c_1}\right), \qquad DE\,(\mathrm{dB}) = 20\log\left(\frac{c_{-1} + c_0 + c_1}{c_{-1} + c_0 - c_1}\right) \tag{3.5}$$

The output swing itself is set by the difference between V_{DD} and $V_{DD} - RI_{SS}/2$, so that $V_{swing} = RI_{SS}/2$. Substituting this into the expressions for V_a, V_b and V_c gives

$$V_a = (c_{-1} + c_0 - c_1)\frac{RI_{SS}}{2} = c_{-1}\frac{RI_{SS}}{2} + c_0\frac{RI_{SS}}{2} - c_1\frac{RI_{SS}}{2} \tag{3.6}$$

$$V_b = (c_{-1} + c_0 + c_1)\frac{RI_{SS}}{2} = c_{-1}\frac{RI_{SS}}{2} + c_0\frac{RI_{SS}}{2} + c_1\frac{RI_{SS}}{2} \tag{3.7}$$

$$V_c = (-c_{-1} + c_0 + c_1)\frac{RI_{SS}}{2} = -c_{-1}\frac{RI_{SS}}{2} + c_0\frac{RI_{SS}}{2} + c_1\frac{RI_{SS}}{2} \tag{3.8}$$

This decomposition shows that the termination current pulled by each tap can be expressed as the product of that tap's coefficient and the termination current $I_{SS}/2$, as shown in Equation (3.9):

$$I_{c_{-1}} = c_{-1}\frac{I_{SS}}{2}, \qquad I_{c_0} = c_0\frac{I_{SS}}{2}, \qquad I_{c_1} = c_1\frac{I_{SS}}{2} \tag{3.9}$$

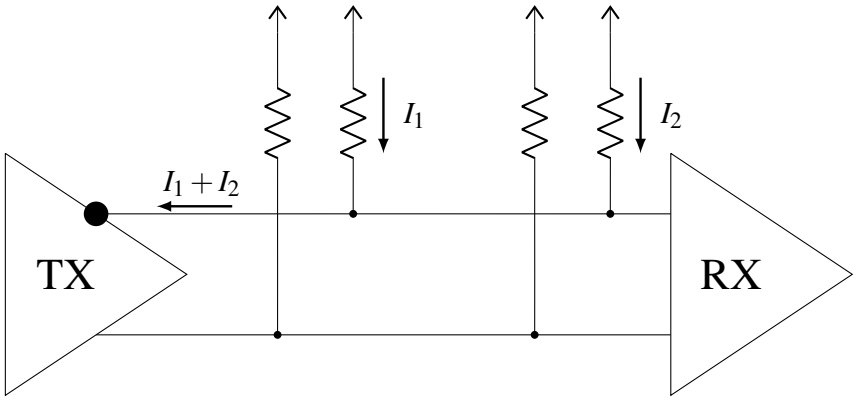


Figure 3.3: TX and RX termination currents

Furthermore, since each tap must sink the current required to generate the voltage swing across both the TX and RX terminations, as seen in Figure 3.3, its tail current is twice the corresponding termination current, as given in Equation (3.10):

$$I_{tail,c_{-1}} = 2I_{c_{-1}} = c_{-1}I_{SS}, \qquad I_{tail,c_0} = 2I_{c_0} = c_0 I_{SS}, \qquad I_{tail,c_1} = 2I_{c_1} = c_1 I_{SS} \tag{3.10}$$

By defining fixed equalization presets, and consequently fixed tap coefficients, the current that each tap must sink to achieve the desired equalization can therefore be calculated directly from Equations (3.9) and (3.10). It should be noted that the tap coefficients are not independent, but must satisfy the normalization constraint

$$|c_{-1}| + |c_0| + |c_1| = 1 \tag{3.11}$$

which ensures that the sum of the absolute values of the coefficients remains equal to unity for all valid equalization presets, or, in current terms, that the total current sunk by the driver remains constant regardless of the equalization setting applied.

3.3.1 Equalization Settings

With the relationships between the equalization coefficients and the current sunk by each tap established, the next step is to determine the specific values of the coefficients corresponding to the equalization presets. The USB 3.2 specification does not provide explicit values for the coefficients, but rather a interval of PS and DE values, from which the coefficients can be arbitrarily defined with respect to the constraint defined in Equation (3.11).

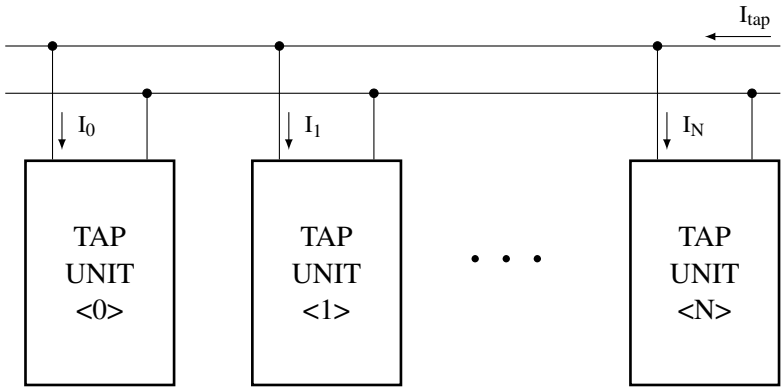


Figure 3.4: Tap current flow

The total current sunk by a tap is configured by setting the enable signals of each individual tap unit that composes it. Each tap unit has 6 controllable current mirrors in parallel, each biased to sink a fixed current of 200 μA. This structure is what allows each tap to sink a maximum current of 1.2 mA and be digitally controlled in steps of 200 μA. The current sunk by each tap is therefore given by the sum of the currents sunk by each of its active tap units, as shown in Figure 3.4. The total current sunk by a tap is therefore given by

$$I_{tap} = I_{mirror} \cdot \sum_{i=0}^{N} \sum_{j=0}^{5} en_{ij} \tag{3.12}$$

where en_{ij} is the boolean value, '0' or '1', of the enable signal of current mirror j in tap unit i, and I_{mirror} is the current sunk by each active current mirror.

The minimum number of tap units per tap is determined by the maximum current that tap must sink to achieve the defined equalization settings.

3.3.1.1 Equalization Presets and Current Requirements

Table 3.2 lists the PS and DE values for each equalization preset, along with the corresponding tap coefficients and tap currents calculated according to the relationships defined by Equation (3.10).

Table 3.2: Equalization Presets

Preset	**PS (dB)**	**DE (dB)**	c_{-1}	c_0	c_1	I_{pre} **(mA)**	I_{main} **(mA)**	I_{post} **(mA)**
EQ0	0	0	0	1	0	0	24	0
EQ1	1.2	2.1	-0.052	0.852	-0.096	1.25	20.4	2.3
EQ2	2.2	3.1	-0.084	0.791	-0.125	2.02	19.0	3.0
EQ3	3.2	7	-0.083	0.686	-0.231	1.99	16.5	5.54
EQ4	2.5	5	-0.079	0.737	-0.184	1.90	17.7	4.42

With the equalization presets chosen, the corresponding tap coefficients can be calculated using the following equations

$$k = 10^{PS/20}, l = 10^{-DE/20} \tag{3.13}$$

$$C_{-1} = \frac{l \times (1-k)}{2 \times (k \times l + 1 - l)} \tag{3.14}$$

$$C_1 = \frac{l-1}{2 \times (k \times l + 1 - l)} \tag{3.15}$$

$$C_0 = 1 - abs(C_{-1}) - abs(C_1) \tag{3.16}$$

The USB specification reports de-emphasis as a positive magnitude. Accordingly, DE_{mag} is used throughout this work. The quantity l represents the ratio V_b/V_a, which is smaller than unity for de-emphasized steady-state symbols; therefore, $l = 10^{-DE_{\text{mag}}/20}$.

The maximum current that each tap is expected to sink in order to achieve all the desired presets can be calculated, along with the number of tap units required to sink that current. With a tap unit being capable of sinking a maximum current of 1.2 mA, the minimum number of tap units required for each tap can be calculated by

$$N_{tap} = \left\lceil \frac{I_{tap,max}}{I_{unit}} \right\rceil \tag{3.17}$$

where $I_{tap,max}$ is the maximum current that the tap must sink to achieve the desired equalization range, and I_{unit} is the maximum current that a single tap unit can sink, with the minimum number of tap units required being the ceiling of the ratio between these two values.

From this calculation, the minimum implementation requires 20 tap units for the main tap, 2 for the pre-cursor tap, and 6 for the post-cursor tap. This corresponds to minimum current-control capacities of 120, 12, and 36 enable signals, respectively. As discussed in Section 3.2, the final main-cursor stage was expanded from 20 to 24 tap units, increasing its available current-mirror segments from 120 to 144. Table 3.3 lists the number of enabled segments required for the nominal equalization presets; the additional 24 main-tap segments remain available as tuning margin.

Because the current banks have a resolution of 200 µA, the analytical current targets in Table 3.2 cannot in general be realised exactly. The enable counts in Table 3.3 therefore correspond to the nearest practical current allocation used as the initial configuration for circuit-level verification.

Table 3.3: Enable Signal Mapping

Preset	# en_{pre}	# en_{main}	# en_{post}
EQ0	0	120	0
EQ1	7	102	12
EQ2	10	95	15
EQ3	10	83	28
EQ4	10	89	22

3.4 Driver Implementation

3.4.1 Tap Unit

The driver is implemented as a segmented, current-mode structure in which each of the three FIR taps described in the previous section is realised by an array of identical physical tap units. Each tap unit is composed of a differential pair together with a six-segment current-mirror bank and contributes a digitally controllable fraction of the total tap current. Because the pre-cursor, main-cursor, and post-cursor stages are realised as separate, independently sized and independently enabled arrays of tap units, the driver as a whole implements the FIR summation of Equation (2.4) directly in the current domain. The corresponding tap-unit schematic, including the differential pair, switchable current mirrors, and local termination elements, is shown in Figure 3.5.

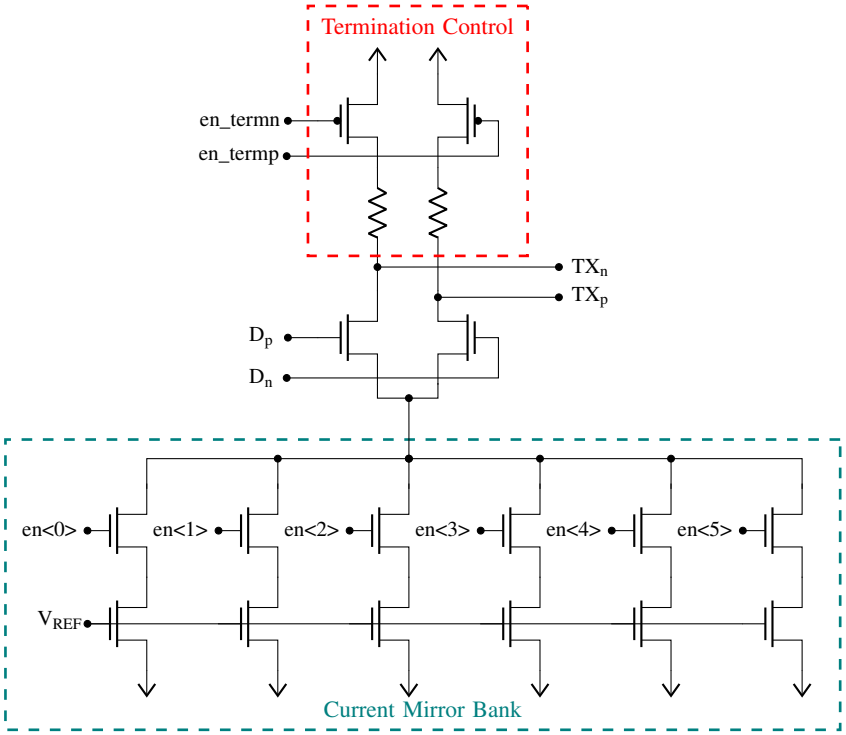


Figure 3.5: Tap unit schematic

Within each tap unit, the current mirror bank acts as the tunable current source and is composed of six current mirrors, each paired with an NMOS device that acts as a switch to enable or disable that mirror. This segmentation allows the current steered by the differential pair to be adjusted digitally by enabling or disabling individual mirrors, rather than through a continuously variable bias. In addition to the current mirror bank, each tap unit incorporates two 1 kΩ resistors, one associated with each of the differential outputs. These resistors are not specific to any single tap; rather, when the resistors from all three taps are combined in parallel, they form the tunable termination network of the driver as a whole. This co-location of the termination resistors within

each tap unit allows the termination to be adjusted using the same segmented, switch-based approach used for the current sources, and keeps the layout of each tap unit self-contained, discussed in Section 3.5.

3.4.2 Tap Unit Sizing and Biasing

The USB 3.2 specification requires a nominal single-ended termination of $50\,\Omega$ at each output, and additionally defines an allowable range for the transmitted output voltage swing. Within this range, the driver described in this work was designed to support the maximum specified differential output swing of $1.2\,\mathrm{V_{pp}}$, in order to maximize the control margin of the driver. Since the differential swing is split equally between the two single-ended outputs, a differential swing, $V_{out,diff}$, of $1.2\,\mathrm{V_{pp}}$ corresponds to a single-ended swing of

$$V_{\mathrm{swing,SE}} = \frac{V_{\mathrm{out,diff,pp}}}{2} = 0.6\ \mathrm{V_{pp}}. \tag{3.18}$$

at each output node.

Each output node of the driver settles to a voltage set by difference between $\mathrm{V_{DD}}$ and the voltage drop across the termination:

$$V_{\mathrm{out,L}} = V_{DD} - \frac{I_{SS}}{2} R_{\mathrm{term}}. \tag{3.19}$$

At the maximum differential output swing, the minimum output-node voltage is therefore approximately 0.6 V. At the same time, the current delivered by each tap unit is not fixed, but is instead digitally programmable, as described in the preceding section, ranging from a minimum drain current of $I_{D,min} = 200\,\mu\mathrm{A}$, corresponding to a single active current mirror, up to a maximum of $I_{D,max} = 1.2\,\mathrm{mA}$, corresponding to all six mirrors active simultaneously.

For correct operation of the driver, three conditions must be satisfied simultaneously, for every combination of output voltage and tap current within the range defined above:

- The device of the differential pair that is turned on by the input data must remain in saturation, so that it behaves as a well-defined current source and the tap current is accurately steered to the intended output.
- The device of the differential pair that is turned off must remain in cutoff, so that it contributes no current to the opposite output and the full tap current is steered to only one side of the differential output; however, this device should not be driven too deep into cutoff, since doing so would require an unnecessarily large input voltage swing, which in turn would degrade the driver's switching speed.
- All active current mirror devices remain in saturation, so that the current sourced by each unit mirror accurately reflects its reference current, independent of the output voltage.

With the output defining the available voltage headroom in which both the active differential pair and active mirror devices must remain in saturation, and the variability of the tap current

defining the range of drain currents that must be achieved, keeping all three of these conditions satisfied simultaneously across the full range of output voltage and tap current makes it essential to identify the worst-case biasing condition, and to define the sizing and biasing of the differential pair and current mirror devices around that corner. The relevant device voltages in the current-steering path are identified in Figure 3.6.

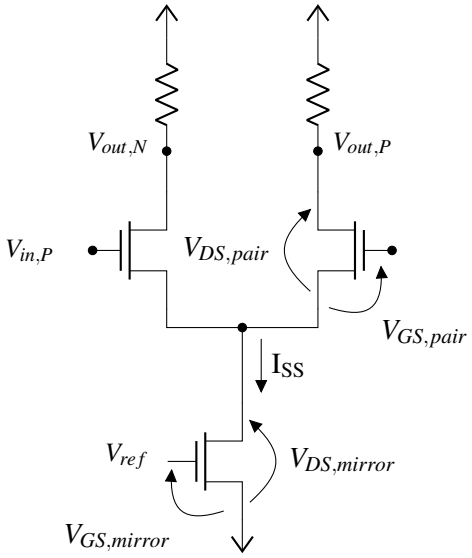


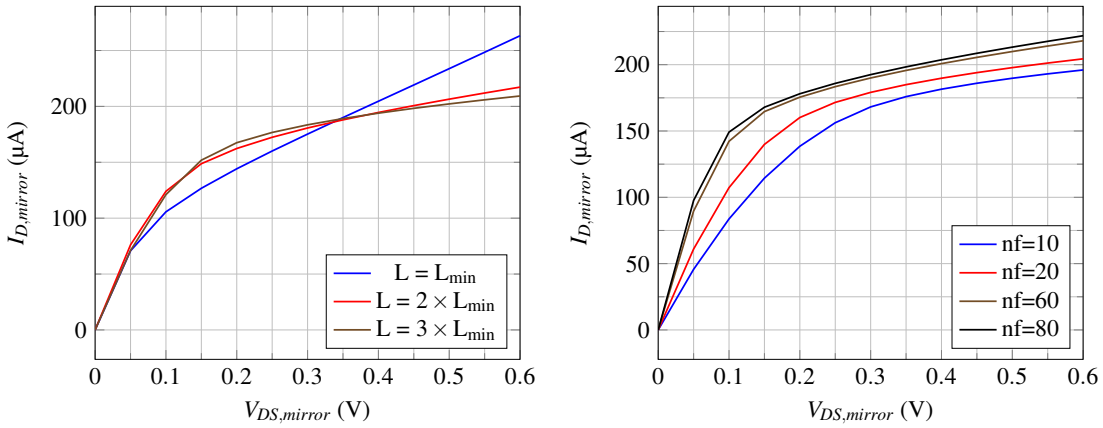
Figure 3.6: Differential pair and current mirror node voltages

Using a diode connected NMOS circuit to generate the biasing voltage, V_{ref}, for a reference current, I_{ref}, of 200 µA, a sweep of the mirror device's length was performed to determine which length would provide a stable current copy across the expected V_{DS} range. The length and finger-count sweeps are combined in Figure 3.7. Its length-sweep panel, Figure 3.7a, plots the mirror drain current, $I_{D,mirror}$, against V_{DS} for three different device lengths. The results show that a mirror length of $L_{mirr} = 3 \times L_{min}$ provides the best tradeoff between device size and current-copying stability across the expected range of $V_{DS,mirror}$, and was therefore selected for the current mirror devices within each tap unit.

With the mirror length defined, its finger count was then swept to determine which device width would provide the closest match to I_{ref} for the selected length. Figure 3.7b shows the results of this sweep, with $I_{D,mirror}$ plotted against its V_{DS} for four different finger counts. The results show that a mirror finger count of $nf = 60$ provides a good trade-off between current matching and device area, and was therefore selected for the current mirror devices within each tap unit.

The total voltage headroom available for the current-steering path is set by the driver's output voltage, and is shared between the current mirror device and the differential pair device connected in series with it, according to

$$V_{DS,mirr} + V_{DS,pair} = V_{out}. \tag{3.20}$$



(a) I_D versus V_{DS} for varying lengths

(b) I_D versus V_{DS} with $L = 3 \times L_{min}$

Figure 3.7: Mirror I_D versus V_{DS} for varying device lengths and finger counts

To ensure that the mirror remains in saturation, that is, so it's overdrive voltage, V_{OV}, is less than its V_{DS}, a Super Low Threshold Voltage (SLVT) device flavor was selected, giving more headroom for the current source devices. The pair's overdrive voltage, $V_{OV,pair}$, is arbitrarily defined based on the device's gm/Id ratio that leads to strong inversion operation. The relationship between g_m/I_D and V_{OV}, including the weak-, moderate-, and strong-inversion regions, is illustrated in Figure 3.8.

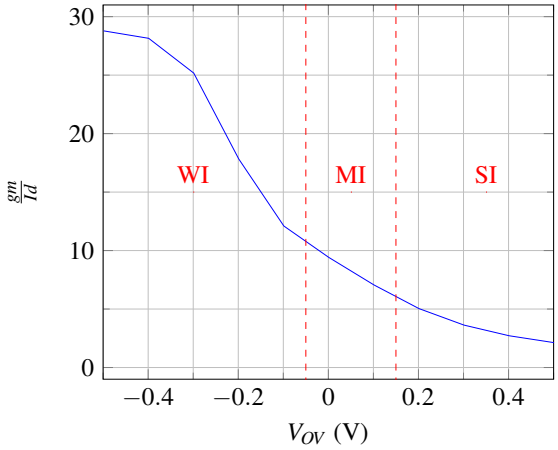


Figure 3.8: $\frac{gm}{Id}$ versus V_{OV}

A value of $V_{OV,\text{pair}} = 0.2$ V was selected as a compromise between transconductance, switching speed, device size, and saturation headroom for the current mirrors. The Regular Threshold Voltage (RVT) device flavor was retained for the differential pair since the required switching speed could be achieved without resorting to SLVT devices. In addition to its lower leakage, the higher threshold voltage reduces residual conduction in the nominally off branch, improving current-steering accuracy and limiting unnecessary static current. The resulting saturation margin and switching performance were verified over the complete current and output-voltage ranges in

the subsequent simulations.

With the mirror sizing done, a sweep of the differential pair drain current, $I_{D,pair}$, and tail voltage, V_{tail} against a range of $V_{in,CM}$ was performed for different finger counts. To define a starting $V_{in,CM}$, three design considerations must be taken into account:

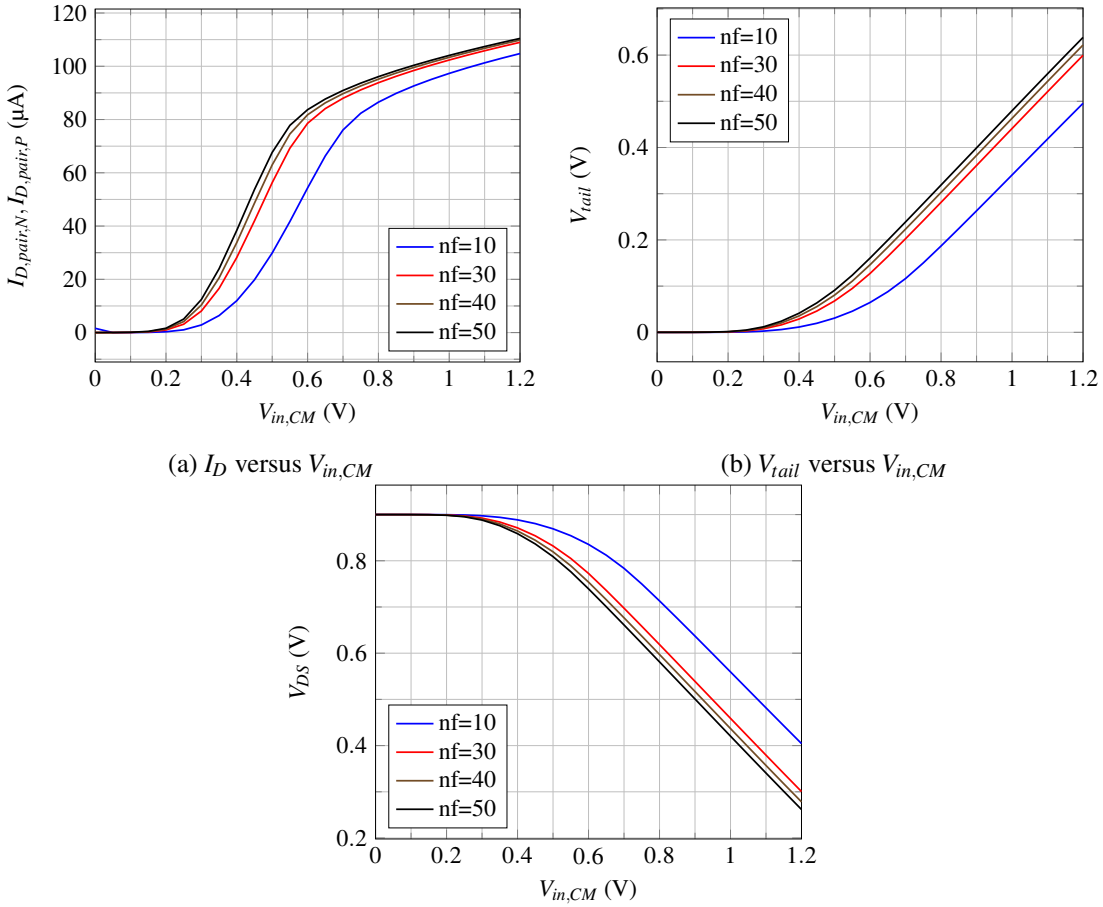
1. For this $V_{in,CM}$ the current steered by the differential pair must be close to the reference current

2. At the same time, for the selected $V_{in,CM}$, the tail voltage must be high enough to ensure that the mirror device is in saturation

3. $V_{in,CM}$ must respect the condition presented in Equation (3.21), which ensures that the condition $V_{DS} \geq V_{OV}$ is met for both current mirror and differential pair devices.

The minimum $V_{in,CM}$ determined in Equation (3.21) ensures that the mirror remains in saturation, while the maximum $V_{in,CM}$ ensures that both mirror and differential pair devices remain in saturation. With these three considerations in mind, a $V_{in,CM}$ can be defined that provides a tradeoff between $I_{D,pair}$ and V_{tail}, as well as keeping both devices in their desired regions during differential operation.

$$V_{GS,pair} + V_{OV,mirror} \leq V_{in,CM} \leq \min(V_{DD} - V_{OUT} + V_{TH,pair}, V_{DD}) \tag{3.21}$$



(a) I_D versus $V_{in,CM}$

(b) V_{tail} versus $V_{in,CM}$

(c) V_{DS} versus $V_{in,CM}$

Figure 3.9: Differential pair I_D, $V_{DS,mirror}$ and V_{tail} versus $V_{in,CM}$ for varying device finger counts

The sweeps in Figure 3.9 show how the differential-pair current and internal node voltages vary with input common-mode voltage for different device finger counts. These results support the selection of an initial common-mode bias before the maximum-current corner is evaluated.

Even though this first design corner is a good start for the differential pair design, another issue arises for the case when the tap unit sinks its maximum current. With the increase of the tail current, V_{DS} of the steering device increases, while V_{DS} of the mirror device decreases. This leads to a higher V_{GS} on the steering device, increasing its V_{OV} and, if not carefully considered, can lead to the steering device leaving saturation since V_{DS} does not increase at the same rate of increase as V_{GS}. This is a corner case that must be considered, and the sizing of the differential pair must be done in a way that ensures that the steering device remains in saturation for the entire current range. This is done by analyzing the system behavior for input swing and maximum current.

By sweeping the differential input voltage around the previously defined $V_{in,CM}$, the current-steering characteristic of the differential pair can be evaluated and an appropriate input swing can be selected. The two input voltages are defined as

$$V_{in,P} = V_{in,CM} + \Delta \tag{3.22}$$

and

$$V_{in,N} = V_{in,CM} - \Delta, \tag{3.23}$$

where Δ represents the voltage excursion of each input from the common-mode level. Therefore, the instantaneous differential input voltage is

$$V_{IN,diff} = V_{in,P} - V_{in,N} = 2\Delta. \tag{3.24}$$

For the opposite logic state, the polarity is reversed, such that $V_{IN,diff} = -2\Delta$. Thus, a value of $\Delta = 0.2$ V corresponds to input levels of 1.0 V and 0.6 V for $V_{in,CM} = 0.8$ V, giving $V_{IN,diff} = \pm 0.4$ V, or 0.8 V_{pp} differential.

Figure 3.10 shows the drain current of both steering devices as a function of the differential input voltage. Around $V_{IN,diff} = 0$, both devices conduct and the tail current is progressively transferred from one branch to the other. As $|V_{IN,diff}|$ increases, the current in the selected device approaches the complete branch current while the current in the opposite device approaches zero. Beyond this transition region, increasing the differential input voltage produces little additional improvement in current steering, since the transfer characteristic has already approached its limiting value.

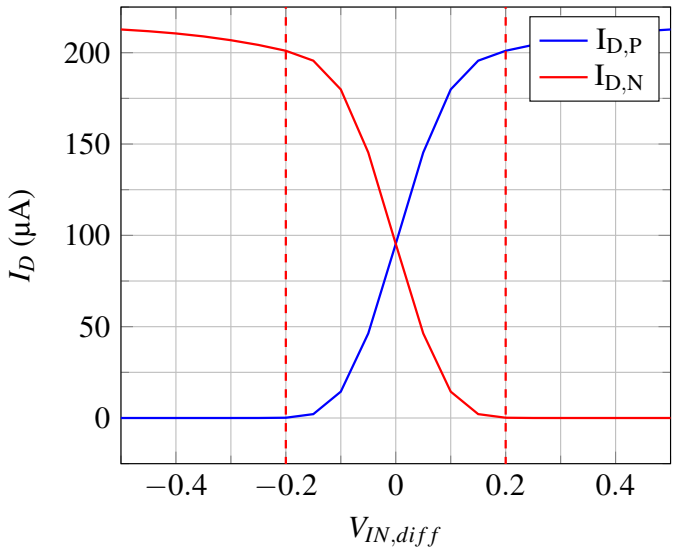


Figure 3.10: I_D of both steering devices

The input swing should therefore be large enough for the two logic levels to reach the edges of the useful current-steering region. If the swing is too small, the operating points remain within the transition region and a significant fraction of the tail current continues to flow through the device that is intended to be off. This reduces the effective current delivered to the selected output branch and introduces an error in the intended tap current. Conversely, a substantially larger swing provides little additional current-steering benefit once nearly complete steering has been reached, while unnecessarily increasing the voltage swing required from the preceding stage.

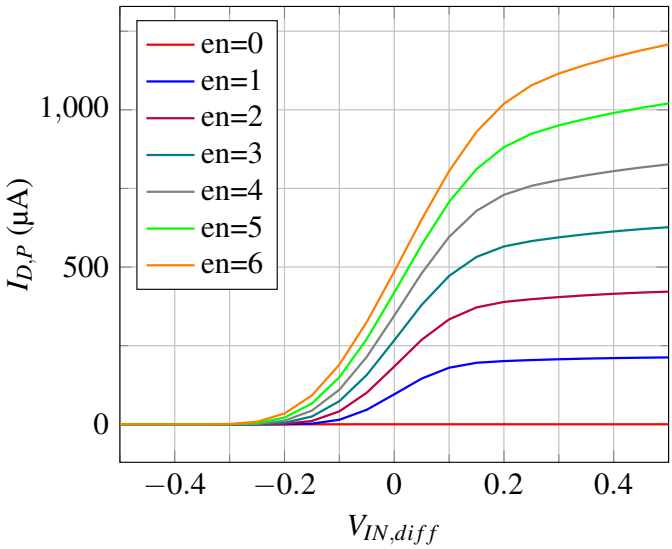


Figure 3.11: $I_{D,P}$ for current enable signals

Figure 3.11 confirms that the same behavior is maintained as the number of enabled current-mirror segments, and therefore the tap current, is increased. Although the absolute branch current increases with the enable code, the differential pair reaches approximately the same current-steering region with respect to $V_{IN,diff}$. The selected input swing can therefore be used over the complete programmable-current range.

Based on this tradeoff, $V_{in,CM}$ was set to 0.8 V and $\Delta = 0.2$ V was selected. Each input therefore swings between 0.6 V and 1.0 V, corresponding to a 0.4 V_{pp} single-ended swing and 0.8 V_{pp} differential swing. This choice places the two logic levels close to the limits of the useful current-steering characteristic, exploiting most of the available transfer region while ensuring that only a small residual current is conducted by the nominally off device. At the same time, it avoids requiring an unnecessarily large input swing from the preceding stage. The final tap-unit device sizing is summarized in Table 3.4.

Table 3.4: Tap unit sizing

Device	Flavor	Length	# fingers
Diff Pair	RVT	L_{min}	40
Mirror	SLVT	$3 \times L_{min}$	60

3.4.2.1 Current Mirror Biasing Circuit

The design of the mirror-based current source requires a reference circuit capable of generating the bias voltage, V_{REF}, used by the current-mirror devices in the tap units. Ideally, this bias should result in a copied current, I_{COPY}, equal to a well-defined reference current, I_{REF}. The general concept is illustrated in Figure 3.12, where a reference generator produces the voltage applied to the current-source device of the target circuit.

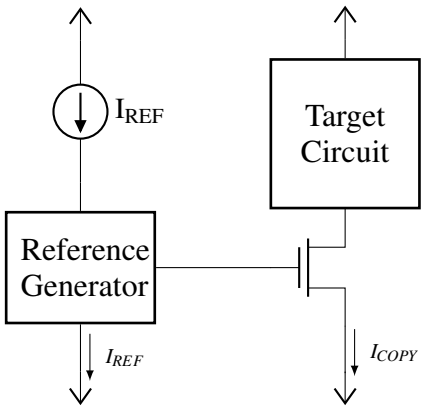


Figure 3.12: Conceptual structure of reference and target circuits. Adapted from [2]

Neglecting channel-length modulation, the drain current of a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) operating in saturation can be approximated by

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2. \tag{3.25}$$

For a device with fixed dimensions and process parameters, this expression shows that the drain current is primarily controlled by V_{GS}. Consequently, a basic current mirror can be constructed using a diode-connected reference transistor to generate the gate-source voltage required to conduct I_{REF} [2]. For matched reference and mirroring devices,

$$V_{REF} = V_{GS} = f^{-1}(I_{REF}), \tag{3.26}$$

and, under ideal conditions, applying the same V_{GS} to the output transistor results in $I_{COPY} \approx I_{REF}$.

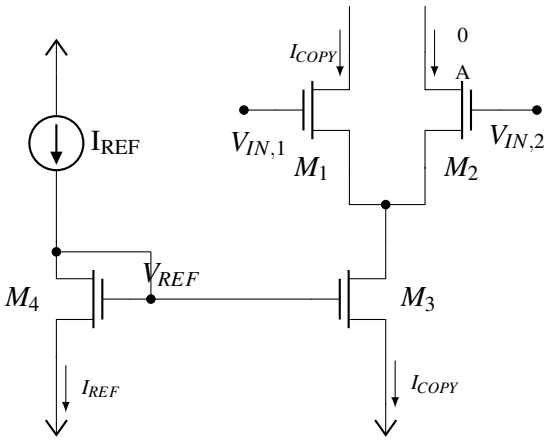


Figure 3.13: CML with a diode-connected reference generator

In the target CML branch shown in Figure 3.13, however, the mirroring transistor M_3 is stacked beneath the differential-pair steering transistor M_1. The current through the two series devices is

necessarily the same,

$$I_{COPY} = I_{D,1} = I_{D,3}, \tag{3.27}$$

but this does not guarantee that $I_{COPY} = I_{REF}$.

The main limitation of the simple diode-connected reference is that the reference and output mirror devices do not necessarily operate at the same drain-source voltage. In the reference branch, the diode-connected device operates with

$$V_{DS,REF} = V_{GS,REF}, \tag{3.28}$$

whereas the V_{DS} of the mirroring device in the target branch is determined by the voltage headroom imposed by the differential-pair steering device and by the transmitter output operating point.

In an ideal square-law model without channel-length modulation, the drain current in saturation is independent of V_{DS} and this difference would not affect the copied current. In a practical short-channel device, however, the saturation current retains a finite dependence on V_{DS}. A first-order representation including channel-length modulation is

$$I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2(1 + \lambda V_{DS}), \tag{3.29}$$

where λ represents the channel-length-modulation coefficient. Therefore, even when the reference and target mirror devices share the same V_{GS}, a difference between their drain-source voltages introduces an error between I_{COPY} and I_{REF}.

The differential-pair device consequently affects the current-copying accuracy indirectly: by being connected in series with the mirror device, it determines part of the voltage distribution in the target branch and therefore modifies V_{DS} of the mirroring transistor. This makes the operating point of the target mirror different from that reproduced by a simple diode-connected reference generator.

To reduce this mismatch, the reference circuit used in this work employs a replica of the relevant target branch, as illustrated in Figure 3.14. Instead of generating V_{REF} using only a diode-connected mirror transistor, a replica steering transistor, M_5, is placed in series with the reference mirror device, M_4. The gate of M_5 is biased at $V_{IN,CM}$, reproducing the nominal bias condition of the steering transistor in the target branch.

The effect of the replica device can be observed directly from the drain-source voltages of the two mirror devices. For the reference branch,

$$V_{DS,4} = V_{IN,CM} - V_{GS,5} - V_{SS}, \tag{3.30}$$

whereas, for the corresponding transmitter branch,

$$V_{DS,3} = V_{IN,1} - V_{GS,1} - V_{SS}. \tag{3.31}$$

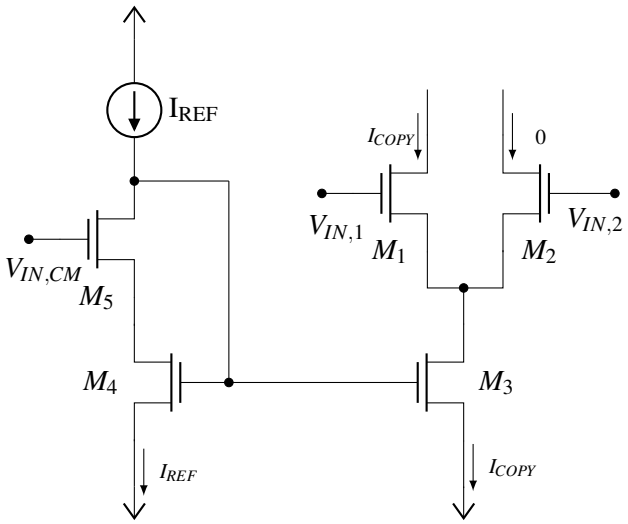


Figure 3.14: CML with a replica reference generator

Since M_5 replicates the steering device M_1, their V_{GS} values are approximately equal when carrying the same branch current. Consequently,

$$V_{DS,3} - V_{DS,4} \approx V_{IN,1} - V_{IN,CM}. \tag{3.32}$$

By reproducing the stacked device structure of the target branch, the replica generator therefore provides a reference operating point that more closely matches the voltage conditions experienced by the current-mirror devices in the transmitter. Since the channel-length-modulation term in Equation 3.29 depends directly on V_{DS}, matching $V_{DS,4}$ and $V_{DS,3}$ reduces the systematic difference between I_{REF} and I_{COPY} relative to the simple diode-connected reference.

3.4.3 Configurable Current Source

As illustrated by the high-level model in Figure 3.15, the current mirror bank within each tap unit provides a digitally tunable tail-current source. As described in Section 3.3, this bank comprises six unit current mirrors, each gated by its own NMOS switch. Each individual current mirror is sized and biased to sink a fixed unit current of 200 µA. Enabling or disabling individual mirrors therefore allows the total tail current of a given tap unit to be adjusted in discrete steps of 200 µA, giving a tunable range from 0 mA, when all mirrors are disabled, up to a maximum of 1.2 mA, when all six mirrors are enabled simultaneously. The resulting tail current can be expressed as

$$I_{SS} = 200\,\mu\text{A} \times n \tag{3.33}$$

where n is the number of enabled current mirrors, with $n \in \{0, 1, \ldots, 6\}$. This unit-cell approach, in which a fixed number of identically sized current mirrors are selectively enabled, was

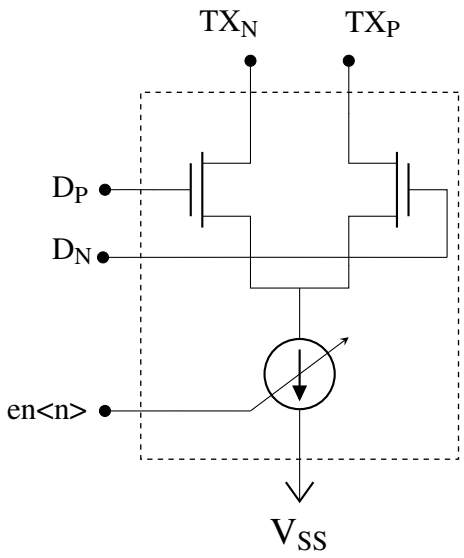


Figure 3.15: Tap Unit current control high-level view

chosen over a continuously variable bias so that the tail current of each tap can be set with predictable, monotonic steps and good matching between tap units, both of which are important for accurately realising the tap coefficient ratios required by the equalization presets.

3.4.4 Configurable Termination

The termination network of the driver is implemented as a resistor bank composed of a parallel array of 32 individual 1 kΩ resistors for each output, giving a total of 64 resistors across the two differential outputs. As with the current source, each resistor is paired with a digitally controlled switch, in this case a PMOS device, which enables or disables that resistor and thereby determines whether it contributes to the termination presented at its associated output. Because the resistors are connected in parallel, enabling additional resistors monotonically reduces the effective termination resistance, allowing the termination to be tuned by digitally controlling how many of the 32 resistors are active. The resulting parallel resistance is given by

$$R_{term} = \frac{1\,\mathrm{k}\Omega}{n} \tag{3.34}$$

where n is the number of enabled resistors, with $n \in \{1, 2, \ldots, 32\}$. This scheme allows the termination resistance to be tuned between a maximum of 1 kΩ, obtained when only a single resistor is enabled, and a minimum of 31.25 Ω, obtained when all 32 resistors are enabled in parallel. This fine granularity of termination control allows the driver's output impedance to be calibrated against PVT variation, and to be matched to the nominal 50 Ω single-ended (100 Ω differential) impedance expected by the USB 3.2 channel, independently of the tap current settings described above.

Equation 3.34 represents the first-order resistor-bank relationship used during the initial design of the termination network. In the implemented circuit, however, the effective impedance of each enabled branch is also influenced by the switching device and by process-dependent parasitic effects. Consequently, the actual number of enabled branches required to obtain a given output impedance is determined from simulation rather than directly from Equation 3.34, as demonstrated by the pre- and post-extraction results in Chapter 4.

3.5 Layout

The modular design approach adopted for the driver, in which each FIR tap is realised as an array of identical, self-contained tap units, also translates directly into flexibility at the layout level. By focusing on a single, well-defined base building block, the tap unit, and by designing its layout as a standard cell with modularity as a primary constraint, it became possible to change certain aspects of the driver's design, such as each indivudal tap drive strength through its tap unit count, without requiring the underlying layout to be redesigned from scratch. This is a significant advantage in an iterative design flow, since it decouples changes at the architectural or sizing level from the physical implementation, allowing the same cell to be reused and re-instantiated as the design is refined.

Central to this approach is the requirement that the tap unit cell be abuttable, meaning that multiple instances of the cell can be placed directly adjacent to one another without gaps or additional routing overhead. To achieve this, all power connections and other shared signals were deliberately positioned along the side edges of the cell, so that matching connections on neighbouring cells align automatically when the cells are abutted. This choice effectively defines a clear physical and electrical boundary for the cell from the very start of the layout process, rather than allowing the cell's footprint to be determined organically as the layout progresses. The resulting top-level organization of tap-unit arrays, reference cells, and boundary cells is illustrated in Figure 3.16.

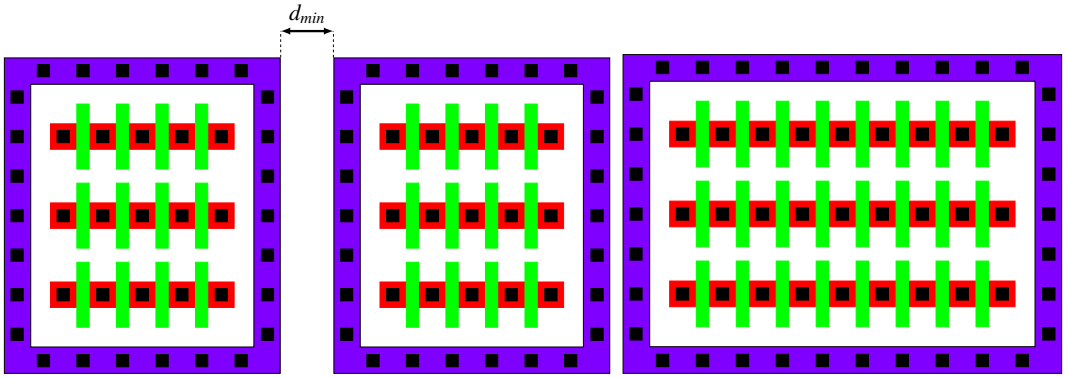
END_CELL	PRE_TAP<0>	POST_TAP<0>	POST_TAP<2>	POST_TAP<4>	MAIN_TAP<0>	MAIN_TAP<2>	MAIN_TAP<4>	MAIN_TAP<6>	MIRR_REF	MAIN_TAP<8>	MAIN_TAP<10>	MAIN_TAP<12>	MAIN_TAP<14>	MAIN_TAP<16>	MAIN_TAP<18>	MAIN_TAP<20>	MAIN_TAP<22>	END_CELL
END_CELL	PRE_TAP<1>	POST_TAP<1>	POST_TAP<3>	POST_TAP<5>	MAIN_TAP<1>	MAIN_TAP<3>	MAIN_TAP<5>	MAIN_TAP<7>	MIRR_DUMMY	MAIN_TAP<9>	MAIN_TAP<11>	MAIN_TAP<13>	MAIN_TAP<15>	MAIN_TAP<17>	MAIN_TAP<19>	MAIN_TAP<21>	MAIN_TAP<23>	END_CELL

Figure 3.16: Driver layout floorplan

Establishing this boundary early in the design process had two important consequences for the remainder of the layout work:

1. It allowed considerably more freedom in how the internal layout of each standard cell was arranged, since the usable area of the cell was clearly and unambiguously defined from the outset; routing and device placement within the cell could be optimized freely, provided the result remained consistent with the fixed boundary and edge connections.

2. Defining the boundary in this way meant that the connection points between adjacent cells had to be given careful consideration from the earliest stages of the layout design, rather than being treated as an afterthought once individual cells were already complete. Since these edge connections directly determine whether cells can be abutted correctly, ensuring their placement and geometry were consistent across all tap unit instances was essential to the success of the modular approach.



(a) Non-abutted layout with separate guard rings (b) Abutted layout with shared guard ring

Figure 3.17: Example of a non-abutted and abutted layout

This layout design methodology allowed for optimized area usage while preserving the modularity of both the layout and the underlying design. In particular, the combination of abuttable cells with shared guard rings between adjacent cells struck a good balance between area efficiency and modularity: because the guard ring of one cell could be shared with its neighbour rather than duplicated, as examplified in Figure 3.17, the continuity of the devices along the abutment edge meant that no unused space was left between adjacent standard cells. This is in contrast to a non-abuttable, more loosely floor-planned approach, where each cell would typically require its own isolated guard ring and margin, d_{min}, leading to wasted area between instances. The result is a layout in which the driver can be scaled or reconfigured, in terms of the number of tap units, current mirror segments, or termination resistors instantiated, by tiling additional copies of the same standard cell, with minimal area penalty and without compromising the electrical isolation and robustness provided by the guard ring structure.

3.5.1 Tap Unit Cell Layout

The final tap unit is composed of two device flavors, RVT and SLVT, as well as two types, NMOS and PMOS. The use of both types brings up the need for two guard rings, one connected to VDD for the PMOS devices and another connected to VSS for the NMOS devices, while the use of two different flavors means that, even with both flavors being of the same type, the DRM does not allow for substrate sharing within the same guard ring. The need for isolating these different flavors, and the fact that they have different lenghts, make it impossible to create a single floorplan that matches all different devices together, therefore making the split into three different floorplans, one for each device type and flavor, the logical design approach. The complete tap-unit schematic, from which these device groups are partitioned, is presented in Figure 3.18.

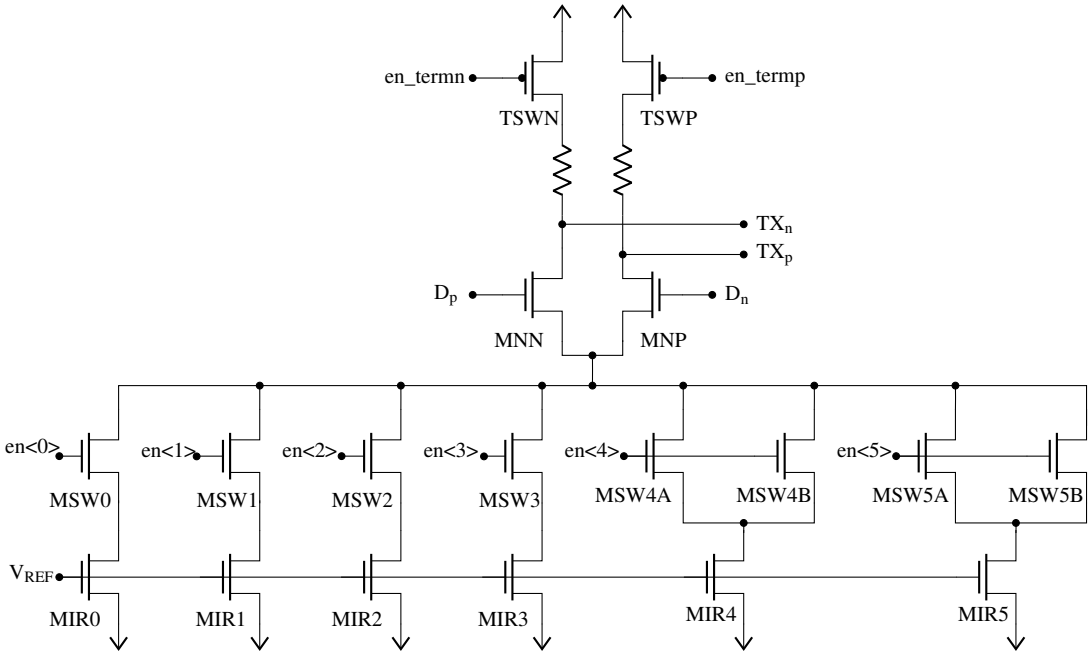


Figure 3.18: Final tap unit schematic

3.5.1.1 Standard Cell Floorplan Design

The floorplan of each basic tap unit was designed starting from the differential pair, since it constitutes the most critical device group within the driver: together with the differential input and output nodes, it forms the most timing- and matching-critical signal path of the entire circuit. Given this criticality, the differential pair devices were arranged in a common-centroid configuration rather than a simpler interdigitated arrangement, since common-centroid placement accounts for two-dimensional process gradient effects across the layout, providing better compensation for process variation than interdigitation alone, as discussed in Section 2.7.1.

For a common-centroid arrangement, both steering devices, MNP and MNN, need to be split into two halves each, and positioned around a central symmetry point, accounting for horizontal

Table 3.5: Tap unit device parameters

ID	Type	W	L	nf	m	Flavor	Description
TSWP	PMOS	W_{min}	L_{min}	16	5	RVT	Termination Switch
TSWN	PMOS	W_{min}	L_{min}	16	5	RVT	Termination Switch
MNP	NMOS	W_{min}	L_{min}	10	4	RVT	Differential Pair
MNN	NMOS	W_{min}	L_{min}	10	4	RVT	Differential Pair
MSW0	NMOS	W_{min}	L_{min}	6	4	RVT	Mirror 0 Switch
MSW1	NMOS	W_{min}	L_{min}	6	4	RVT	Mirror 1 Switch
MSW2	NMOS	W_{min}	L_{min}	6	4	RVT	Mirror 2 Switch
MSW3	NMOS	W_{min}	L_{min}	6	4	RVT	Mirror 3 Switch
MSW4A	NMOS	W_{min}	L_{min}	2	4	RVT	Mirror 4 Switch
MSW4B	NMOS	W_{min}	L_{min}	4	4	RVT	Mirror 4 Switch
MSW5A	NMOS	W_{min}	L_{min}	2	4	RVT	Mirror 5 Switch
MSW5B	NMOS	W_{min}	L_{min}	4	4	RVT	Mirror 5 Switch
MIRR0	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 0
MIRR1	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 1
MIRR2	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 2
MIRR3	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 3
MIRR4	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 4
MIRR5	NMOS	W_{min}	$3 \times L_{min}$	2	30	SLVT	Mirror 5

and vertical axis. A final fingers count of 40 for both steering devices made it logical to split into two halves of 20 each. Taking in consideration the geometry of the layout, and that the differential pair floorplan will dictate the area headroom for all other device groups, the decision was made to split each 20 finger half into two quarters of 10 fingers, leading to a multiplicity of 4 for both MNP and MNN.

With the differential pair matching defined, the remaining RVT NMOS devices of same length, namely the current mirror switches, were added to the floorplan. These devices act as switches, operating in cutoff or triode, making their biasing point less critical. This operating condition makes these devices considerably less sensitive to matching, and therefore can be arranged by only taking the layout geometry into account.

Another benefit that these switches provide is that they can also be used to shield the differential pair from effects such as WPE and LOD, instead of having to use purely dummy devices. Using the switches as dummies drastically decreased the area of this floorplan, making it needed to only add pure, deactivated dummy devices on the far side edges and top of floorplan boundary. The switches were positioned at the bottom so it does not interfere with the differential routing planned to be postioned at the very center of the floorplan, since the current mirror devices were to be positioned at the bottom end of the standard cell floorplan.

As mentioned above, the addition of the switches to the floorplan was done with a focus on layout geometry and area. To this end, measuring the floorplan width with finger counts instead of physical dimensions makes it easier to find the most optimal arrangement that results in a balanced tradeoff between area and geometry. The following considerations were taken into account when

<table>
<tr><td colspan="8">DUMMY</td></tr>
<tr><td>D</td><td>SW4B</td><td colspan="2">MNP</td><td colspan="2">MNN</td><td>SW5B</td><td>D</td></tr>
<tr><td>D</td><td>SW4B</td><td colspan="2">MNP</td><td colspan="2">MNN</td><td>SW5B</td><td>D</td></tr>
<tr><td>D</td><td>SW4B</td><td colspan="2">MNN</td><td colspan="2">MNP</td><td>SW5B</td><td>D</td></tr>
<tr><td>D</td><td>SW4B</td><td colspan="2">MNN</td><td colspan="2">MNP</td><td>SW5B</td><td>D</td></tr>
<tr><td>D</td><td>SW4A</td><td>SW3</td><td>SW0</td><td>SW1</td><td>SW2</td><td>SW5A</td><td>D</td></tr>
<tr><td>D</td><td>SW4A</td><td>SW3</td><td>SW0</td><td>SW1</td><td>SW2</td><td>SW5A</td><td>D</td></tr>
<tr><td>D</td><td>SW4A</td><td>SW3</td><td>SW0</td><td>SW1</td><td>SW2</td><td>SW5A</td><td>D</td></tr>
<tr><td>D</td><td>SW4A</td><td>SW3</td><td>SW0</td><td>SW1</td><td>SW2</td><td>SW5A</td><td>D</td></tr>
</table>

Figure 3.19: RVT devices floorplan

deciding how to position the current mirror switches:

1. The even number of switching devices makes it possible to split them into equal groups and mirror them across the vertical symmetry axis of the differential pair, resulting in a symmetrical arrangement of all active devices.

2. There is still a need for dummy devices to be added to the side edges of the floorplan to keep the standard cell abuttable, examplified in Figure 3.20. These dummy devices will be used to connect adjacent cells to VSS, and isolate the terminal of the active switching devices, which is only possible if these dummy devices have at least two fingers.

3. Each switch has 24 fingers, which makes it reasonable to split in even multiples of 6 or 4

4. The considerably longer width of the current mirror devices has to be accounted for when deciding how the switches will be split, since the driver is floorplanned to have the tap units horizontally abuttable. If the differential pair floorplan is to narrow, this will be amplified by the current mirror bank and result in a narrow standard cell, making local cell routing and matching of the mirror devices more difficult.

The final floorplan of the RVT devices, shown in Figure 3.19, has a width of 32 fingers. Devices SW0, SW1, SW2 and SW3 have 6 fingers each, with multiplicity of 4. In order to occupy the lowest area possible, devices SW4 and SW5 are split into two sub devices, SW4A and SW5A, with 2 fingers and multiplicity of 4, and SW4B and SW5B, with 4 fingers and multiplicity of 4. This arrangement allowed for the minimum width of 32 fingers to be realized, which is just enough width for the current mirror SLVT floorplan to be designed without leaving too much unused area.

The current-mirror devices were likewise arranged in a common-centroid configuration. Unlike the differential pair, however, this choice was not driven primarily by the need to compensate for process gradients, but rather by the geometric constraints imposed by the width of the RVT floorplan. Within the available cell width, a conventional interdigitated arrangement would have resulted in an impractical aspect ratio and insufficient routing space. The common-centroid arrangement therefore provided a compact solution while preserving symmetry and adequate matching between the current-mirror devices.

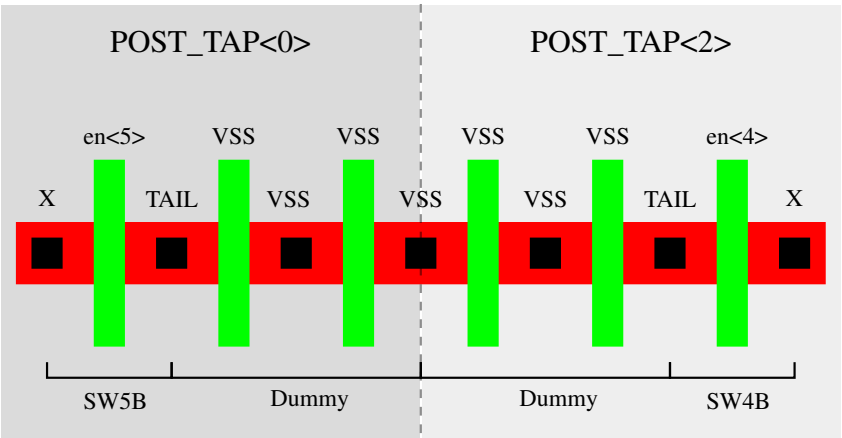


Figure 3.20: Example of adjacent cell abutting using dummies

Each logical current-mirror device is implemented using 30 identical two-finger sub-devices, corresponding to the $n_f = 2$ and $m = 30$ configuration listed in Table 3.5. These sub-devices are distributed symmetrically throughout the SLVT floorplan, shown in Figure 3.21. Dummy devices were inserted between groups of active mirror devices to provide additional routing space for the connections between the enable switches and their corresponding current mirrors. Additional dummy devices were placed along the outer edges of the floorplan to maintain a uniform device environment, reduce edge-dependent effects, and preserve symmetry. Since all active current-mirror devices must remain in saturation and are biased by the common replica-based reference generator, maintaining similar physical surroundings and operating conditions among the mirror devices is important for limiting systematic current mismatch.

With the RVT and SLVT device floorplans defined, the PMOS termination-switch devices were then arranged. These devices perform a switching function analogous to that of the RVT NMOS current-mirror switches, but are used to enable or disable the 1 kΩ resistor branches that form the configurable on-die termination network. Each PMOS switch is implemented with 16 fingers and a multiplicity of five, resulting in a total effective finger count of 80. Because these devices operate primarily as switches for passive termination elements, their matching requirements are less stringent than those of the differential pair or current-mirror devices. The PMOS floorplan was therefore optimized primarily for area efficiency and compatibility with the standard-cell geometry. Since the PMOS devices use the same channel length as the RVT NMOS switches, their arrangement could be designed to match the width of the RVT floorplan, avoiding unused lateral area and maintaining a uniform cell boundary for abutment.

Each of the two 1 kΩ resistors are physically implemented as two parallel resistances, with two more, one at each side, as dummy devices. This makes the total resistors placed in the floorplan to be eight, located below the PMOS devices, outside the guard ring. The PMOS floorplan is shown in Figure 3.22.

This part of the tap unit floorplan is located at the top of the standard cell, right above the differential pair and the differential output nets. This positioning was chosen to minimize the

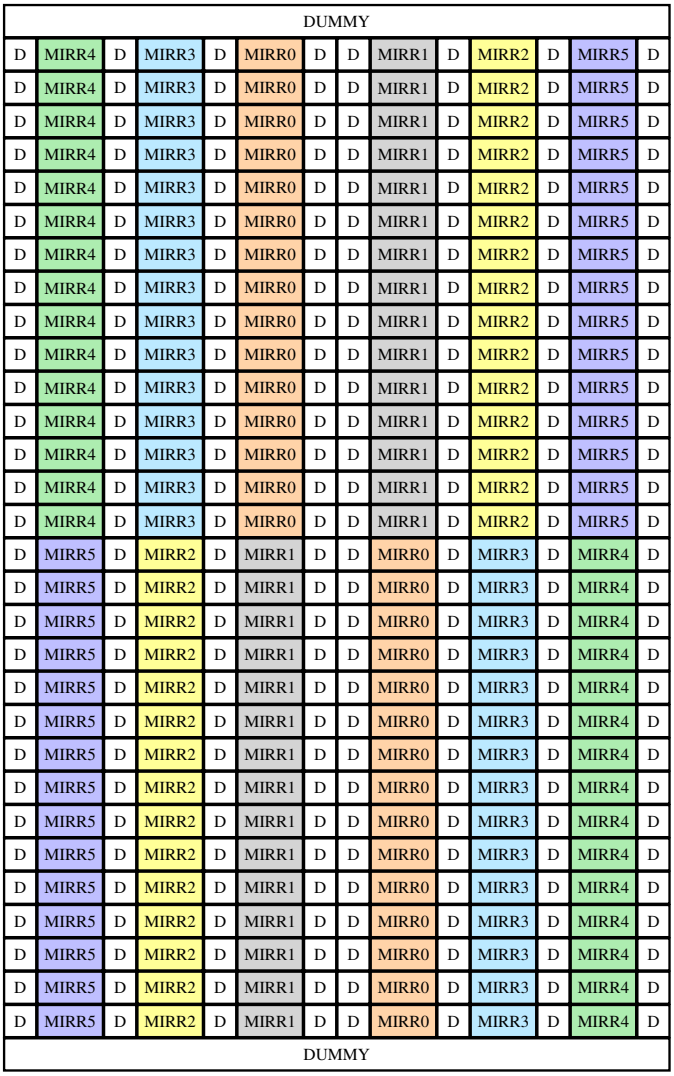


Figure 3.21: SLVT devices floorplan

length of the routing between the termination devices and the differential output nets, which carries the highest current density among the output nets, making IR Drop and EM concerns more relevant for these nets. Figure 3.23 shows a high-level representation of the tap unit floorplan, highlighting the relative positioning of the differential pair, current mirrors, termination and switching devices.

3.5.2 Standard Cell Routing

Following the floorplanning of the differential pair and current mirror bank, the routing of each basic tap unit standard cell was carried out with particular attention to the differing sensitivity of each signal type to parasitic effects, EM, and crosstalk.

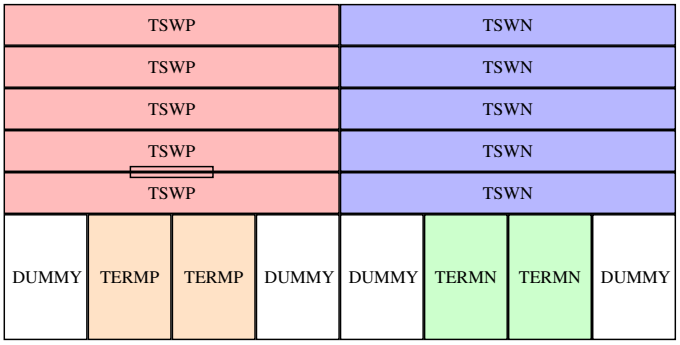


Figure 3.22: PMOS devices and resistors floorplan

3.5.2.1 Enable Signal Routing

The nets carrying the enable signals for the current mirror and termination enabling devices are routed using polysilicon and the M1 metal layer, at the minimum width allowed by the technology. Since these signals are static during normal operation, and given the relatively short length of the associated routing paths, IR drop is not a significant concern for these nets; similarly, the low current density expected on these paths does not raise EM concerns. Routing the enable nets on the lowest available layers frees up the higher metal layers for the differential pair and current mirror routing, which are considerably more sensitive to parasitic resistance and capacitance, and are subject to stricter EM and IR drop requirements, as discussed further below.

Within the standard cell floorplan, the enable nets are routed predominantly along the top and bottom edges of the cell, while the differential input and output nets are routed through the middle of the cell. This placement was chosen with crosstalk in mind, in order to avoid coupling from the switching differential pair inadvertently toggling an enable signal to an incorrect level, and to minimize the parasitic loading introduced on the differential signal path by the fixed voltage levels of the nearby enable nets, which have their signal at either VDD or VSS, posing the highest voltage potential possible, during operation.

The enable pins themselves are placed along the bottom edge of the cell, with the exception of the termination enable pins, which partially overlap with the differential input and output nets. Since the differential input and output nets are routed mainly on the Metal 4 (M4) layer, the termination enable nets connecting to the bottom of the cell use the Metal 5 (M5) layer instead. This layer assignment was the result of routing congestion on layers M2 through M4 in the middle region of the cell. Among the remaining available layers, M5 resulted in the smallest and fewest overlapping regions with the other nets in the cell.

The current mirror bank enable nets are routed using M1 within the current mirror bank floorplan area itself, and Metal 3 (M3) for the connections to the input ports located at the bottom of the cell. M3 was preferred over M2 for this purpose due to the heavy use of M1 already present within the current mirror bank floorplan area, resulting in less overlap with other nets when M3 is used instead. The only net that overlaps with the mirror bank enable nets is the reference voltage net, V_{ref}, which does so at eight points across the layout area. This overlap is not considered a

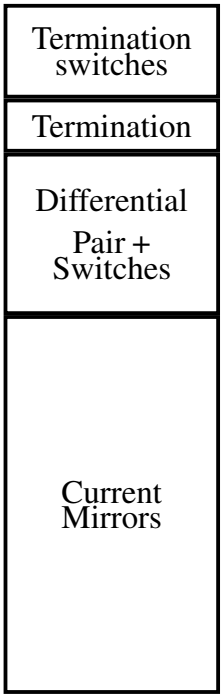


Figure 3.23: High-level representation of the tap unit floorplan

significant concern, since the V_{ref} signal is quasi-static, and combined with the static nature of the enable signals themselves, the resulting parasitic coupling capacitance is not expected to cause significant performance degradation, as both the V_{ref} and enable signal levels are established at system start-up, well before the beginning of data transmission, providing sufficient time for these parasitic capacitances to fully charge.

3.5.2.2 Reference Voltage Routing

The V_{ref} net is routed primarily using the M4 layer, since it must connect to adjacent tap unit cells as well as to the current mirror reference biasing circuit described in Section 3.4.2.1. Electromigration is not a significant concern for this net, since its sole function is to set the gate voltage of the current mirror devices, resulting in a low expected current density. Nonetheless, these nets were sized with a width 67% larger than the minimum width allowed for the M4 layer, in order to account for the resistance of the routing path and its associated IR drop, and to keep the biasing voltage seen by the current mirror devices as close as possible to the reference value, minimizing the resulting error between the reference current and its mirrored copies.

3.5.2.3 Differential Signal Routing

The differential signal nets were the most critical nets in the standard cell, and consequently received the greatest attention during the planning and execution of the routing.

The first priority in routing these nets was to ensure that all differential nets of a given type, that is, the input pair or the output pair, were routed as symmetrically as possible with respect

to one another, in order to avoid introducing timing skew between the two differential branches, consistent with the matching and symmetry principles discussed in Section 2.7.2.

The second priority was to reduce parasitics on these nets while respecting the electromigration constraints of the metal layers used. To this end, the differential pair devices are implemented as four parallel sub-devices, each composed of ten fingers. As a result, the input signal nets, D_p and D_n, are each split into ten parallel routing paths per sub-device, for a total of forty parallel paths connecting to the gate of each differential pair device.

The output nets, TX_p and TX_n, are similarly split into five parallel routing paths per sub-device, connecting to the drain terminals of the current-steering devices, for a total of twenty parallel paths per output node. This parallelization means that the total current sunk by each tap is distributed across twenty parallel paths, which reduces the current density on each individual path and allows the routing metal width to be reduced accordingly, while still respecting the electromigration constraints imposed by the technology.

The differential nets are routed using the highest metal layers available for internal chip-level connections, M4 and M5, in order to reduce parasitic coupling and crosstalk, avoid overlap with other nets in the cell, and avoid the routing congestion present on the lower metal layers. Although the differential routing was among the first aspects of the cell routing to be designed, its layout was defined with the routing requirements of the remaining, lower-priority nets on the lower metal layers already taken into consideration, in order to avoid conflicts later in the routing process.

3.5.2.4 Electromigration Considerations

From an EM standpoint, the nets connecting the differential pair to the termination bank and to the current mirror bank were identified as the most critical in the design.

For the termination nets, the worst-case voltage drop considered across the termination network was 0.6 V, corresponding to the maximum single-ended output swing. Although the ideal resistor-only model of Section 3.4.4 would require approximately 20 parallel 1 kΩ elements to obtain a 50 Ω single-ended termination, this ideal relationship does not accurately represent the current distribution in the implemented termination bank. The effective impedance of an enabled termination branch is also affected by the switching device, whose simulated on-state resistance modifies the number of branches required to reach the target impedance.

For EM sizing, a value of approximately 10 simultaneously active termination paths was adopted as a conservative approximation of the implemented bank behaviour observed during circuit simulation. To avoid underestimating the current stress on the local termination routing, the total current associated with the maximum single-ended swing was conservatively assumed to be distributed among these 10 paths:

$$I_{\text{path,EM}} \approx \frac{12\ \text{mA}}{10} = 1.2\ \text{mA}. \tag{3.35}$$

The resulting 1.2 mA value should therefore be interpreted as a conservative per-path current used for EM dimensioning, rather than as the exact current predicted by the ideal resistor-only model.

The width of the metal used to route these nets was chosen to keep the resulting current density within the EM margins specified by the technology. Since the transmitted signal has a bit period of 100 ps, the track width was sized using the peak current, I_{peak}, as the relevant EM limit, rather than the RMS current, I_{RMS}. This choice follows directly from the foundry's DRM, which specifies that for any signal with a period shorter than a certain value, which is higher than the bit period of 100 ps, the EM limit for the corresponding metal routing may be evaluated using I_{peak} rather than I_{RMS}, which would otherwise impose a considerably tighter constraint on the required metal width.

The output nets, TX_p and TX_n, correspond to the points in the layout where the currents associated with the transmitter termination and the receiver termination are combined, as discussed in Section 3.3, and therefore exhibit the highest current density in the entire design. Since each tap unit is capable of sinking a maximum current of 1.2 mA, the output metal tracks were sized to accommodate this current, in addition to satisfying the EM limits imposed by the DRM. Because these nets carry the same class of high-speed digital signal as the termination nets described above, the I_{peak}-based EM limit was likewise used to determine the required M4 and M5 widths for the output nets.

Beyond these general current-density and EM considerations, the routing in the immediate vicinity of the differential pair devices was designed specifically to avoid introducing parasitic Miller capacitance between the gate and drain terminals of the differential pair, which would otherwise degrade the performance of the driver.

Chapter 4

Verification and Results

4.1 Verification Methodology

Following the general IC design flow described in Section 2.6, this section outlines the specific tools and verification methodology adopted for the schematic design, layout implementation, and electrical validation of the transmitter driver presented in this dissertation.

Both the schematic design and the physical layout of the driver were carried out using Synopsys Custom Compiler. The design targets the GlobalFoundries 55 nm LPx CMOS technology, with device models, design rules, and electrical constraints obtained from the corresponding foundry-provided PDK and DRM.

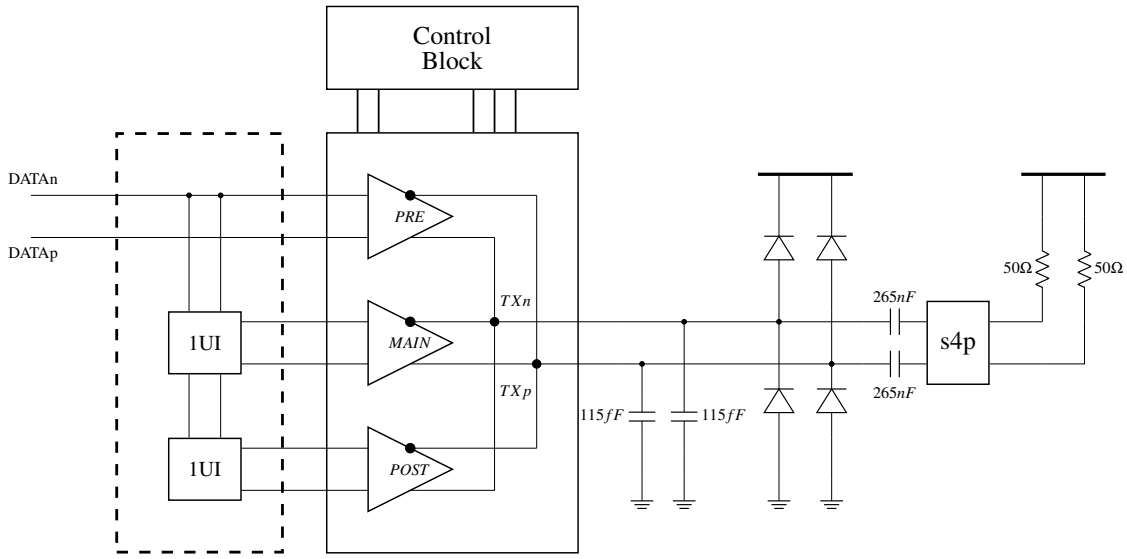


Figure 4.1: Driver testbench schematic

Figure 4.1 shows the schematic of the testbench used to verify the TX driver.

The taps are configured through a dedicated control block, implemented as a VerilogA logic module that generates the corresponding enable signals for each tap according to the desired settings.

The differential input data, DATAp and DATAn, is generated by a bit-pattern generator with 15 ps rise and fall times, with DATAn produced as the complement of DATAp. A logic '0' bit is represented by a voltage level $V_{low} = V_{cm} - \Delta$, and a logic '1' bit by $V_{high} = V_{cm} + \Delta$, where V_{cm} and Δ are defined as in the previous chapter, with $V_{cm} = 0.8$ V and $\Delta = 0.2$ V. The main-cursor and post-cursor tap inputs are obtained by delaying the pre-tap data using delay elements provided by the analoglib library, each configured for a delay of 1 UI, corresponding to 100 ps at the target data rate.

At the driver output, each of the TXp and TXn nodes is loaded with a 115 fF capacitor, and is additionally protected by a double-diode ESD structure provided by the PDK. Both output lines are AC-coupled through 265 nF capacitors, as specified by the USB 3.2 Gen 2 standard.

The transmission channel itself is modeled as a four-port S-parameter block, implemented using the corresponding analoglib component and populated with an s4p file supplied by the USB-IF, allowing the channel losses and reflections present in a real USB 3.2 SuperSpeed link to be reproduced in simulation. At the channel output, two 50 Ω resistors are connected to emulate the termination presented by the RX.

Rise and fall times, output swing and output impedance are all measured with the channel model disconnected, so as to isolate these metrics from channel-induced distortion, while eye-diagram measurements are performed with the channel model connected, in order to capture its and equalization impact on the transmitted signal.

The verification methodology followed a two-stage evaluation process, consistent with the schematic-to-layout design flow described in Section 2.6. In the first stage, pre-extraction results, obtained directly from schematic-level simulation without the inclusion of layout-induced parasitics, were evaluated against the normative USB 3.2 specifications summarized in Section 3.1, across the relevant PVT corners, Typical-Typical (TT), Slow-Slow (SS) and Fast-Fast (FF), as well as across resistor corners accounting for variation in the poly and diffusion resistor models used within the design. This stage served to validate the functional correctness and nominal performance of the design prior to layout.

In the second stage, following the completion of the physical layout and parasitic extraction, the post-extraction results were compared with the corresponding pre-extraction results in order to quantify the performance changes introduced by layout parasitics. The extracted implementation was also evaluated against the same relevant USB 3.2 numerical electrical limits and PVT and resistor-corner sweeps used in the pre-extraction stage. This allowed the implemented driver to be assessed against the selected standard requirements once layout-induced effects were taken into account. Any target not met at this stage was addressed through iterative refinement of the layout and, where necessary, the schematic, consistent with the design flow described in Section 2.6.

4.2 Driver Performance

The normative output swing specification is measured with the system operating without equalization, i.e., with the driver configured in the EQ0 preset. For the tap enable mapping defined in

Section 3.3.1.1, the 120 enabled current mirrors for the main tap were not enough to keep to swing within the normative range for all three PVT corners, with an output swing of 0.677 V for the SS corner, as shown in Figure 4.2.

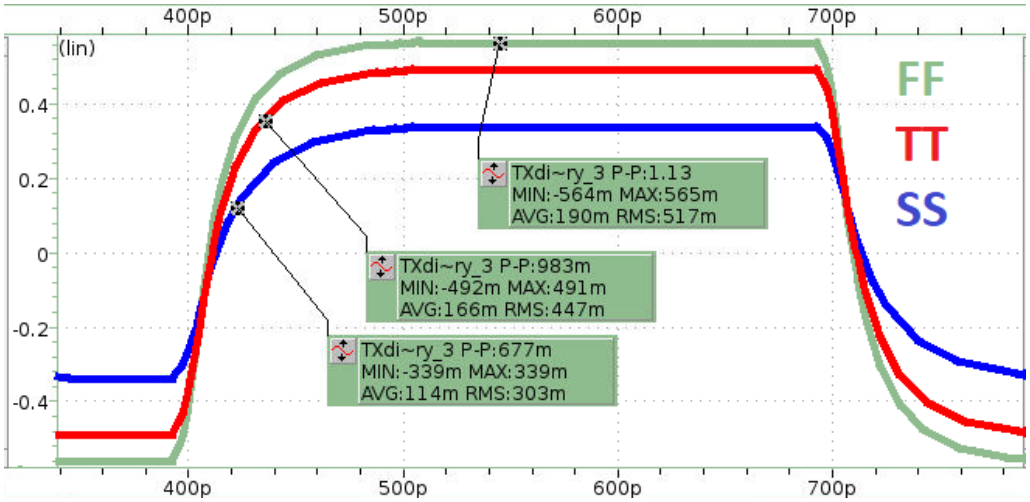


Figure 4.2: Starting EQ0 output differential voltage for all three PVT corners

This raised the need to add more units to the main tap in order to achieve the required swing range, which was done by adding four more units to the main tap, resulting in a total of 144 enabled current mirrors. The output swing for all three PVT corners with the modified main tap is shown in Figure 4.3, where it can be seen that the output swing is now within the normative range for SS and TT, and slightly above the normative range for FF, at 1.3 V, which is acceptable since it can be lowered by deactivating some of the current mirrors.

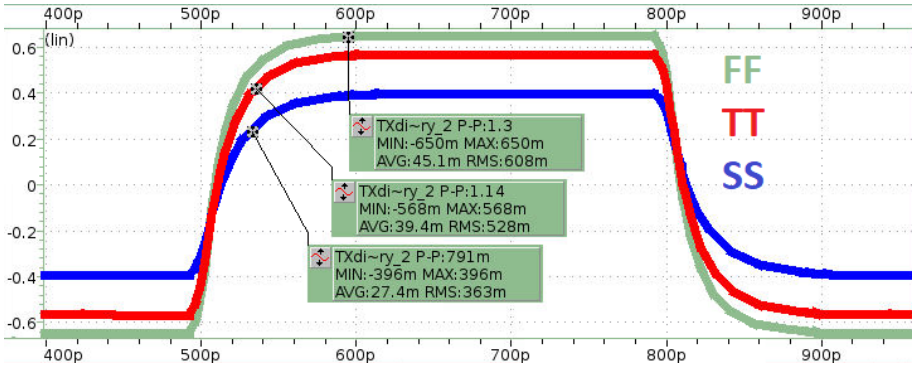


Figure 4.3: Modified EQ0 output differential voltage for all three PVT corners

This difference between the calculated and simulated output swing values can be attributed to the fact that the calculations in Section 3.3.1.1 were based on idealized assumptions, such as perfect current mirror matching, which do not hold in practice, and can change the effective equalization performed by the driver. Table 4.2 summarizes the PS and DE values obtained for each equalization preset across the three simulated PVT corners, and shows the difference between the simulated and target values.

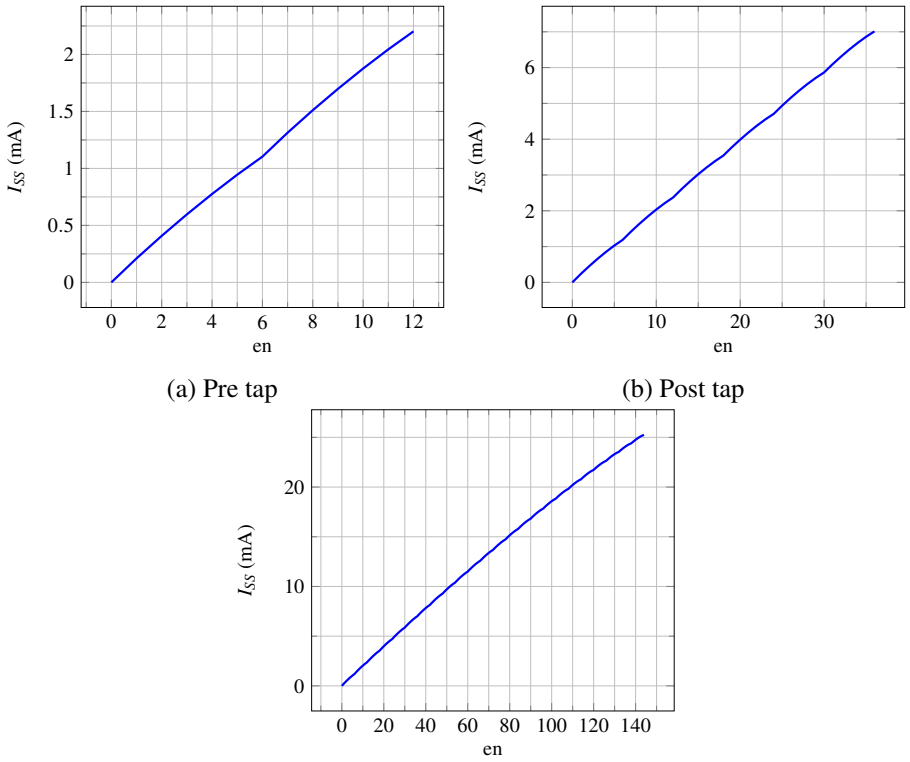
Table 4.1: Equalization results across PVT corners, expressed as % deviation from target

Preset	**TT**		**FF**		**SS**	
	PS	**DE**	**PS**	**DE**	**PS**	**DE**
EQ1	+8.33%	-10%	+4.17%	-9.52%	+20%	-12.38%
EQ2	-4.09%	-7.74%	-5.91%	-9.68%	+2.27%	-6.77%
EQ3	-5.94%	-5.29%	-6.56%	-6.29%	-0.62%	-3%
EQ4	+0.4%	-7.4%	-1.6%	-8.2%	+6.4%	-4.6%

Table 4.2: Equalization results for PVT corners

Preset	**Process**	**VDD**	**Temp**	**PS**	**DE**	**Target PS**	**Target DE**
EQ1	TT	1.2	25	1.30	1.89	1.2	2.1
EQ1	FF	1.26	-40	1.25	1.90	1.2	2.1
EQ1	SS	1.14	125	1.44	1.84	1.2	2.1
EQ2	TT	1.2	25	2.11	2.86	2.2	3.1
EQ2	FF	1.26	-40	2.07	2.80	2.2	3.1
EQ2	SS	1.14	125	2.25	2.89	2.2	3.1
EQ3	TT	1.2	25	3.01	6.63	3.2	7
EQ3	FF	1.26	-40	2.99	6.56	3.2	7
EQ3	SS	1.14	125	3.18	6.79	3.2	7
EQ4	TT	1.2	25	2.51	4.63	2.5	5
EQ4	FF	1.26	-40	2.46	4.59	2.5	5
EQ4	SS	1.14	125	2.66	4.77	2.5	5

Figure 4.4 shows the sweep of each tap I_{SS} for their enable signals, where it can be seen that the current sunk suffer from slight variations with the increase of enabled current mirrors. These current variations adds up and results in the differences between the calculated and simulated PS and DE values shown in Table 4.2. The table shoes that EQ1 does not meet the normative equalization range. This can be improved by adjusting the number of enabled current mirrors, as shown further in Section 4.3.



(a) Pre tap

(b) Post tap

(c) Main tap

Figure 4.4: Sweep of each tap I_{SS} for their enable signals

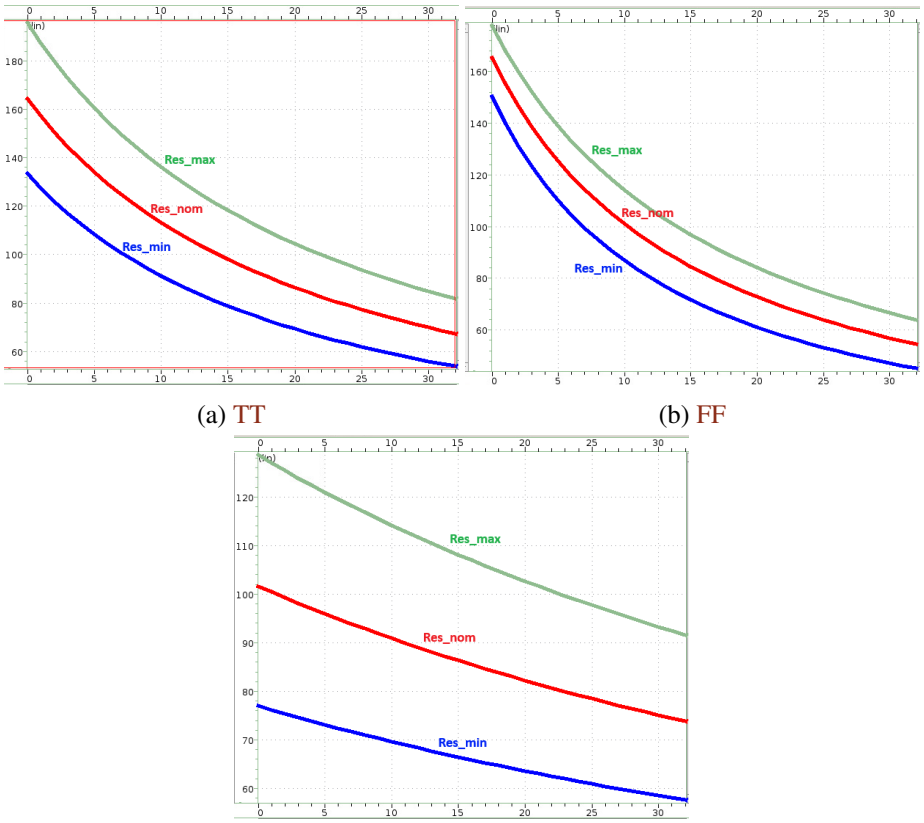
4.2.1 Output Impedance Across PVT and Resistor Corners

In addition to the standard PVT corners considered in the verification methodology described in Section 4.1, the use of technology-provided polysilicon resistances for the termination network, discussed in Section 3.2, introduces an additional set of corners associated with the resistor process variation itself: a minimum resistance corner (Res_min), a nominal resistance corner (Res_nom), and a maximum resistance corner (Res_max). Since these resistor corners are, in principle, independent of the standard PVT corners, they must be analyzed jointly with the PVT corners rather than in isolation, in order to correctly capture the full range of expected output impedance variation.

The differential output impedance of the driver is particularly sensitive to these resistor corners, since the termination network is composed entirely of passive resistors, as described in Section 3.2, and its resistance therefore varies directly with the underlying resistor process corner, independently of the transistor-level PVT corner.

Figure 4.5 shows the resulting differential output impedance across the combined PVT and resistor corner space, as a function of the number of termination bank resistors activated, ranging from 0 to the full 32 resistors available. The results for each PVT process corner are shown in Figure 4.5.

Across all evaluated PVT corners, the differential output impedance remains within the normative range of 72 Ω to 120 Ω specified by the USB 3.2 Gen 2 standard, discussed in Section 3.1, confirming that the termination bank provides sufficient tuning range and resolution to maintain compliant impedance matching under the combined effect of PVT and resistor process variation.



(a) TT (b) FF

(c) SS

Figure 4.5: Differential output impedance across corners and bank configurations

Table 4.3 summarizes the number of termination bank resistors that must be enabled in order to achieve the maximum and minimum output impedance values across the combined PVT and resistor corner space, as well as the corresponding number of resistors required to achieve the nominal differential impedance of 100 Ω.

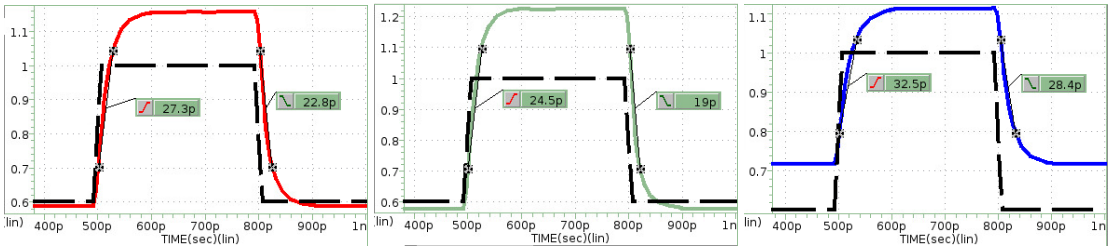
Table 4.3: $Z_{out,diff}$ across PVT and resistor corners

PVT	Res	$\#en_{max}$	$\#en_{nom}$	$\#en_{min}$
TT	Res_min	3	8	18
TT	Res_nom	9	15	28
TT	Res_max	15	22	NA
SS	Res_min	NA	NA	6
SS	Res_nom	NA	1	NA
SS	Res_max	6	23	NA
FF	Res_min	4	7	15
FF	Res_nom	6	10	20
FF	Res_max	9	14	26

4.2.2 Signal Integrity

Rise and fall time performance of the transmitter driver, discussed in the context of slew rate in Section 2.2.2.5, was characterized using the modified EQ0 preset, corresponding to the full main-tap configuration with all 144 main-tap current mirrors enabled and no pre-cursor or post-cursor equalization applied, driving a load capacitance of 115 fF.

The pre-extraction rise and fall waveforms across the three PVT corners are presented in Figure 4.6.



(a) TT (b) FF (c) SS

Figure 4.6: Rise and fall times across PVT corners

Table 4.4: Rise and fall times across PVT corners

Corner	Rise	Fall
TT	27.3 ps	22.8 ps
FF	24.5 ps	19 ps
SS	32.5 ps	28.4 ps

Across all three corners, a consistent asymmetry is observed between the rise and fall times, with the fall time being shorter than the rise time in every case. This asymmetry is a direct consequence of the CML topology adopted for the driver, discussed in Section 2.4.1, which implements an actively switched Pull-Down Network (PDN) together with a passive Pull-Up Network (PUN).

Despite this asymmetry, the rise and fall times remain well within the nominal UI of 100 ps across all evaluated PVT corners, indicating that the driver retains sufficient timing margin for correct operation at the target 10 Gb/s data rate.

The channel model used to evaluate the transmitted and received eye diagrams is provided by the USB-IF, and its insertion loss profile, S_{21}, is shown in Figure 4.7. The channel is supplied as a Touchstone file, named `Mock_Host_Cascaded_Model_TypeC_rspl.s4p`, available at [30]. This channel introduces an insertion loss of 21.6 dB at 5 GHz, corresponding to the Nyquist frequency of the 10 Gb/s data rate. The input data source is a PRBS Linear Feedback Shift Register (LFSR) mode PN32, that generates the pseudo random bit sequence used to generate the transitions. A transient analysis was done for a runtime of 10000 × UI, with a step of 10 ps.

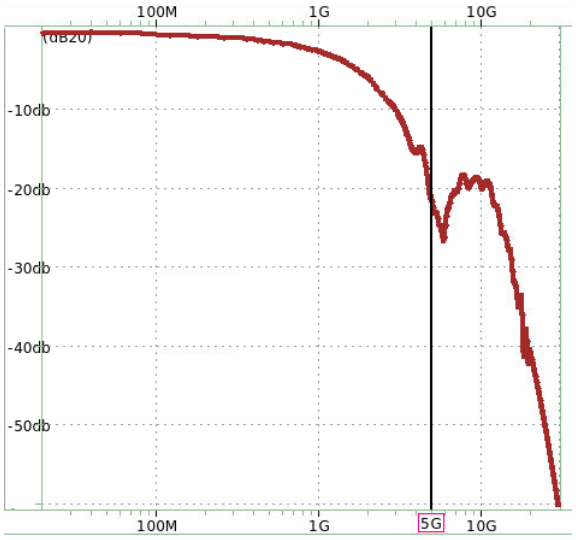


Figure 4.7: Channel model S_{21}

The following eye diagrams were obtained at the TT and Res_nom corner, for each of the five equalization presets defined in Section 3.3. For each preset, both the TX eye, measured directly at the driver output, and the RX eye, measured after propagation through the reference channel, are evaluated. The corresponding metrics are summarized in Table 4.5.

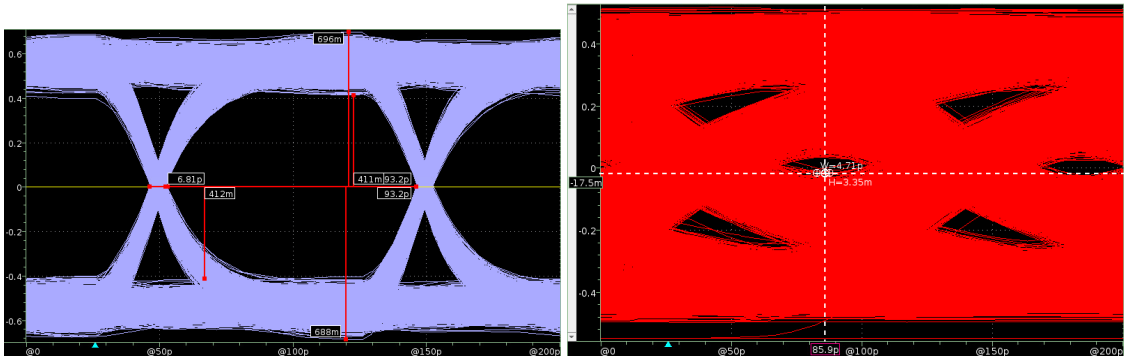
The improvement of the received signal when transmitter equalization is applied can be clearly observed in the eye diagrams. The RX eye opening increases substantially from EQ0 through EQ3, demonstrating the effectiveness of the transmitter FFE in compensating for the reference-channel loss. The trend is not strictly monotonic for the strongest preset, however, since EQ4 produces a slightly smaller received eye than EQ3, indicating that EQ3 is the best preset for this particular channel model. The corresponding eye-diagram metrics are summarized in Table 4.5.

The results demonstrate a substantial improvement in received eye opening as transmitter equalization is increased from EQ0 toward EQ3. The trend is not strictly monotonic through EQ4, however, since the largest pre-extraction RX eye is obtained with EQ3.

For EQ0, the TX eye has a height of 823 mV and a width of 93.2 ps, whereas the RX eye is almost completely closed, with only 3.35 mV of height and 4.71 ps of width. This confirms that

the loss of the selected reference channel cannot be adequately compensated without transmitter equalization.

The corresponding unequalized TX and RX eye diagrams are presented in Figure 4.8.

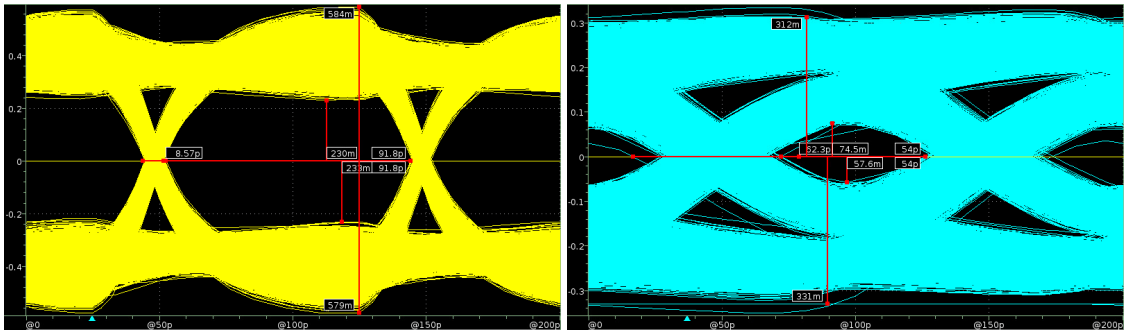


(a) TX (b) RX

Figure 4.8: Eye diagrams for EQ0 at corner TT

For EQ1, the RX eye increases to 132.1 mV in height and 54 ps in width. EQ2 further increases the opening to 162 mV and 60.2 ps.

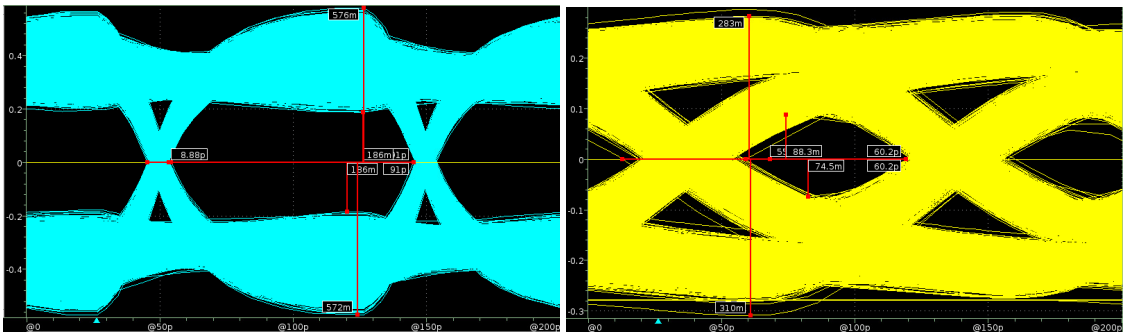
Figure 4.9 shows the TX and RX eye diagrams for the EQ1 preset.



(a) TX (b) RX

Figure 4.9: Eye diagrams for EQ1 at corner TT

Figure 4.10 shows the TX and RX eye diagrams for the EQ2 preset.

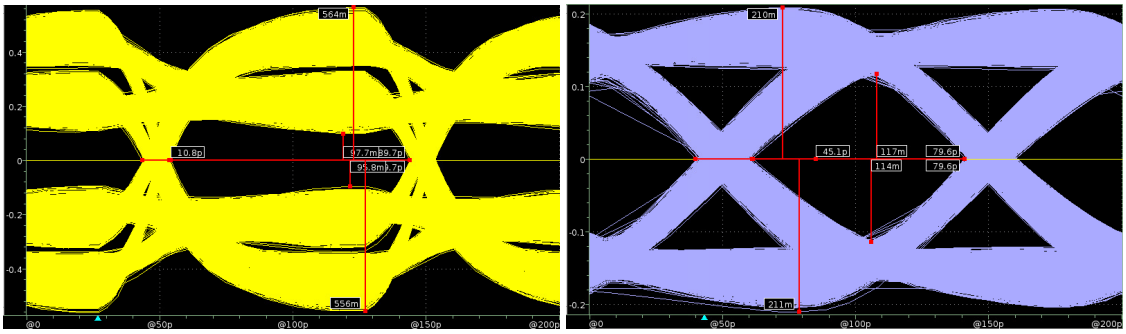


(a) TX (b) RX

Figure 4.10: Eye diagrams for EQ2 at corner TT

The largest pre-extraction eye is obtained with EQ3, which produces an RX eye height of 231 mV and a width of 79.6 ps. EQ4 also provides a widely open received eye, but its height of 201.9 mV and width of 73.1 ps are both slightly smaller than those obtained with EQ3. EQ3 is therefore selected as the representative equalization preset for the subsequent pre-extraction corner analysis.

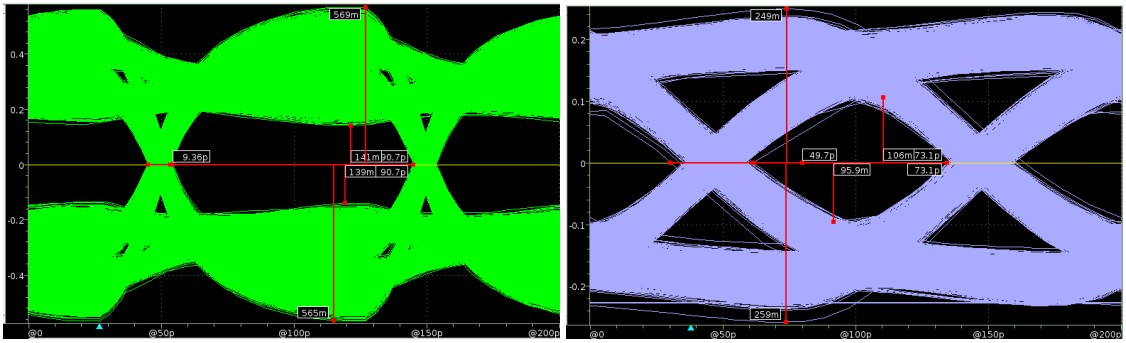
Figure 4.11 shows the TX and RX eye diagrams for EQ3, the preset yielding the largest pre-extraction received eye.



(a) TX (b) RX

Figure 4.11: Eye diagrams for EQ3 at corner TT

Figure 4.12 shows the TX and RX eye diagrams for the stronger EQ4 preset.



(a) TX (b) RX

Figure 4.12: Eye diagrams for EQ4 at corner TT

It should be noted that these preset coefficient values were used primarily as a general target during the design of the equalizer, rather than as an exact target to be matched precisely by the fabricated tap weights.

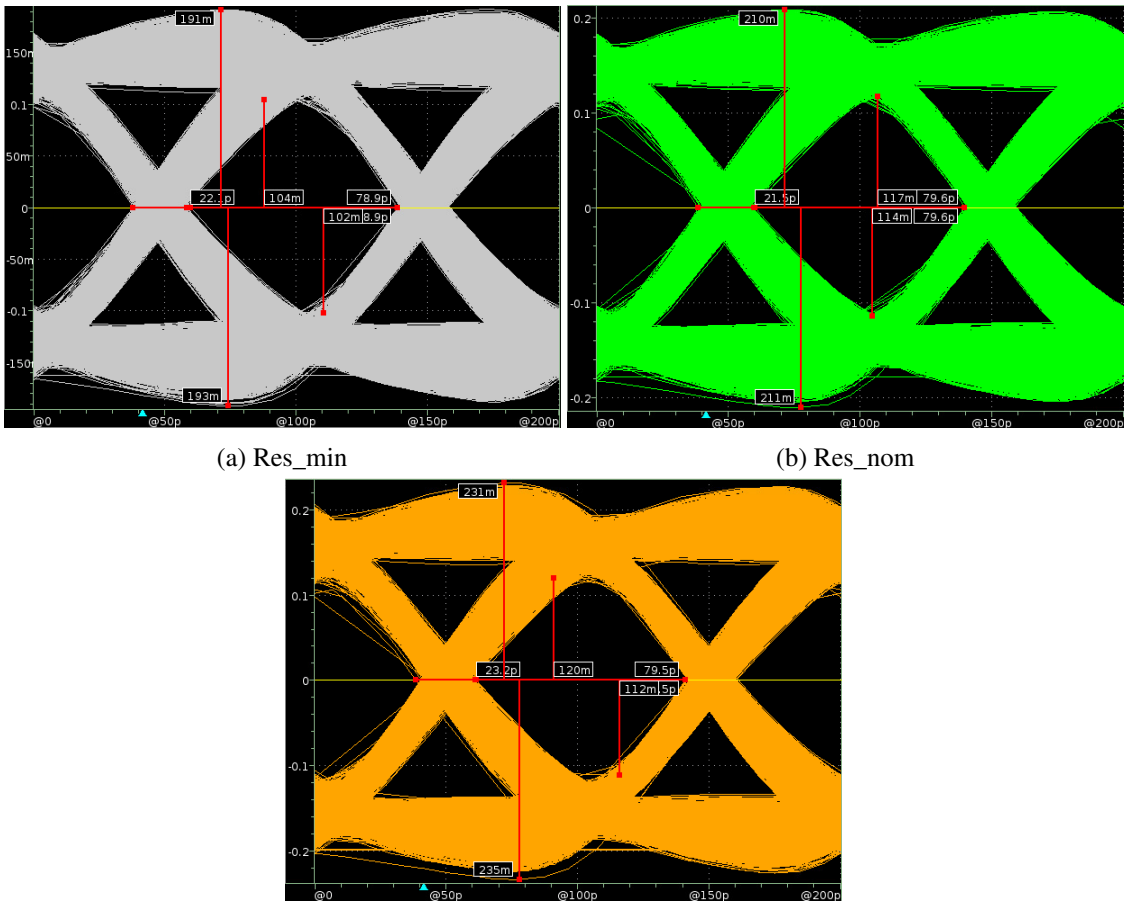
Table 4.5: TX and RX eye diagram metrics at TT for different equalization presets

EQ Preset	**TX Height**	**TX Width**	**RX Height**	**RX Width**
EQ0	823 mV	93.2 ps	3.35 mV	4.71 ps
EQ1	463 mV	91.8 ps	132.1 mV	54 ps
EQ2	372 mV	91 ps	162 mV	60.2 ps
EQ3	193.5 mV	89.7 ps	231 mV	79.6 ps
EQ4	280 mV	90.7 ps	201.9 mV	73.1 ps

The results presented above show that not all of the defined equalization presets are equally well suited to the specific channel used in this evaluation, with a considerable performance difference between EQ1 and EQ2 from EQ3 and EQ4. This outcome shows that, from the five defined equalization presets, EQ3 is the best suited to the specific channel used in this evaluation, producing the largest received eye opening and therefore the best SI performance. For different channel loss profiles, however, the best equalization preset may differ from EQ3.

To evaluate the impact of PVT and resistor variation on the SI of the transmitted signal, EQ3 was selected as the representative equalization preset, since it produces the largest RX eye among the five pre-extraction configurations evaluated at TT. The resulting RX eye height and width for the TT, FF, and SS process corners are reported in Tables 4.6, 4.7, and 4.8, respectively.

The EQ3 received eye diagrams at TT for the three resistor corners are compared in Figure 4.13.



(a) Res_min

(b) Res_nom

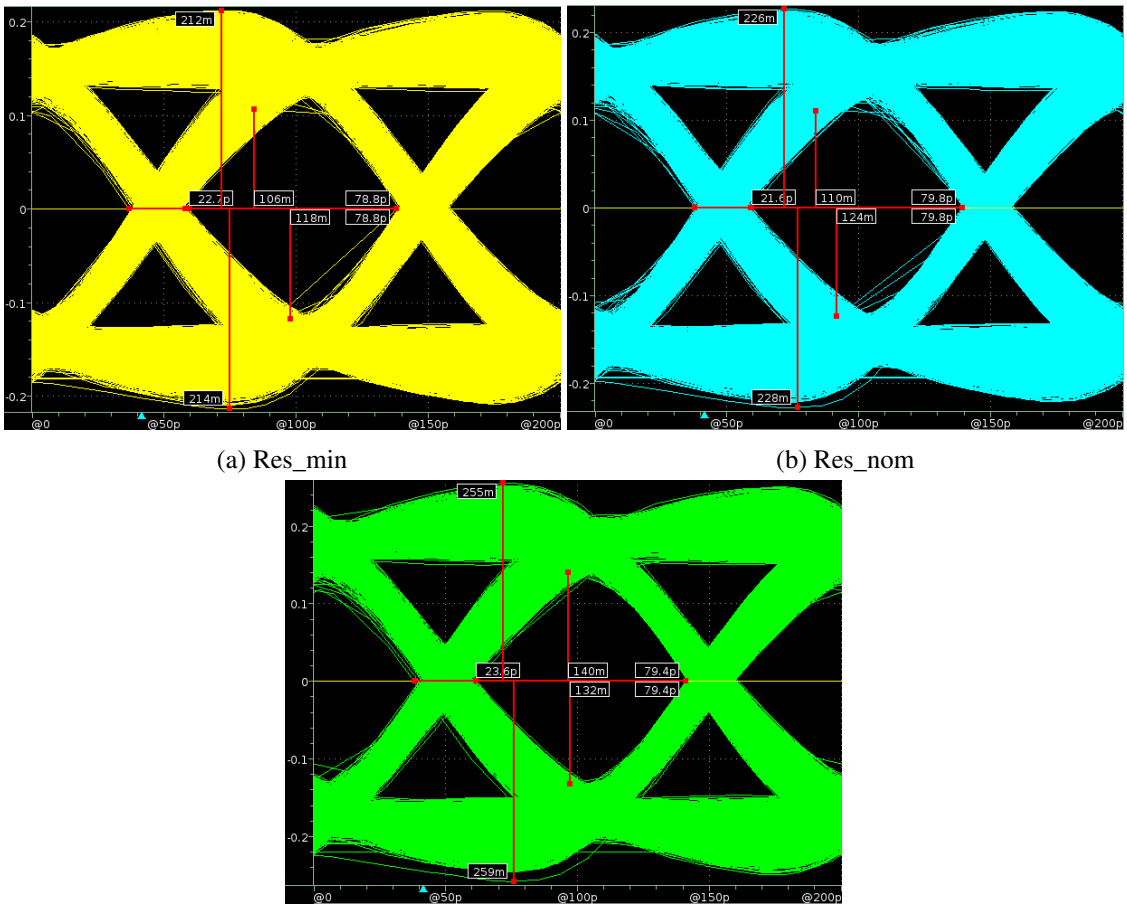
(c) Res_max

Figure 4.13: EQ3 eye diagrams for resistor corners at TT

Table 4.6: EQ3 RX eye diagram metrics across resistor corners at TT

Corner	Height	Width
Res_min	206 mV	78.9 ps
Res_nom	231 mV	79.6 ps
Res_max	232 mV	79.5 ps

The corresponding EQ3 resistor-corner eye diagrams at FF are shown in Figure 4.14.



(a) Res_min

(b) Res_nom

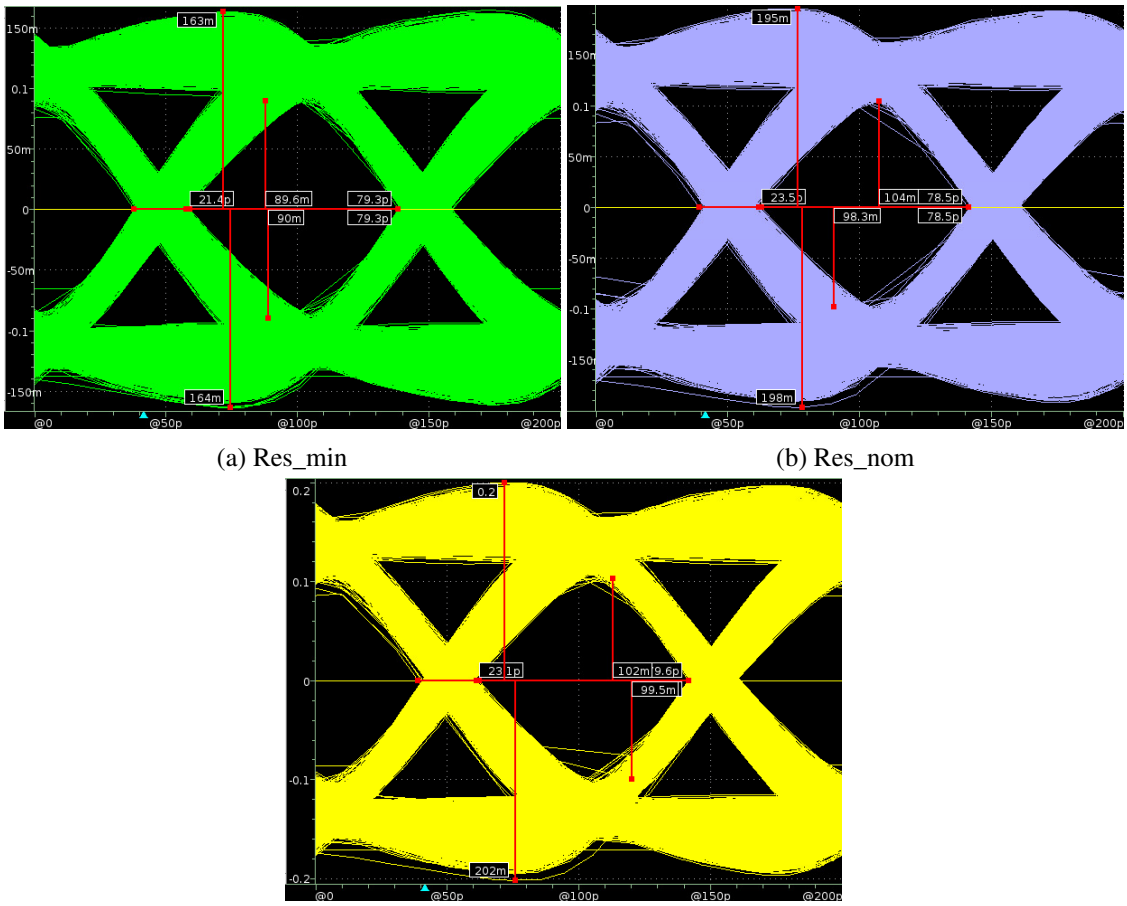
(c) Res_max

Figure 4.14: EQ3 eye diagrams for resistor corners at FF

Table 4.7: EQ3 RX eye diagram metrics across resistor corners at FF

Corner	Height	Width
Res_min	224 mV	78.8 ps
Res_nom	234 mV	79.8 ps
Res_max	272 mV	79.4 ps

The corresponding EQ3 resistor-corner eye diagrams at SS are shown in Figure 4.15.



(a) Res_min

(b) Res_nom

(c) Res_max

Figure 4.15: EQ3 eye diagrams for resistor corners at SS

Table 4.8: EQ3 RX eye diagram metrics across resistor corners at SS

Corner	Height	Width
Res_min	179.6 mV	79.3 ps
Res_nom	202.3 mV	78.5 ps
Res_max	201.5 mV	79.6 ps

4.2.3 Power Consumption

The power consumption of the transmitter driver was evaluated at the EQ3 equalization preset across all three PVT corners considered, with the supply current and corner-dependent supply voltage used to compute the resulting static power dissipation, following the general relationship

$$P = I_{DD} \cdot V_{DD}. \tag{4.1}$$

As expected, the power consumption follows the combined trend of supply current and supply voltage across corners, reaching its maximum at the FF corner, where both the current sunk by

the driver and the supply voltage are at their highest, and its minimum at the SS corner, where both quantities are at their lowest. The resulting power consumption values are summarized in Table 4.9.

Table 4.9: Driver power consumption for EQ3

Corners	I_{DD}	V_{DD}	**P**
TT, Res_nom	21.2 mA	1.2 V	25.5 mW
TT, Res_min	21.4 mA	1.2 V	25.6 mW
TT, Res_max	21.1 mA	1.2 V	25.3 mW
FF, Res_nom	21.8 mA	1.26 V	27.4 mW
FF, Res_min	21.9 mA	1.26 V	27.6 mW
FF, Res_max	21.7 mA	1.26 V	27.3 mW
SS, Res_nom	20.2 mA	1.14 V	23 mW
SS, Res_min	20.4 mA	1.14 V	23.3 mW
SS, Res_max	20 mA	1.14 V	22.9 mW

4.3 Post Extraction Results

The complete driver, including all tap unit instances, the current-mirror biasing cell, and the termination network, occupies a total layout area of 176 µm × 82 µm. To close the guard ring around the full extent of the driver, additional dummy cells had to be inserted along the boundary of the layout, beyond those already present within the individual tap unit standard cells described in Section 3.5.1, in order to maintain a continuous, uninterrupted guard ring around the complete macro. The resulting full driver layout is shown in Figure 4.16.

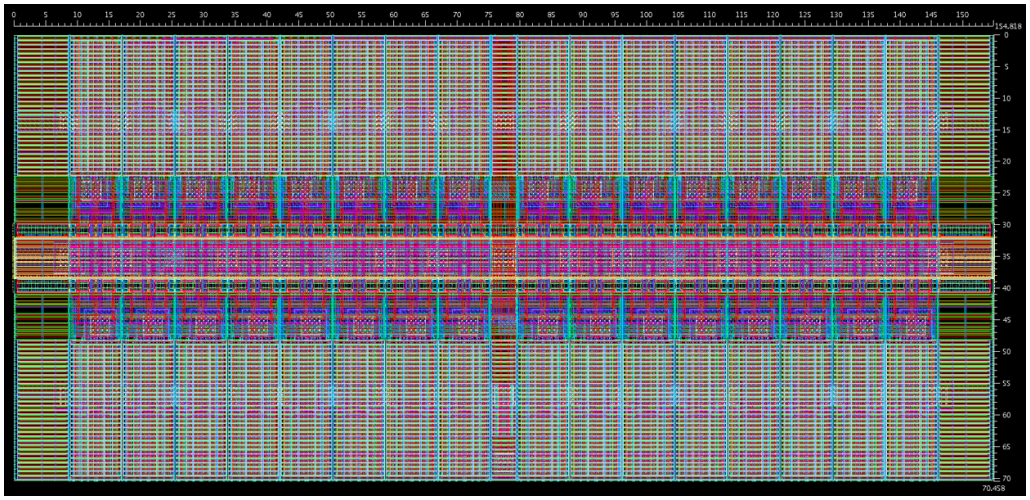


Figure 4.16: Complete driver layout

4.3.1 Extracted Parasitic Capacitance and Coupling Analysis

The extracted total parasitic capacitances were found to be highly symmetrical across all differential net pairs in the design. The output nets, TX_p and TX_n, exhibit a difference of only 0.98% in total parasitic capacitance between the two branches. The pre-cursor tap nets, PRE_p and PRE_n, exhibit a difference of 0.39%, the post-cursor tap nets, $POST_p$ and $POST_n$, exhibit a difference of 1.29%, and the main-cursor tap nets, $MAIN_p$ and $MAIN_n$, exhibit a difference of 0.41%.

Table 4.10: Differential nets parasitic capacitances

Net	C_{par}
TX_p	340.17 fF
TX_n	336.87 fF
PRE_p	17.9 fF
PRE_n	17.97 fF
$POST_p$	55.09 fF
$POST_n$	55.8 fF
$MAIN_p$	233.49 fF
$MAIN_n$	232.55 fF

These results provide a quantitative confirmation of the effectiveness of the symmetry-first layout methodology adopted for the driver, in which the differential signal nets were prioritized above all other routing during the layout design process. The consistently low mismatch observed across all four differential net pairs, all below 1.3%, demonstrates that this prioritization was effective in achieving a high degree of parasitic symmetry between the complmeentary branches of the differential signal path, which is essential for minimizing DCD and timing skew, as discussed in Section 2.2.2.4.

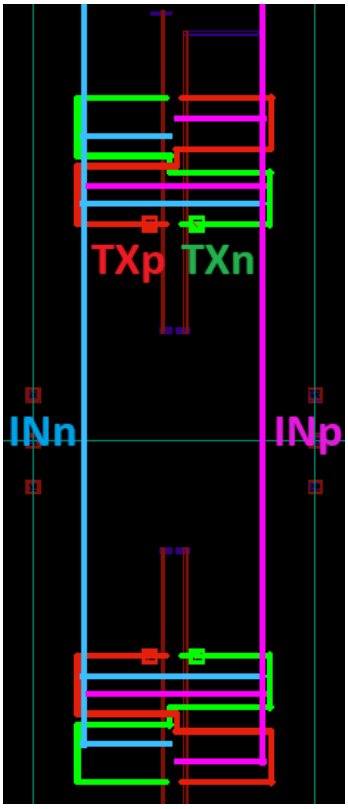
4.3.1.1 Dominant Net-to-Net Coupling

Beyond the overall symmetry of the total parasitic capacitance on each net, the extracted results were further analyzed to identify the dominant coupling capacitances between individual nets, summarized in Table 4.11.

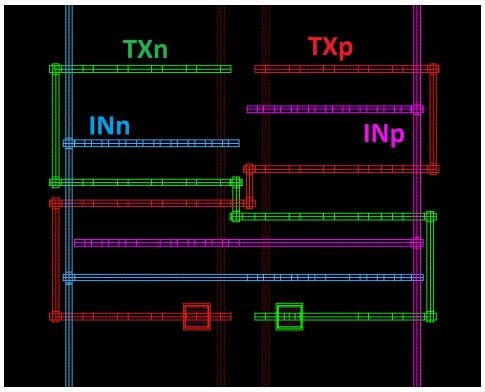
Table 4.11: Dominant coupled net and coupling capacitance for each differential net

Net	**Highest Coupled Net**	**Coupling Cap**	**% of net C_{par}**
TX_p	$MAIN_n$	82.45 fF	24.24%
TX_n	$MAIN_p$	80.25 fF	23.82%
PRE_p	TX_p	6.78 fF	37.91%
PRE_n	TX_n	6.75 fF	37.58%
$POST_p$	TX_p	20.36 fF	36.96%
$POST_n$	TX_n	20.27 fF	36.33%
$MAIN_p$	TX_n	80.25 fF	34.37%
$MAIN_n$	TX_p	82.45 fF	35.45%

These results are a direct consequence of the metal layers used, and placement of the differential nets within the layout. The use of M5 for TX_p and TX_n, and M4 for the tap input nets make them prone to coupling with each other when overlapped. Figure 4.17 shows a close-up view of the differential pair, highlighting the overlap between the differential output nets and the tap input nets.



(a) Pre-cursor tap input and output routing



(b) Post-cursor tap input and output routing

Figure 4.17: Differential pair input and output routing

At the center of the differential pair, an overlap between TX_p and TX_n is observed, due to the common-centroid matching chosen for these devices during floorplanning. It was known that these nets would have at least one point of overlap, and the routing of the tap unit was designed to minimize the overlapping of these input nets. The extracted parasitics show that, even though this overlap was minimized as much as possible, the existence of this overlap on every single tap unit cell adds up to a significant portion of the total parasitic capacitance on these nets, with TX_p and TX_n amounting to around 14% of the total parasitic on each other, the second highest for both.

On the side edges, the tap input nets are placed, with seven overlapping points between input and output nets of the same polarity.

4.3.2 Post Extraction Performance

With no equalization applied, corresponding to the EQ0 preset, the driver achieves a post-extraction differential output voltage swing of $1.28\,V_{pp}$ at the FF corner, $1.1\,V_{pp}$ at the TT corner, and $0.908\,V_{pp}$ at the SS corner. Figure 4.18 shows the resulting differential output voltage swing across all three PVT corners for the EQ0 preset.

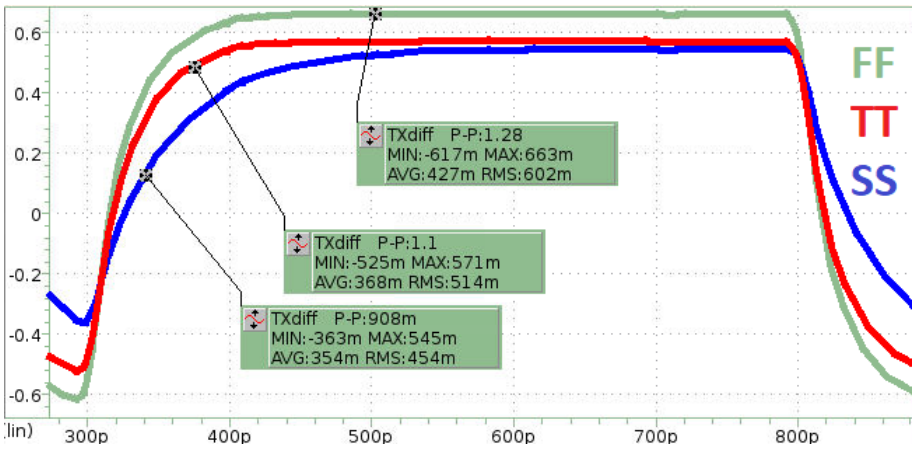


Figure 4.18: Differential output voltage swing across PVT corners for EQ0

The post-extraction PS and DE results for the four equalized presets (EQ1–EQ4), evaluated across all three PVT corners, are summarized in Table 4.12. Table 4.13 expresses the same results as the percentage deviation from the nominal target defined for each preset. For consistency with the target values used throughout this chapter, DE is reported as a positive magnitude in the following tables.

Table 4.12: Post-extraction equalization results across PVT corners

Preset	Corner	PS	DE
EQ1	TT_Rnom	1.317	1.426
EQ1	FF_Rnom	1.246	1.468
EQ1	SS_Rnom	1.39	1.41
EQ2	TT_Rnom	2.159	2.346
EQ2	FF_Rnom	2.065	2.344
EQ2	SS_Rnom	1.996	0.315
EQ3	TT_Rnom	3.121	6.114
EQ3	FF_Rnom	3.086	6.100
EQ3	SS_Rnom	2.972	3.848
EQ4	TT_Rnom	2.554	4.119
EQ4	FF_Rnom	2.499	4.110
EQ4	SS_Rnom	2.390	1.867

The post-extraction results show that the layout does not affect all equalization metrics or PVT corners equally. At the TT and FF corners, PS remains comparatively close to its target for all four presets, whereas the DE magnitude is systematically reduced. At the SS corner, the

Table 4.13: Post-extraction equalization results across PVT corners, expressed as percentage deviation from target

Preset	**TT**		**FF**		**SS**	
	PS	**DE**	**PS**	**DE**	**PS**	**DE**
EQ1	+9.75%	−32.10%	+3.83%	−30.10%	+15.83%	−32.86%
EQ2	−1.85%	−24.32%	−6.12%	−24.40%	−9.29%	−89.85%
EQ3	−2.45%	−12.66%	−3.56%	−12.86%	−7.11%	−45.02%
EQ4	+2.17%	−17.61%	−0.05%	−17.80%	−4.40%	−62.66%

degradation becomes substantially more pronounced for the stronger equalization presets. For EQ1, for example, the DE magnitude decreases to 1.41 dB, corresponding to a 32.86% reduction relative to its 2.1 dB target. More severe DE reductions are observed for EQ2–EQ4 at SS, reaching 89.85% for EQ2, 45.02% for EQ3, and 62.66% for EQ4.

These results indicate that parasitic extraction modifies the effective relative weighting of the three FFE tap currents. The effect is substantially more pronounced for DE than for PS and becomes particularly severe at the SS corner. However, the sensitivity is not identical for all presets, indicating that the impact of the extracted parasitics depends on the particular current allocation between the pre-cursor, main-cursor, and post-cursor paths. The equalization results alone do not uniquely identify a single tap as the sole source of the error. Instead, they demonstrate that the nominal tap-current allocation derived at schematic level is not preserved uniformly across PVT after extraction.

The resulting error also demonstrates the importance of the digitally segmented current-source architecture. Since the pre-cursor, main-cursor, and post-cursor currents can be adjusted independently, the tap-enable mapping can be retuned after extraction without modifying the underlying analog driver.

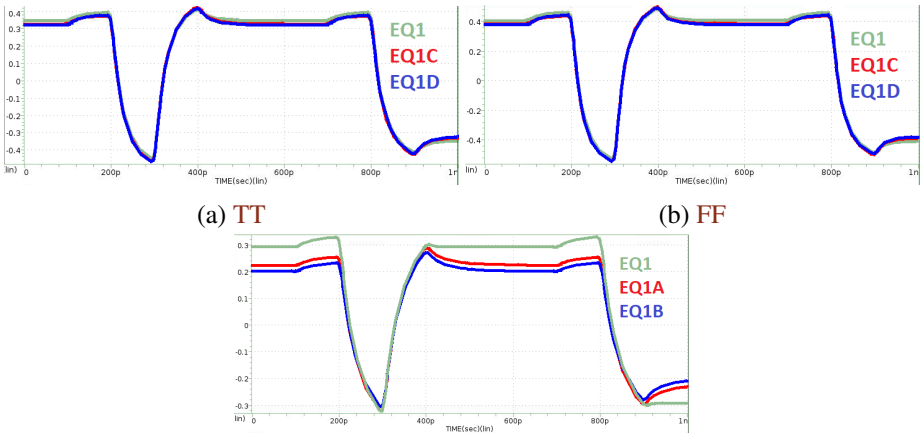
EQ1 was selected as a representative case to evaluate the available post-layout tuning range. Among the evaluated presets, EQ1 exhibits the largest DE deviation at the TT and FF corners, while still showing a significant reduction at SS. It therefore provides a useful case for evaluating whether the segmented current-source architecture can recover the required equalization through digital reconfiguration and whether the required correction remains valid across PVT.

For the tuning sweep, the starting EQ1, 7-102-12, configuration was first evaluated as a reference operating point. At TT, this configuration produces 1.317 dB of PS and 1.426 dB of DE. Increasing the post-cursor contribution to obtain 6-102-16, named EQ1C, significantly reduces the equalization error, producing 1.148 dB of PS and 2.124 dB of DE. However, its PS remains below the 1.2 dB normative minimum. Increasing the pre-cursor enable count to obtain 7-102-16, named EQ1D, produces 1.382 dB of PS and 2.153 dB of DE. Although this configuration has a larger deviation from the nominal PS target, both equalization metrics fall within their required ranges, and 7-102-16 is therefore selected for the TT corner.

The same behaviour is observed at FF. The 6-102-16 configuration gives 1.1 dB of PS and 2.09 dB of DE, providing very small DE error but insufficient PS. The 7-102-16 configuration

instead produces 1.32 dB of PS and 2.12 dB of DE, placing both quantities within the normative range. Consequently, 7-102-16 is also selected at FF

The SS corner requires a substantially different current allocation. Starting from 7-102-12, the extracted driver produces 1.39 dB of PS and 1.41 dB of DE. The 4-90-20, named EQ1A, candidate recovers the DE magnitude to 2.152 dB, but its 1.082 dB PS remains below the normative minimum. Reducing the main-cursor enable count further results in the 4-82-20 configuration, named EQ1B, which produces 1.203 dB of PS and 2.526 dB of DE. Although its DE exceeds the nominal EQ1 target, both quantities are within their normative specification ranges, making 4-82-20 the selected configuration for the SS corner. The effect of the evaluated EQ1 tap-enable mappings on the output waveform at each PVT corner is illustrated in Figure 4.19.



(a) TT (b) FF

(c) SS

Figure 4.19: Starting and tuned differential output for PVT corners

Table 4.14: Post-extraction EQ1 target deviation before and after tap reconfiguration

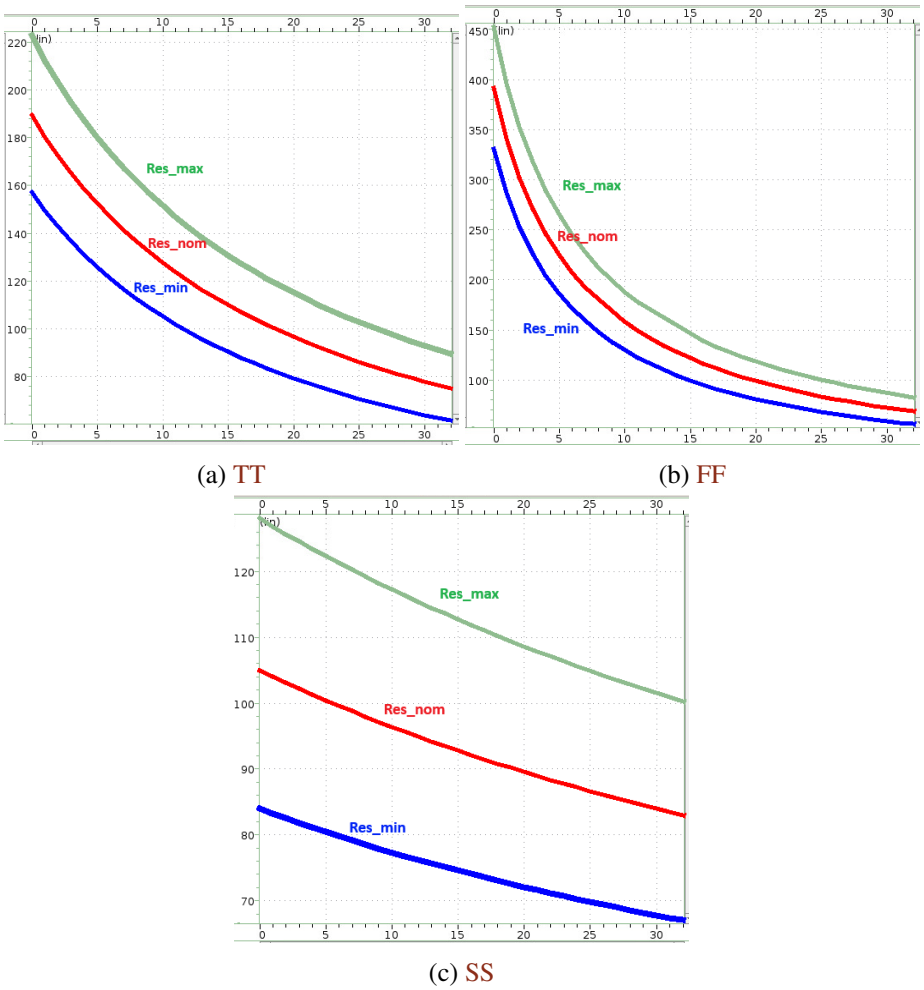
Corner	Mapping	PS [dB]	DE [dB]	ΔPS	ΔDE	Result
TT	EQ1	1.317	1.426	9.75%	−32.10%	Reference
TT	EQ1C	1.148	2.124	−4.33%	+1.14%	PS below minimum
TT	**EQ1D**	**1.382**	**2.153**	+15.17%	+2.52%	**Selected**
FF	EQ1	1.246	1.468	+3.83%	−30.10%	Reference
FF	EQ1C	1.100	2.099	−8.33%	−0.05%	PS below minimum
FF	**EQ1D**	**1.320**	**2.124**	+10.00%	+1.14%	**Selected**
SS	EQ1	1.39	1.41	+15.83%	−32.86%	Reference
SS	EQ1A	1.082	2.152	−9.87%	+2.48%	PS below minimum
SS	**EQ1B**	**1.203**	**2.526**	+0.23%	+20.27%	**Selected**

The tuning results show that minimizing the numerical deviation from the nominal EQ1 target is not by itself sufficient for selecting the final mapping. At both TT and FF, the 6-102-16 mapping gives the smallest overall target error, but its PS remains below the normative minimum. The 7-102-16 mapping is therefore preferred because it places both PS and DE inside the allowed ranges.

The same trade-off occurs at SS, where 4-90-20 gives a smaller target error but insufficient PS, whereas 4-82-20 provides a specification-compatible operating point.

Most importantly, the final mapping is not identical across PVT: TT and FF are both corrected using 7-102-16, while SS requires the substantially different 4-82-20 mapping. The segmented current-source architecture therefore provides sufficient tuning range to recover the required equalization after extraction, but the results also demonstrate that a single fixed EQ1 enable mapping is not robust across the full evaluated PVT range. A practical implementation would consequently benefit from a calibration procedure or from pre-characterized corner-dependent configuration codes.

The post-extraction differential output-impedance sweeps across termination-bank settings and process corners are plotted in Figure 4.20.



(a) TT (b) FF

(c) SS

Figure 4.20: Post extraction differential output impedance across corners and bank configurations

The post-extraction termination results show that a compliant differential output impedance can be obtained for every evaluated combination of PVT and resistor corner by selecting an appropriate termination-bank enable count. However, the complete 72 Ω to 120 Ω tuning interval is not reproduced at every combined corner, as indicated by the unavailable entries in Table 4.15. The

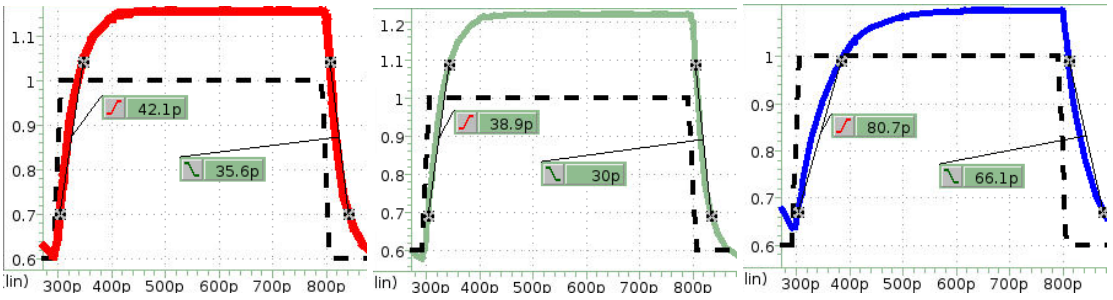
Table 4.15: Post extraction $Z_{out,diff}$ across PVT and resistor corners

PVT	Res	#en$_{max}$	#en$_{nom}$	#en$_{min}$
TT	Res_min	6	12	24
TT	Res_nom	13	19	32
TT	Res_max	19	27	NA
SS	Res_min	NA	NA	22
SS	Res_nom	NA	6	NA
SS	Res_max	9	32	NA
FF	Res_min	12	15	NA
FF	Res_nom	16	20	NA
FF	Res_max	20	26	NA

extracted implementation should therefore be interpreted as providing at least one specification-compatible termination setting at each evaluated corner, rather than providing the complete normative impedance range under every condition.

The required enable count also changes substantially with process and resistor variation. For example, at TT and Res_nom, the nominal impedance point moves from 15 enabled termination elements before extraction to 19 after extraction. This shift further demonstrates that the programmable termination network is required not only to accommodate process variation, but also to compensate for the resistance and interconnect parasitics introduced by the physical implementation.

The post-extraction rise and fall waveforms for the three PVT corners are shown in Figure 4.21.



(a) TT (b) FF (c) SS

Figure 4.21: Rise and fall times across PVT corners

Table 4.16: Post extraction rise and fall times across PVT corners

Corner	Rise	Fall
TT	42.1 ps	35.6 ps
FF	38.9 ps	30 ps
SS	80.7 ps	66.1 ps

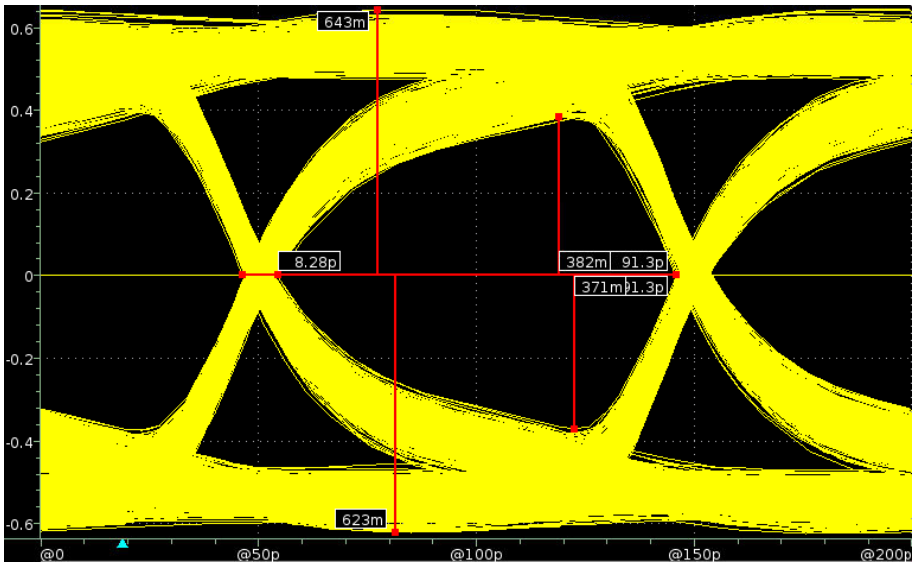
The post-extraction rise and fall times increase at all three PVT corners, consistent with the

additional capacitive loading introduced by the extracted interconnect and device parasitics discussed in Section 4.3.1. At TT, the rise time increases from 27.3 ps before extraction to 42.1 ps after extraction, while the fall time increases from 22.8 ps to 35.6 ps. These correspond to increases of approximately 54.2% and 56.1%, respectively.

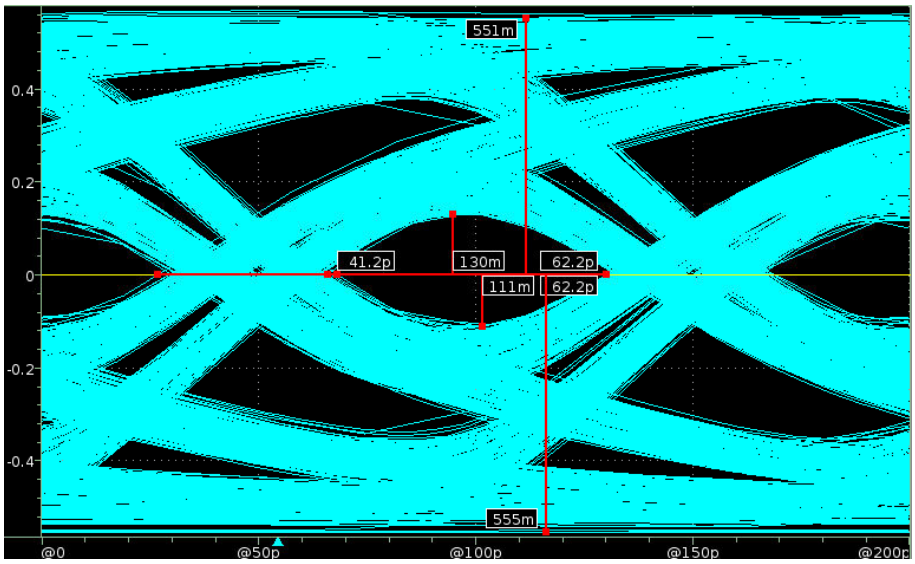
The extracted SS corner exhibits the slowest transitions, with a rise time of 80.7 ps and a fall time of 66.1 ps. Thus, although all measured transition times remain below the nominal 100 ps UI, the reduction in transition-speed margin after extraction is significant, particularly at SS. The persistent difference between rise and fall times is consistent with the asymmetric loading of the active PDN and passive PUN paths discussed previously.

At TT, the post-extraction eye diagrams demonstrate a clear dependence of the received signal quality on the applied transmitter equalization. Table 4.17 summarizes the corresponding TX and RX eye metrics.

The post-extraction EQ0 TX and RX eye diagrams are presented in Figure 4.22.



(a) TX

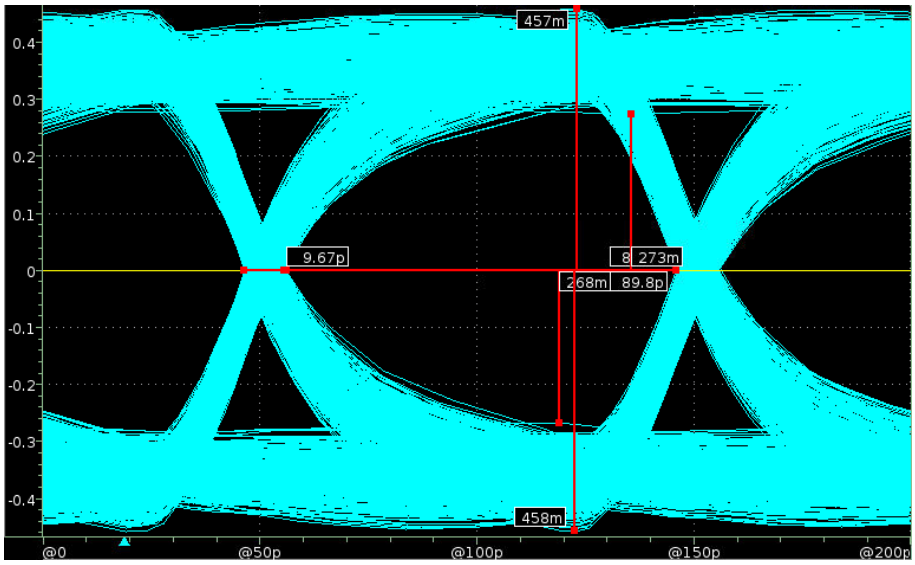


(b) RX

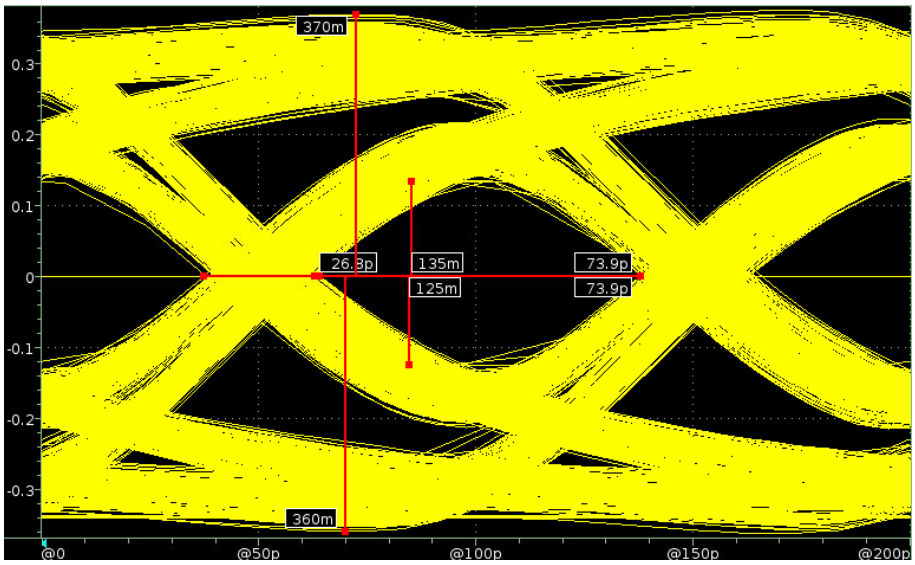
Figure 4.22: Post extraction eye diagrams for EQ0 at corner TT

For EQ0, the RX eye has a height of 241 mV and a width of 62.2 ps. Although a vertical opening remains after transmission through the reference channel, EQ0 provides the smallest horizontal opening of all evaluated post-extraction configurations.

The corresponding post-extraction EQ1 eye diagrams are shown in Figure 4.23.



(a) TX

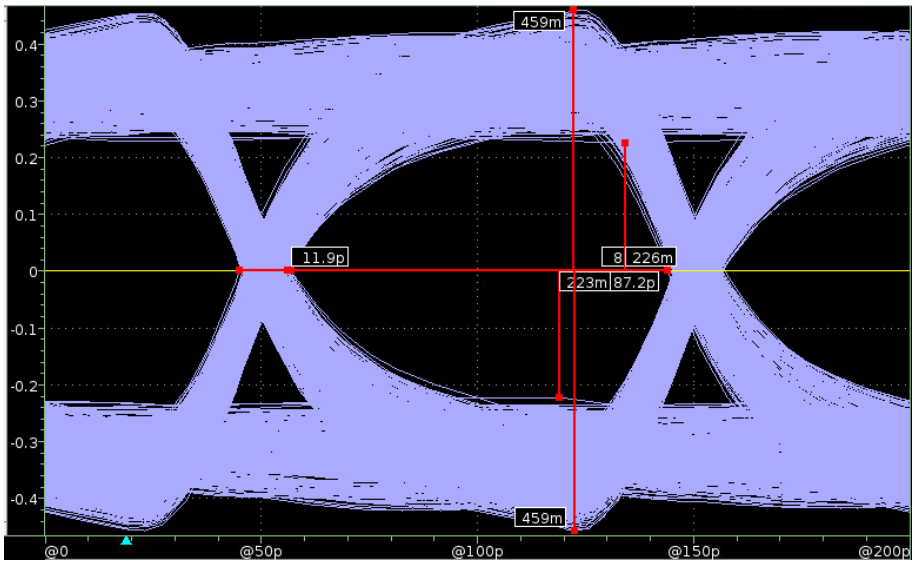


(b) RX

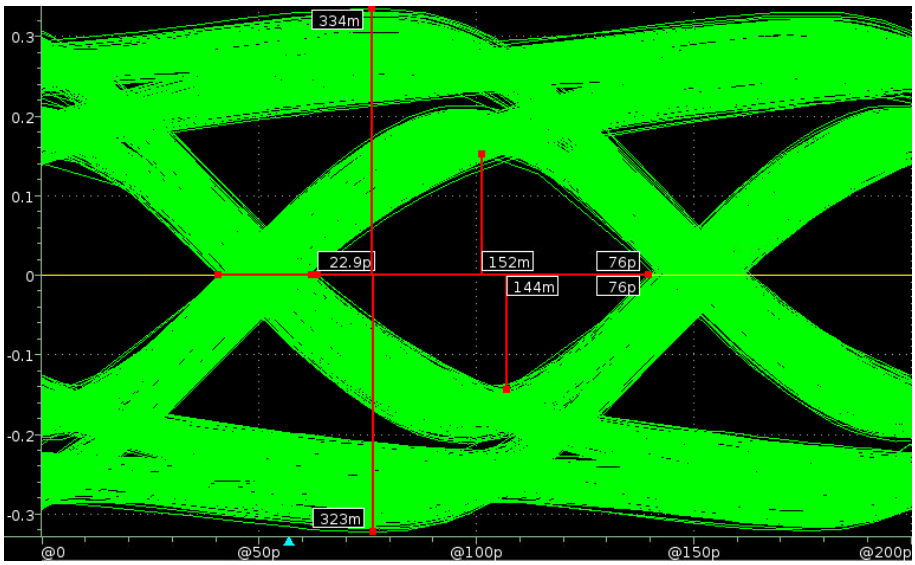
Figure 4.23: Post extraction eye diagrams for EQ1 at corner TT

Applying equalization progressively increases the RX eye opening. EQ1 produces an RX eye height of 260 mV and a width of 73.9 ps, while EQ2 increases these values to 296 mV and 76 ps, respectively.

The corresponding post-extraction EQ2 eye diagrams are shown in Figure 4.24.



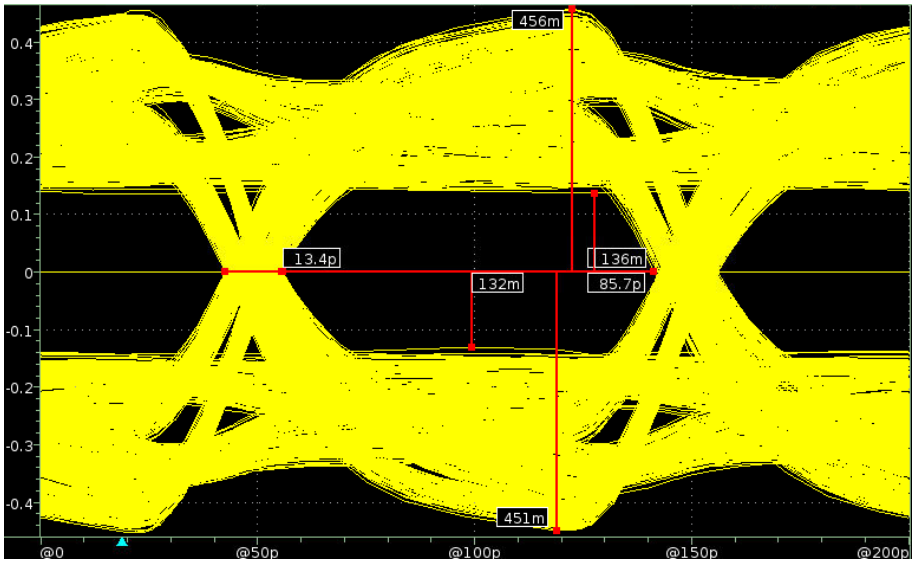
(a) TX



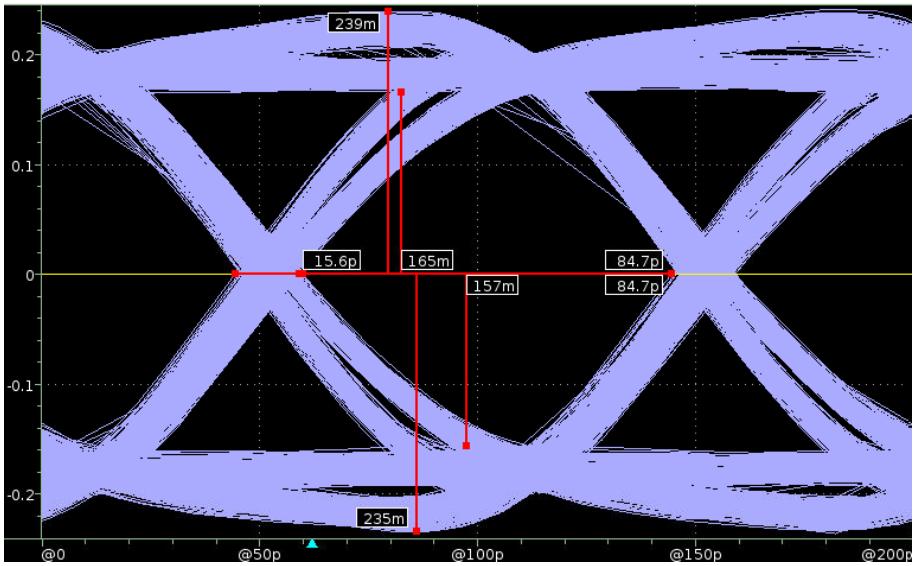
(b) RX

Figure 4.24: Post extraction eye diagrams for EQ2 at corner TT

The corresponding post-extraction EQ3 eye diagrams are shown in Figure 4.25.



(a) TX

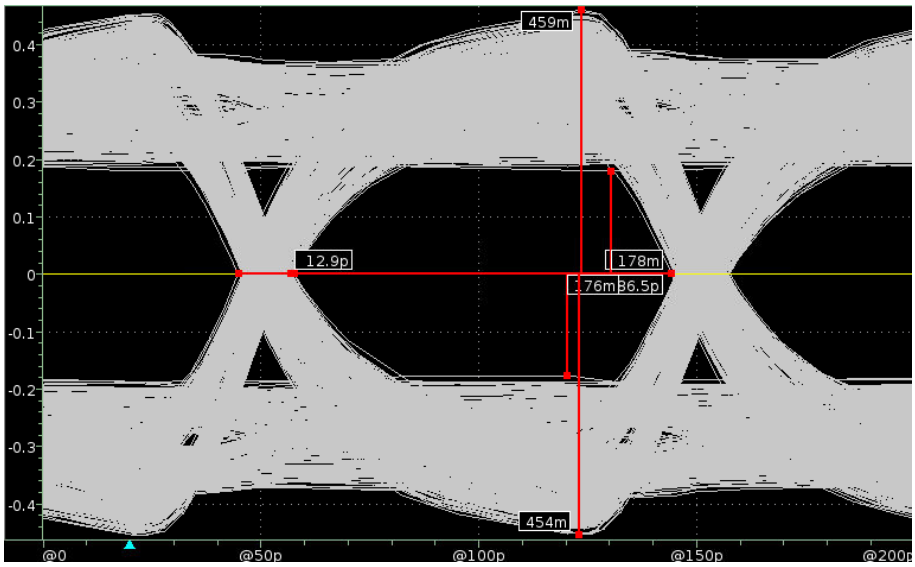


(b) RX

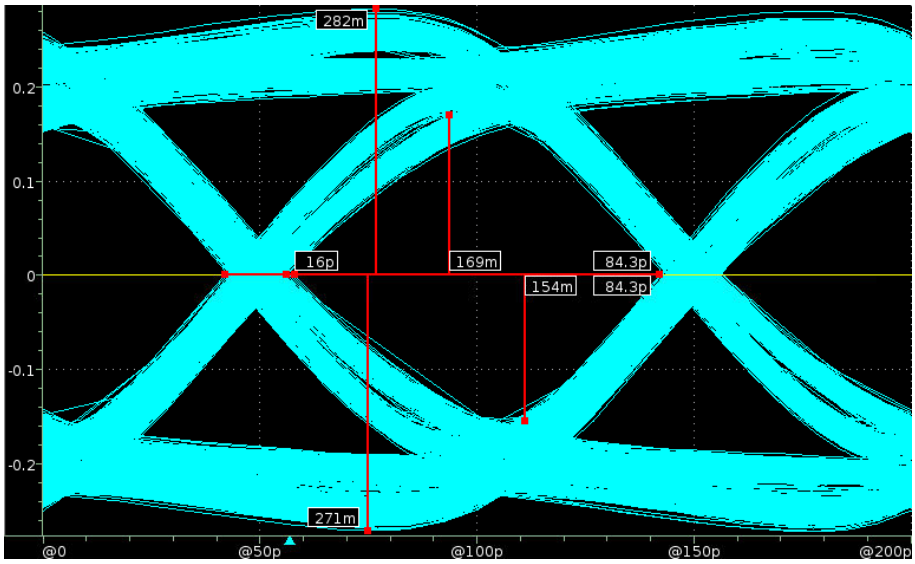
Figure 4.25: Post extraction eye diagrams for EQ3 at corner TT

The strongest eye openings are obtained with EQ3 and EQ4. EQ3 provides an RX eye height of 322 mV and an eye width of 84.7 ps, whereas EQ4 provides an eye height of 323 mV and a width of 84.3 ps. The two configurations therefore provide essentially equivalent post-extraction performance: EQ4 gives the largest vertical opening by only 1 mV, while EQ3 provides the largest horizontal opening by 0.4 ps.

The corresponding post-extraction EQ4 eye diagrams are shown in Figure 4.26.



(a) TX



(b) RX

Figure 4.26: Post extraction eye diagrams for EQ4 at corner TT

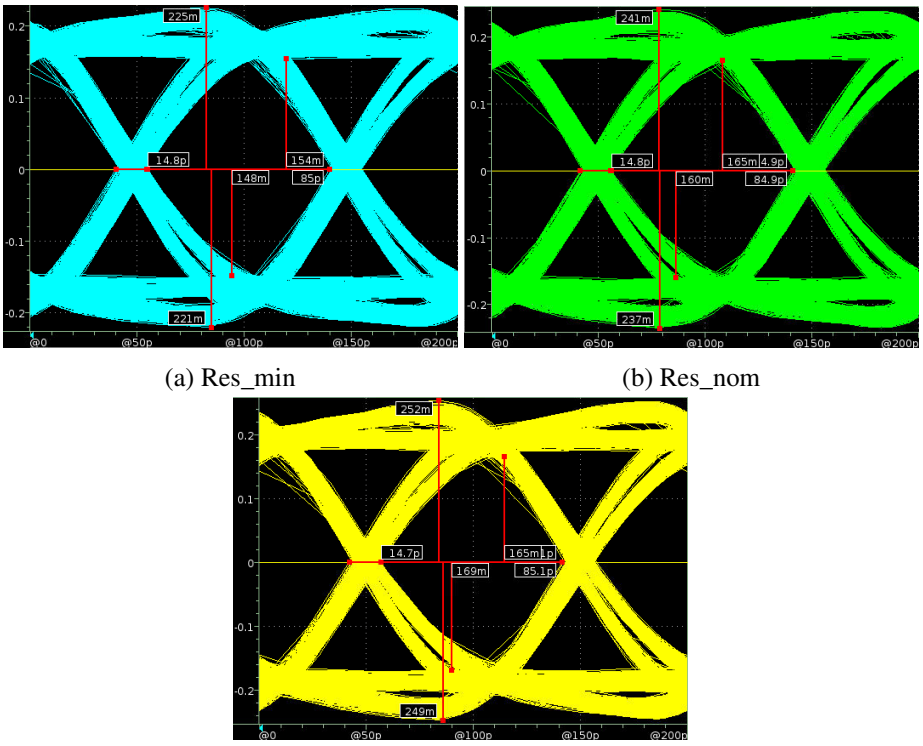
EQ3 and EQ4 provide nearly identical post-extraction performance. EQ4 gives the marginally larger vertical eye opening (323 mV versus 322 mV), whereas EQ3 provides both the marginally larger eye width (84.7 ps versus 84.3 ps) and the slightly smaller crossover width (15.6 ps versus 16.0 ps). Considering these metrics together, EQ3 is retained as the representative configuration for the detailed post-extraction corner analysis.

Across the resistor-corner sweeps, the EQ3 RX eye width remains relatively stable compared with the variation in eye height. At TT, the measured eye width varies only from 84.9 ps to 85.1 ps, while the eye height varies from 302 mV to 334 mV. At FF, the width remains between 86.4 ps and 86.9 ps, while the height ranges from 359 mV to 396 mV. At SS, the width remains between 83.1 ps and 84.9 ps, whereas the height ranges from 202.3 mV to 221 mV.

Table 4.17: Post extraction eye metrics at TT for different equalization presets

EQ Preset	TX Height	TX Width	RX Height	RX Width
EQ0	699 mV	91.3 ps	241 mV	62.2 ps
EQ1	541 mV	89.8 ps	260 mV	73.9 ps
EQ2	449 mV	87.2 ps	296 mV	76 ps
EQ3	268 mV	85.7 ps	322 mV	84.7 ps
EQ4	354 mV	86.5 ps	323 mV	84.3 ps

The post-extraction EQ3 received eye diagrams at TT for the three resistor corners are presented in Figure 4.27.



(a) Res_min

(b) Res_nom

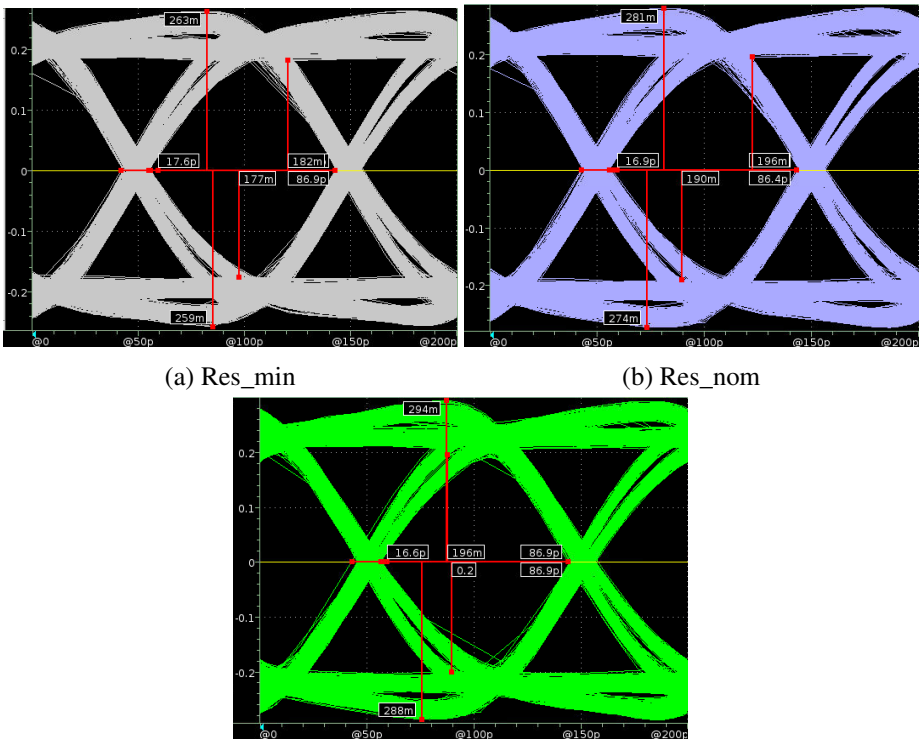
(c) Res_max

Figure 4.27: Post extraction EQ3 eye diagrams for resistor corners at TT

Table 4.18: Post extraction EQ3 RX eye diagram metrics across resistor corners at TT

Corner	Height	Width
Res_min	302 mV	85 ps
Res_nom	325 mV	84.9 ps
Res_max	334 mV	85.1 ps

The corresponding post-extraction EQ3 eye diagrams at FF are shown in Figure 4.28.



(a) Res_min (b) Res_nom

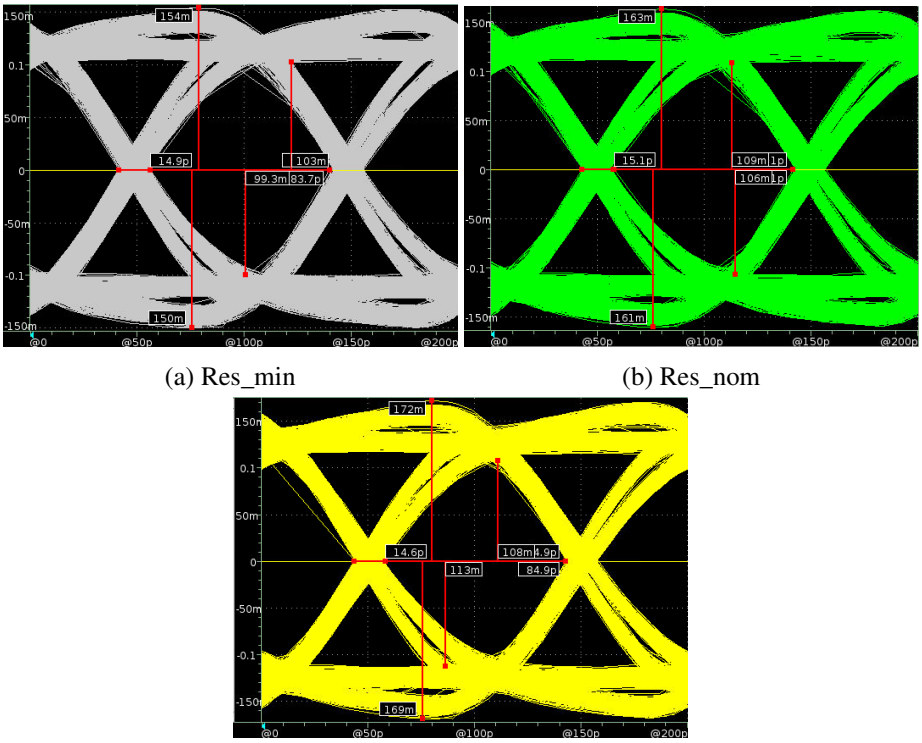
(c) Res_max

Figure 4.28: Post extraction EQ3 eye diagrams for resistor corners at FF

Table 4.19: Post extraction EQ3 RX eye diagram metrics across resistor corners at FF

Corner	Height	Width
Res_min	359 mV	86.9 ps
Res_nom	386 mV	86.4 ps
Res_max	396 mV	86.9 ps

The corresponding post-extraction EQ3 eye diagrams at SS are shown in Figure 4.29.



(a) Res_min (b) Res_nom

(c) Res_max

Figure 4.29: Post extraction EQ3 eye diagrams for resistor corners at SS

Table 4.20: Post extraction EQ3 RX eye diagram metrics across resistor corners at SS

Corner	Height	Width
Res_min	202.3 mV	83.7 ps
Res_nom	215 mV	83.1 ps
Res_max	221 mV	84.9 ps

These results indicate that, for the evaluated EQ3 configuration, resistor variation has a stronger influence on the vertical eye opening than on the horizontal timing opening. The larger difference between the FF and SS eye heights further shows that transistor-level PVT variation has a more pronounced effect on received voltage margin than the resistor-corner variation within a given PVT condition.

4.4 Pre and Post Extraction Comparison

Table 4.21 compares the principal pre- and post-extraction results for the representative EQ3 configuration at the TT corner. EQ3 is retained as the reference case because it provides the largest pre-extraction RX eye and remains one of the two strongest post-extraction equalization configurations. For consistency, the DE values in this comparison are reported as positive magnitudes.

Table 4.21: Pre- versus post-extraction results at the TT corner (EQ3, unless noted)

Metric	**Pre-extraction**	**Post-extraction**	Δ
PS [dB]	3.01	3.121	+3.7%
DE magnitude [dB]	6.63	6.114	−7.8%
Output rise time [ps]	27.3	42.1	+54.2%
Output fall time [ps]	22.8	35.6	+56.1%
RX eye height [mV]	231	322	+39.4%
RX eye width [ps]	79.6	84.7	+6.4%
RX crossover width [ps]	45.1	15.6	−65.4%
$Z_{out,diff}$, $\#en_{nom}$ (Res_nom)	15	19	+26.7%
Power consumption [mW] (Res_nom)	25.5	24.42	−4.2%

The comparison shows that parasitic extraction significantly modifies the transmitter behaviour, but that the effect is not a uniform degradation of all metrics. The clearest direct penalty is observed in the output transition times. At TT, the rise time increases from 27.3 ps to 42.1 ps, while the fall time increases from 22.8 ps to 35.6 ps, corresponding to increases of approximately 54.2% and 56.1%, respectively. The similar relative degradation of the two transitions indicates that parasitic extraction introduces an overall increase in the effective time constants of the high-speed output paths, expected with the increase that the parasitics introduces to the RC time constant, τ, of the nets. The rise–fall asymmetry already present before extraction remains.

The equalization coefficients are also modified after extraction. For EQ3 at TT, the PS changes from 3.01 dB to 3.121 dB, while the DE magnitude decreases from 6.63 dB to 6.114 dB. More generally, Table 4.13 shows that the effect becomes much stronger for DE at the SS corner. Together with the EQ1 retuning results, this demonstrates that extraction alters the effective relative weighting of the tap currents and that the optimum enable mapping can become corner-dependent. The available results do not, however, isolate this behaviour to one individual tap current alone.

The termination network is similarly affected by the physical implementation. For TT and Res_nom, the enable count associated with the nominal output impedance changes from 15 before extraction to 19 after extraction. The programmable termination bank therefore provides useful margin for compensating not only resistor-process variation but also the additional resistance introduced by the extracted interconnect.

Despite the slower transmitter transitions, the received EQ3 eye does not degrade. The RX eye height increases from 231 mV to 322 mV, while the eye width increases from 79.6 ps to 84.7 ps. At the same time, the measured crossover width decreases substantially, from 45.1 ps before extraction to 15.6 ps after extraction, corresponding to a reduction of approximately 65.4%.

A similar behaviour is observed for EQ4, whose crossover width decreases from 49.7 ps before extraction to 16.0 ps after extraction, a reduction of approximately 67.8%. The post-extraction EQ3 crossover is also relatively insensitive to the evaluated process corner, varying from 15.1 ps at SS to 16.9 ps at FF.

The substantial reduction in crossover width indicates a more concentrated distribution of the received transition crossings in the extracted simulations. However, this observation should not be

interpreted as a direct measurement of DJ or total jitter, since the normative jitter decomposition was not performed. Furthermore, because extraction modifies several circuit properties simultaneously, including the transition times, effective tap weighting, and termination operating point, the reduction cannot be uniquely attributed to a single parasitic mechanism.

Finally, the EQ3 power consumption at TT, Res_nom changes only moderately, from 25.5 mW before extraction to 24.42 mW after extraction. This 4.2 % difference is small compared with the changes observed in transition time and equalization accuracy and does not materially alter the power-performance conclusions of the design.

4.4.1 Evaluation Against USB 3.2 Gen 2 Electrical Requirements

Table 4.22 summarizes the principal simulated transmitter metrics against the corresponding USB 3.2 Gen 2 electrical limits. The post-extraction values correspond to the latest results presented in this chapter. Where programmability is required to obtain a compliant operating point, this is explicitly indicated.

Table 4.22: Summary of the evaluated USB 3.2 Gen 2 transmitter electrical requirements, pre- versus post-extraction

Requirement	**USB 3.2 Specification**	**Pre-extraction**	**Post-extraction**	**Status**
Unit Interval	99.97 ps to 100.03 ps	100 ps	100 ps	Compliant
Differential output voltage swing (EQ0)	0.8 V_{pp} to 1.2 V_{pp}	1.17 V_{pp} (TT)	1.10 V_{pp} (TT)	Compliant
DC differential output impedance	72 Ω to 120 Ω	Compliant code available	Compliant code available	Compliant with corner-specific code[1]
Preshoot (EQ3, TT)	1.2 dB to 3.2 dB	3.01 dB	3.121 dB	Compliant
De-emphasis magnitude (EQ3, TT)	2.1 dB to 7 dB	6.63 dB	6.114 dB	Compliant
Eye diagram height (EQ3, TT, RX)	70 mV to 1200 mV	231 mV	322 mV	Within numerical limit; normative eye procedure not fully evaluated

[1] The complete 72 Ω to 120 Ω interval is not available for every combined PVT and resistor corner. However, at least one termination-bank configuration within the normative range is available for every evaluated corner.

The results in Table 4.22 demonstrate that the simulated transmitter provides sufficient programmable range to obtain operating points within the evaluated USB 3.2 Gen 2 numerical limits for output swing, equalization, and output impedance after parasitic extraction. The simulated EQ3 receiver eye height also lies within the numerical eye-height range defined by the standard, although the complete normative transmitter-eye measurement procedure was not reproduced. The

EQ1 tuning study further shows, however, that the required equalization code can depend on PVT, rather than being represented by one universal post-layout mapping.

These results should not be interpreted as a complete transmitter compliance demonstration. In particular, the normative TX-eye and jitter methodology defined by the USB 3.2 specification was not reproduced in full. This methodology includes the prescribed Gen 2 compliance patterns, reference channel, compliance receiver equalization and jitter-transfer processing, measurement over 10^6 consecutive UI, and extrapolation of the jitter results to a BER of 10^{-12}. Furthermore, not all normative transmitter electrical parameters were evaluated in the present work. The results presented here therefore demonstrate agreement with the relevant numerical limits for the evaluated simulated metrics and sufficient post-layout tuning range, rather than formal USB-IF compliance.

4.5 Comparison with the State of the Art

The results presented in the previous sections establish the post-layout performance of the proposed transmitter against the USB 3.2 Gen 2 electrical requirements. To place these results in a broader context, Table 4.23 compares the implemented driver with representative CML and CML-based transmitters operating at similar data rates.

A direct numerical ranking is not appropriate because the reported circuits target different applications, channel conditions, voltage swings, circuit boundaries, and verification methodologies. In particular, some publications report fabricated-silicon measurements whereas the results of the present work are obtained from post-layout extracted simulations. The comparison is therefore intended primarily to identify architectural and performance trade-offs rather than to claim superiority based on a single figure of merit.

Table 4.23: Comparison with representative CML and CML-based high-speed transmitters.

Work	Tech.	Rate	Equalization	Supply	Channel / loss	Eye / swing	Consumption	Verification
Rylyakov and Rylov [19]	90 nm	10 Gb/s	4-tap FFE	1.2/1.65 V	16-in backplane	0.9 Vppd	174 mW	Silicon
Weiss *et al.* [20]	130 nm	10 Gb/s	3-tap FIR	1.5 V	24.3 dB @ 4.5 GHz[a]	BER $< 10^{-12}$	–	Silicon
Chen *et al.* [12]	130 nm	10 Gb/s	–	3.3 V	Microstrip model	1.6 Vppd	10 mA	Post-layout sim.
Yi *et al.* [21]	65 nm	9 Gb/s	VM pre-emphasis + CML	1.0 V	12-in FR-4	40 mVppd	2.8 mW TX	Silicon
Lv and Zhang [22]	28 nm	12.5 Gb/s	Current pre-emphasis	1.05 V	–	604 mVppd	28.9 mW	Post-layout sim.
Choi *et al.* [23]	55 nm	10 Gb/s	2-tap + cap. peaking	1.2 V	12.3 dB @ 5 GHz	160 mV / 0.92 UI[b]	36.25 mA[c]	Post-layout sim.
This work	**55 nm**	**10 Gb/s**	**3-tap FFE**	**1.2 V**	**21.6 dB @ 5 GHz**	**322 mV / 0.847 UI[d]**	**24.42 mW**	**Post-layout sim.**

[a] Motherboard-channel attenuation reported at the 4.5 GHz Nyquist frequency for the 9 Gbps experiment.
[b] Differential received eye height and normalized eye width reported after the 12.3 dB channel.
[c] Includes the pre-driver and timing circuitry and is therefore not directly comparable with power numbers whose circuit boundaries differ.
[d] EQ3 post-extraction receiver eye. The 84.7 ps eye width corresponds to 0.847 UI at 10 Gbps.

The most direct architectural comparison is with the work of Weiss *et al.* [20], since both transmitters employ current-domain multi-tap FIR equalization around 10 Gbps. Their implementation demonstrates experimentally that summing independently weighted CML tap currents is effective for compensating strongly attenuating electrical channels. The present design follows the same general current-domain equalization principle but uses explicit pre-cursor, main-cursor, and post-cursor segmentation derived from the USB preshoot and de-emphasis requirements. In

addition, both the tap-current sources and output termination are digitally segmented to provide adjustment after physical implementation. The principal advantage of the earlier work is its experimental validation, including direct BER measurements, which is not available for the present design.

The four-tap transmitter of Rylyakov and Rylov [19] provides another useful comparison because it also combines programmable current weighting with a CML output stage. At 10 Gbps, that implementation reports 174 mW total transmitter consumption under its 125 °C test condition. The substantially lower numerical power of the present transmitter, 24.42 mW at the nominal condition, reflects in part the benefits of a more scaled technology and a different circuit boundary and supply arrangement. It should therefore not be interpreted as a technology-normalized power improvement. Nevertheless, it shows that a programmable multi-tap current-mode architecture can be implemented at 10 Gbps with a power consumption in the tens-of-milliwatt range while retaining a large programmable output swing.

The work of Lv and Zhang [22] is closer in terms of simulated power. Their 28 nm, 12.5 Gbps driver consumes 28.9 mW, compared with 24.42 mW for EQ3 in the present implementation. This corresponds to approximately 2.31 $\mathrm{pJ\,bit^{-1}}$ for their reported operating point and 2.44 $\mathrm{pJ\,bit^{-1}}$ for the present design. These figures are of the same order despite the different technology nodes and output requirements. However, the comparison remains approximate because the two circuits employ different equalization structures, output swings, and channel environments.

The implementation by Choi *et al.* [23] provides the closest technology-level comparison. Both designs operate at 10 Gbps in 55 nm CMOS from a 1.2 V supply. Choi *et al.* use two current-mode taps together with reconfigurable capacitive peaking to reduce the static-current penalty associated with stronger equalization. Their post-layout simulations produce a 160 mV eye height and a 0.92 UI eye width after a channel with 12.3 dB insertion loss at 5 GHz. In comparison, the present EQ3 configuration produces a 322 mV receiver eye height and a 84.7 ps, or 0.847 UI, eye width after the USB-IF reference channel, whose insertion loss is 21.6 dB at the same Nyquist frequency. The larger channel loss in the present verification makes a direct eye-opening comparison inappropriate, but it demonstrates that the three-tap programmable structure retains substantial voltage and timing margin under the more strongly attenuating reference channel used in this work.

The Choi implementation draws 36.25 mA from the 1.2 V supply when the pre-driver and timing circuits are included, corresponding to approximately 43.5 mW. Although this value is numerically higher than the 24.42 mW reported for EQ3 in the present transmitter, the circuit boundaries are not equivalent. The difference should therefore not be interpreted as a direct power-efficiency advantage. Instead, the two implementations illustrate different strategies: capacitive peaking reduces the current required for additional high-frequency boost, whereas the present architecture uses digitally controllable current-domain taps to maximize configurability and to implement the required USB preshoot and de-emphasis presets directly.

The transmitter of Yi *et al.* [21] demonstrates the opposite end of the power–swing trade-off. Its measured transmitter consumes only 2.8 mW at 9 Gbps, substantially below the present

design. However, this result is obtained with only 40 mV peak-to-peak differential channel swing, no conventional TX-end termination, and a receiver specifically designed for an input sensitivity of approximately 9 mV. Its voltage-mode pre-emphasis path also eliminates the low-frequency de-emphasis current of a conventional current-mode equalizer. The result therefore demonstrates that substantially higher energy efficiency is possible when the transmitter and receiver are co-designed for very-low-swing signaling, but it does not represent a directly equivalent solution to the USB transmitter requirements addressed here.

The main distinction of the present work is consequently not a record absolute power or data rate. Instead, it is the combination of a conventional current-steering output structure with three independently programmable FFE taps, programmable termination, and an explicit post-layout investigation across transistor and resistor process corners. The extracted results show why this configurability is important. At the representative EQ3 operating point, extraction changes the rise and fall times from 27.3 ps and 22.8 ps to 42.1 ps and 35.6 ps, respectively, and changes the termination-bank setting required to obtain the nominal differential output impedance. The equalization mapping also becomes PVT-dependent for some presets. These effects would not be visible from a nominal schematic-only comparison.

From this perspective, the principal contribution of the proposed architecture is its ability to recover specification-compatible operating points after the physical implementation modifies the nominal circuit behaviour. The segmented current banks provide sufficient range to retune equalization, while the segmented termination network provides corresponding impedance adjustment. This flexibility is particularly relevant for an interface such as USB 3.2 Gen 2, for which output swing, preshoot, de-emphasis, and output impedance must be simultaneously controlled.

The comparison must nevertheless be qualified by the verification status. Rylyakov and Rylov, Weiss *et al.*, and Yi *et al.* report measurements from fabricated silicon, including BER measurements in the first two cases. In contrast, the present work is verified at schematic and post-extraction simulation levels. Consequently, the results demonstrate the performance and programmability of the implemented layout but do not yet establish measured silicon performance. Package parasitics, fabrication mismatch, laboratory measurement uncertainty, and the complete normative DJ, RJ, TJ, and BER characterization remain to be evaluated experimentally.

Chapter 5

Conclusions and Future Work

5.1 Summary of the Developed Work

This dissertation presented the design, physical implementation, and verification of a 10 Gbps CML transmitter targeting the USB 3.2 Gen 2 electrical requirements. The work followed the transmitter from specification analysis to transistor-level design, layout, parasitic extraction, and post-layout verification in the GlobalFoundries 55 nm LPx CMOS technology.

The design was motivated by the SI limitations that arise in multi-gigabit serial links. At 10 Gbps, the channel introduces substantial frequency-dependent attenuation and therefore significant ISI. The background study in Chapter 2 established the relationship between channel loss, eye closure, jitter, TX-line behaviour, and transmitter-side equalization, and provided the basis for selecting a current-mode, three-tap FFE architecture.

The proposed transmitter, described in Chapter 3, implements independently configurable pre-cursor, main-cursor, and post-cursor taps. Their currents are generated by segmented current-mirror banks and are digitally programmable, allowing the transmitted waveform to be adjusted without changing the underlying analog topology. The required tap coefficients were derived directly from the USB preshoot and de-emphasis definitions and converted into current targets and enable-code mappings. A programmable termination network was integrated in the same modular structure so that output impedance can also be adjusted across process and resistor variations.

A major part of the work concerned the physical implementation of the transmitter. The tap unit was developed as a modular, abuttable layout cell using common-centroid placement, symmetric routing, dummy devices, continuous guard-ring structures, and routing strategies intended to reduce mismatch and layout-dependent effects such as WPE and LOD. The complete extracted transmitter, including the tap arrays, biasing circuitry, and termination network, occupies 176 µm $\times$ 82 µm.

Chapter 4 evaluated the transmitter bfeore and after parasitic extraction and across the relevant PVT and resistor corners. The results show that the physical implementation has a significant effect on the electrical behaviour of the driver. For the representative EQ3 configuration at the TT, Res_nom corner, the post-extraction preshoot and de-emphasis magnitudes are 3.121 dB and

6.114 dB, respectively. The output rise and fall times increase to 42.1 ps and 35.6 ps, compared with 27.3 ps and 22.8 ps before extraction.

Despite the slower output transitions, the received eye remains widely open after propagation through the USB-IF reference channel. EQ3 produces an RX eye height of 322 mV and a width of 84.7 ps, while EQ4 produces 323 mV and 84.3 ps, respectively. The two presets therefore provide essentially equivalent post-extraction performance. EQ4 gives a marginally larger vertical opening, whereas EQ3 provides both a marginally larger horizontal opening and a slightly smaller crossover width, 15.6 ps compared with 16.0 ps. EQ3 was consequently retained as the representative configuration for the detailed corner analysis. The nominal post-extraction power consumption of EQ3 is 24.42 mW at TT and Res_nom.

The pre- and post-extraction comparison demonstrates that layout parasitics cannot be treated as a secondary effect at this data rate. At the representative TT operating point, the rise and fall times increase by approximately 54.2% and 56.1%, respectively, showing a comparable degradation of both output transitions. The EQ3 de-emphasis magnitude decreases by approximately 7.8%, and the termination-bank setting required to obtain the nominal output impedance also changes after extraction. These results demonstrate that parasitic extraction affects not only the transition speed but also the effective equalization weighting and termination operating point.

The equalization results further demonstrate the importance of the segmented architecture. The nominal analytical tap mappings are not preserved uniformly after extraction, with de-emphasis becoming particularly sensitive at the SS corner. The EQ1 retuning study shows that the final compliant mapping is itself PVT-dependent: TT and FF are corrected using the 7-102-16 precursor/main-cursor/post-cursor enable mapping, whereas SS requires the substantially different 4-82-20 mapping. The programmable current banks therefore provide sufficient tuning range to recover valid equalization operating points, but a single universal post-layout equalization code is not sufficient across the complete evaluated PVT space.

When compared with representative CML transmitters reported in the literature, the developed architecture occupies a different point in the power–equalization–robustness trade-off. Earlier current-mode transmitters demonstrate experimentally the effectiveness of multi-tap FFE, while hybrid and capacitive equalization techniques show that lower transmitter power can be obtained by reducing static equalization current or by operating with very small channel swings. The contribution of the present work is instead the combination of a three-tap current-domain equalizer with programmable termination and sufficient digital adjustment range to compensate the changes introduced by physical implementation and PVT variation. The state-of-the-art comparison in Section 4.5 therefore supports the positioning of the proposed transmitter as a configurable, layout-aware 10 Gbps CML implementation rather than as an absolute minimum-power or maximum-data-rate design.

Overall, the work demonstrates that the proposed segmented CML architecture is capable of operating at the target data rate and provides sufficient digital control range to obtain USB 3.2 Gen 2-compatible output swing, equalization, output impedance, and eye-opening operating points after physical implementation. The results validate the usefulness of both the programmable cir-

cuit architecture and the modular layout methodology, while also showing that calibration becomes an important part of achieving robust operation across PVT. Complete compliance validation still requires the normative jitter and BER methodology together with experimental characterization in silicon.

5.2 Fulfillment of the Objectives

The objectives introduced in Chapter 1 were addressed throughout the dissertation as summarized in Table 5.1. In particular, the work progressed from the theoretical analysis of the link and equalization problem to a complete post-layout transmitter implementation, allowing the original specifications to be evaluated against extracted circuit behaviour rather than only against an ideal schematic.

Table 5.1: Summary of the fulfillment of the dissertation objectives.

Objective	Status	Reference
Study the SI mechanisms affecting high-speed serial links, including ISI, jitter, channel loss, and equalization.	Completed	Chapter 2
Develop a segmented CML transmitter implementing a three-tap FFE.	Completed	Chapter 3
Derive tap coefficients and current requirements from the USB 3.2 Gen 2 PS and DE targets.	Completed	Section 3.3
Design the differential pairs, current mirrors, biasing, and programmable current sources required by the transmitter.	Completed	Chapter 3
Develop a modular and abuttable physical implementation accounting for layout-dependent effects and differential routing symmetry.	Completed	Section 3.5
Verify the TX before and after parasitic extraction across PVT and resistor corners.	Completed	Chapter 4
Evaluate the implemented TX against the relevant USB 3.2 Gen 2 electrical requirements.	Completed with digital trim	Section 4.4.1

The final objective is qualified as “completed with digital trim” because the post-extraction results demonstrate specification-compatible operating points, but the required digital configuration is not universal across all evaluated conditions. This is particularly evident in the equalization and termination results.

For EQ1, after extraction, the same current-mirror allocation does not provide adequate equalization at every PVT corner. The final selected mapping is 7-102-16 for both TT and FF, whereas the SS corner requires 4-82-20. Similarly, although the termination network does not reproduce the complete 72 Ω to 120 Ω tuning interval for every combination of transistor and resistor corners, at least one compliant termination setting is available for each evaluated condition.

The objective is therefore considered fulfilled at the level of programmable post-layout verification: the implemented architecture provides sufficient adjustment range to recover the evaluated electrical requirements. A practical silicon implementation would, however, require the appropriate equalization and termination codes to be determined through calibration or pre-characterized operating tables.

5.3 Limitations

Although the post-extraction results demonstrate the functionality of the proposed transmitter and the ability of its digital programmability to recover suitable operating points, several limitations remain in the present implementation and verification methodology.

The first limitation is that the results presented in this work are entirely simulation-based. The transmitter was verified at schematic and post-extraction levels, but it was not fabricated and experimentally characterized. Consequently, the reported performance does not include measurement uncertainty, fabrication mismatch, package effects, or other silicon-specific non-idealities that can only be quantified after tape-out. The presented results should therefore be interpreted as post-layout design verification rather than as measured compliance of a fabricated transmitter.

A second limitation is the absence of a complete jitter-compliance characterization. The eye crossover width was used as an indicator of the received transition dispersion, and the strongest equalization presets show a substantial reduction in crossover width after extraction. At the TT corner, the EQ3 crossover width decreases from 45.1 ps before extraction to 15.6 ps after extraction, while the EQ4 crossover width decreases from 49.7 ps to 16.0 ps. The post-extraction EQ3 crossover width also remains relatively stable across the evaluated PVT corners, with values of 15.6 ps, 16.9 ps, and 15.1 ps for TT, FF, and SS, respectively. These measurements indicate a comparatively concentrated distribution of the received transition crossings after extraction.

However, eye crossover width is only an indicator of timing dispersion and cannot be directly equated with the normative jitter quantities. The complete normative TX-eye and jitter procedure, including the required compliance patterns, reference receiver equalization, DJ/RJ/TJ characterization, measurement over 10^6 consecutive UI, and jitter extrapolation to a BER of 10^{-12}, was not performed.

A third limitation is the dependence of post-extraction equalization accuracy on PVT and layout parasitics. The initial tap-current allocations were derived from ideal relationships between the pre-cursor, main-cursor, and post-cursor coefficients, whereas the extracted implementation modifies their effective relative weighting. At the TT and FF corners, preshoot generally remains comparatively close to its target, while de-emphasis exhibits a larger reduction. This effect becomes substantially more pronounced at the SS corner, particularly for the stronger equalization presets. For example, the untuned EQ1 configuration produces 1.41 dB of de-emphasis magnitude at SS, compared with its nominal 2.1 dB target, corresponding to a reduction of approximately 32.9%. Even larger relative DE errors are observed for some of the stronger presets at this corner.

The segmented current-source architecture provides sufficient tuning range to correct this behaviour, but the EQ1 tuning results demonstrate that the required correction is itself corner-dependent. The selected post-extraction mapping is 7-102-16 for both TT and FF, whereas the SS corner requires the substantially different 4-82-20 mapping. Consequently, a single fixed equalization code cannot be assumed to provide the required behaviour across the complete evaluated PVT space. The present implementation does not include an automatic equalization-calibration loop or a stored corner-dependent configuration procedure, and therefore relies on the appropriate digital code being selected externally.

The available simulations also do not uniquely identify a single physical mechanism or individual FFE tap as the cause of the post-extraction equalization error. Parasitic extraction simultaneously modifies the resistance and capacitance associated with the differential signal paths, the current-source routing, and the shared output nodes. The observed equalization changes should therefore be interpreted as the combined effect of the physical implementation on the relative tap contributions. A more detailed sensitivity analysis in which the parasitic contributions of the individual tap paths are introduced or removed independently would be required to quantify their separate contributions.

The programmable termination network presents a related limitation. Its segmented implementation provides sufficient flexibility to obtain an output impedance within the specified range for every evaluated combination of PVT and resistor corner, but the digital code corresponding to a target impedance changes substantially after parasitic extraction and across resistor corners. Furthermore, the complete 72 Ω to 120 Ω tuning interval is not available at every combined corner, even though at least one compliant operating point is available in each evaluated case. A single fixed termination code therefore cannot guarantee the target impedance over all operating conditions. In a fabricated implementation, the appropriate code would need to be determined from the actual device characteristics or selected through a calibration procedure.

The physical implementation also introduces a trade-off between modularity, programmability, routing parasitics, and occupied area. The repeated tap-unit architecture simplifies layout reuse, matching, and digital adjustment, but requires a comparatively large number of devices, control signals, and parallel interconnects. The post-extraction increase in transition times illustrates the electrical cost of this physical complexity. At the representative TT condition, the rise time increases from 27.3 ps to 42.1 ps, while the fall time increases from 22.8 ps to 35.6 ps, corresponding to increases of approximately 54.2% and 56.1%, respectively. Further reduction of routing resistance and capacitance could improve the high-speed response, but such changes must be balanced against area, routing congestion, EM constraints, and preservation of the modular layout methodology.

Finally, the SI verification was performed using one specific USB-IF reference channel. The selected channel is intentionally demanding, with 21.6 dB insertion loss at the 5 GHz Nyquist frequency, and provides a useful benchmark for evaluating the equalization capability of the transmitter. Nevertheless, it represents only one link condition rather than the complete range of channels that can occur in a practical system. The equalization setting that maximizes the received eye

can change with channel loss, package parasitics, connector discontinuities, and receiver characteristics. Verification over a broader set of channel models and, ultimately, with a complete transmitter–channel–receiver link would therefore be required to establish the robustness of the proposed equalization strategy under a wider range of system conditions.

5.4 Future Work

The most direct continuation of this work is the fabrication and experimental characterization of the proposed transmitter. A silicon prototype would allow the simulated current-source accuracy, output swing, differential output impedance, transition times, equalization settings, eye diagrams, and power consumption to be compared directly with measured results. This would also make it possible to quantify the correlation between the schematic, post-extraction, and fabricated implementations and to identify additional non-idealities associated with device mismatch, package parasitics, supply noise, and measurement interfaces that are not captured completely by the present simulation methodology.

A fabricated implementation should also undergo a complete physical-reliability sign-off and validation. In addition to functional and signal-integrity measurements, the final chip should be evaluated for EM and supply-network IR drop under the worst expected operating conditions. This is particularly relevant for the transmitter output stage, programmable current-source banks, termination network, and their associated supply and ground interconnects, where relatively large currents are concentrated in a limited routing area. Excessive current density can accelerate electromigration and compromise long-term interconnect reliability, while resistive voltage drops in the power-distribution network can modify the effective bias conditions, output swing, tap-current ratios, and consequently the equalization behaviour of the transmitter.

Such verification should include current-density checks in metal segments and vias, static and dynamic IR-drop analysis, and thermal or self-heating assessment under representative and worst-case switching activity. These analyses should be performed across the relevant PVT conditions and should account for the simultaneous activation of the current-source and termination segments associated with the most demanding transmitter configurations. Where necessary, the power-routing width, via multiplicity, and local supply distribution should be reinforced to satisfy the technology reliability limits. Combined with silicon measurements and, where applicable, accelerated stress testing, these checks would provide confidence that the transmitter not only meets its electrical specifications at initial operation but also maintains them throughout the intended operating lifetime.

Experimental characterization should also include a complete USB 3.2 Gen 2 transmitter compliance procedure. In particular, the normative decomposition of the transmitter jitter into DJ, RJ, and TJ should be performed using the appropriate reference receiver and statistical methodology defined by the standard. A direct BER-based evaluation should likewise be included. Although the eye crossover measurements obtained in this work provide a useful indication of the received

transition dispersion, they cannot replace these normative jitter and error-rate measurements. Silicon characterization would therefore provide the final step required to determine whether the post-layout electrical compatibility demonstrated in this dissertation translates into complete experimental compliance.

A particularly important extension is the development of a calibration strategy for both equalization and termination. The post-extraction EQ1 results demonstrate that the required current-source configuration is PVT-dependent. The selected compliant mapping is 7-102-16 at both TT and FF, whereas the SS corner requires the substantially different 4-82-20 mapping. The existing segmented architecture already provides the actuation mechanism required for calibration, since the pre-cursor, main-cursor, and post-cursor current-source mirrors can be controlled independently.

A practical implementation could therefore employ either a start-up calibration procedure or a set of pre-characterized configuration tables to select the appropriate tap-current codes according to the actual operating conditions. The same calibration process could determine the termination-bank code required to place the differential output impedance within its target range. Ultimately, an automatic calibration loop would remove the need to assume a single universal equalization or termination code and would make the transmitter substantially more robust to process, supply-voltage, temperature, and passive-device variation.

The post-extraction results also identify opportunities for further physical optimization. The substantial increase in output rise and fall times shows that the parasitic resistance and capacitance associated with the high-speed differential paths remain important limitations of the current layout. Furthermore, the change in equalization accuracy after extraction indicates that the physical implementation modifies the relative contribution of the three FFE taps. Future layout iterations should therefore aim to reduce and balance the parasitic loading of the differential output and tap-current paths, while also reducing the series resistance associated with the programmable termination network.

These modifications should be guided by a more detailed parasitic sensitivity analysis. In the present work, the extracted network includes the parasitic contributions of all tap and output paths simultaneously, making it difficult to identify the individual contribution of each physical mechanism to the observed equalization error. A future analysis could selectively introduce the extracted resistance and capacitance associated with the pre-cursor, main-cursor, post-cursor, and shared output paths in order to quantify their separate impact on preshoot, de-emphasis, transition time, and received eye opening. This would allow subsequent layout changes to target the dominant parasitic contributions rather than relying on qualitative assumptions regarding the origin of the post-extraction error.

The current-source segmentation and digital activation strategy could also be investigated further. Alternative distributions of enabled current-mirror segments across the available tap units could be compared to determine whether a more spatially distributed activation sequence improves current matching or reduces sensitivity to local layout-dependent effects. Such an investigation should be supported by extracted and mismatch-aware simulations, since the present results are

not sufficient to conclude whether activation ordering is a dominant source of the measured tap-current error. Any improvement would also need to be balanced against the additional decoder complexity, routing density, and layout area required by a more elaborate control scheme.

Another useful extension would be to incorporate post-extraction equalization optimization directly into the transmitter design flow. In the present methodology, the initial digital preset mappings are derived from analytical tap coefficients and subsequently corrected after extraction. The strong corner-dependence observed for EQ1 demonstrates that the final physical implementation can require substantially different configurations from those predicted by the ideal current relationships.

A future optimization flow could therefore include the extracted transmitter, the target channel, and the relevant PVT condition directly in an automated search for the digital current-source and termination codes. The optimization criterion should not be limited to minimizing the numerical error relative to the nominal preshoot and de-emphasis targets. As demonstrated by the EQ1 tuning procedure, the configuration having the smallest target error is not necessarily the preferred configuration if one of the resulting quantities lies outside its normative range. The optimization should therefore treat the USB preshoot, de-emphasis, swing, and impedance intervals as explicit constraints while optimizing a system-level metric such as received eye height, eye width, power consumption, or a combination of these quantities.

The channel space considered during verification could likewise be expanded. The present work uses one demanding USB-IF reference channel with 21.6 dB insertion loss at the 5 GHz Nyquist frequency. Future verification should include channels with different insertion-loss profiles, impedance discontinuities, connector and package parasitics, and different overall link lengths. This would allow the required equalization settings to be characterized as a function of the actual channel rather than for a single reference case, and would provide a more complete evaluation of the useful tuning range offered by the segmented transmitter.

Finally, the transmitter could be extended toward a complete PHY implementation. Integration with the preceding high-speed data path, digital control and calibration logic, realistic package and interface models, and a representative receiver would enable end-to-end link verification. Such a platform would allow the interaction between transmitter equalization, channel loss, receiver equalization, jitter, and BER to be evaluated jointly, rather than treating the transmitter as an isolated block.

The next development stage is therefore centered on converting the post-layout-verified transmitter into a calibrated, reliability-verified, and experimentally validated system. The results of this dissertation indicate that the segmented architecture provides sufficient adjustment range after physical implementation, but they also show that this programmability must be used actively to compensate for layout- and PVT-dependent behaviour. Automatic calibration, targeted physical optimization, EM and IR-drop verification, broader channel characterization, and silicon measurement therefore constitute the principal steps toward a complete and robust USB 3.2 Gen 2 transmitter implementation.

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Appendix A

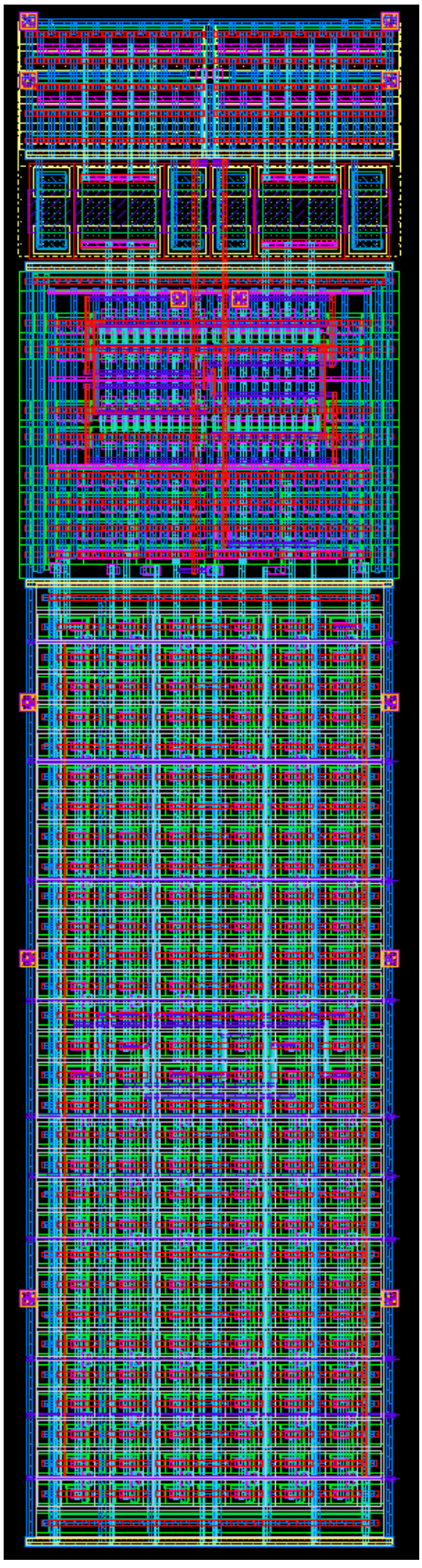


Figure A.1: Layout of the tap unit standard cell

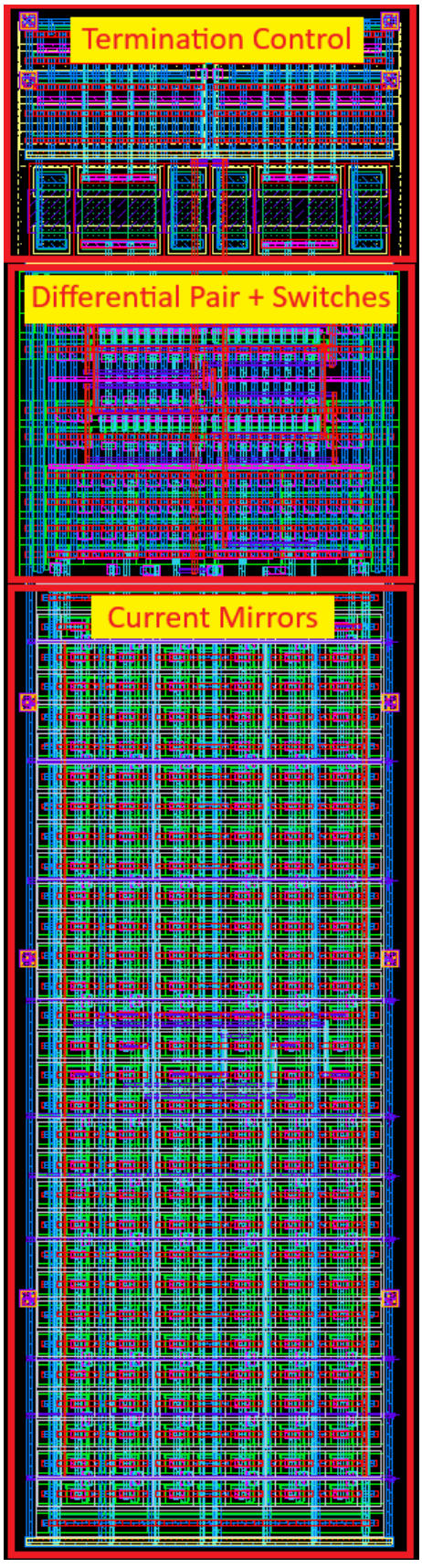


Figure A.2: Annotated layout of the tap unit standard cell

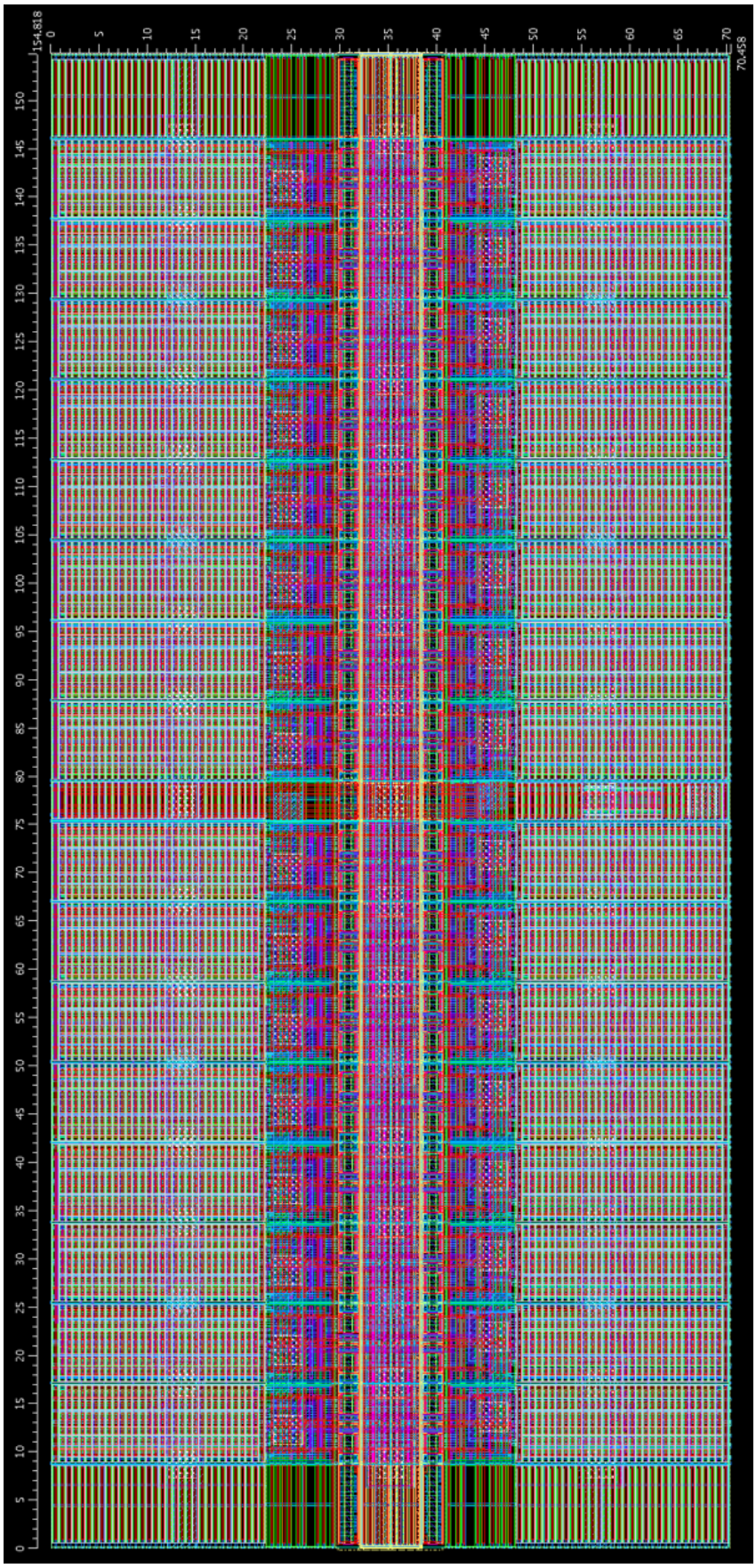


Figure A.3: Global layout of the driver

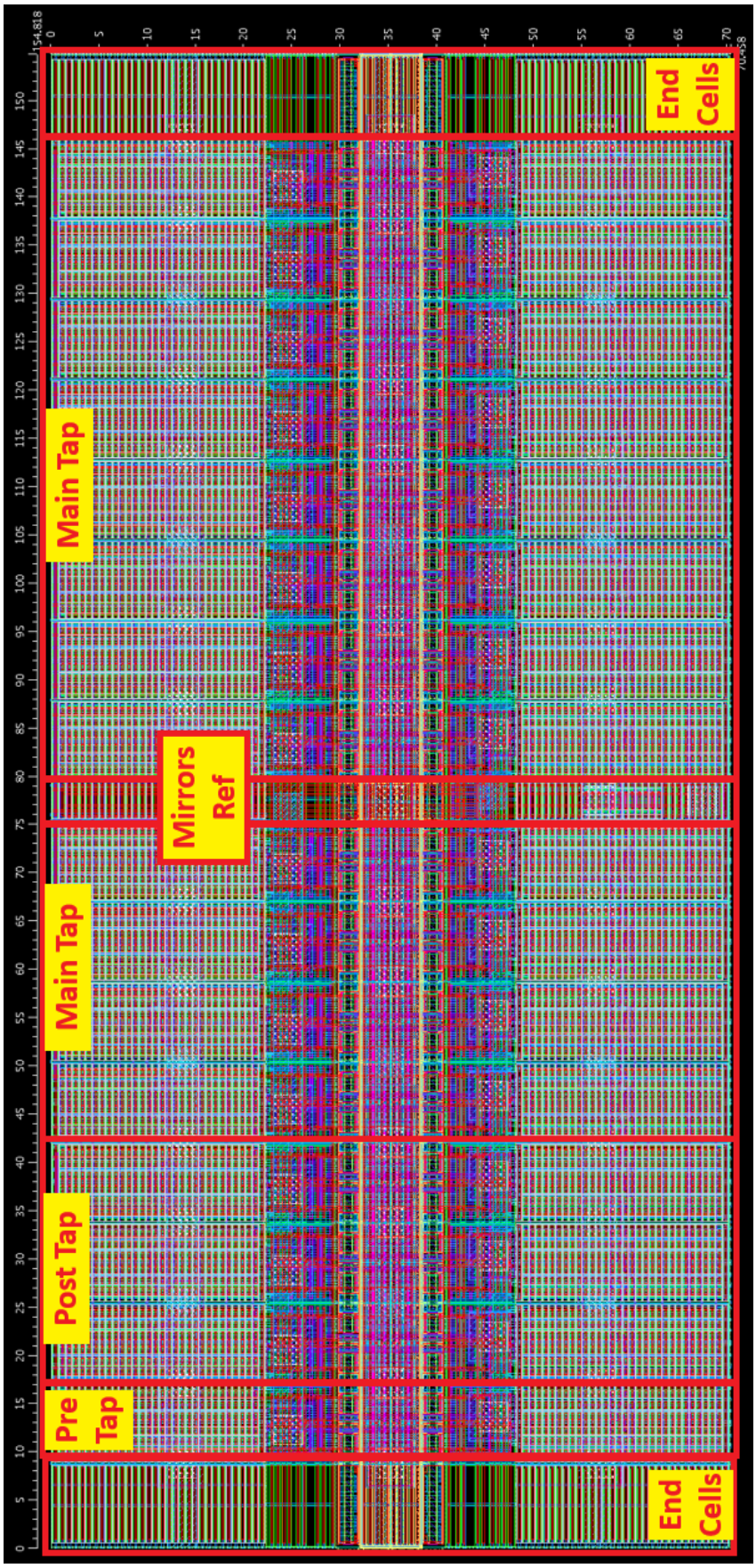


Figure A.4: Annotated global layout of the driver



Figure A.5: Driver layout floorplan